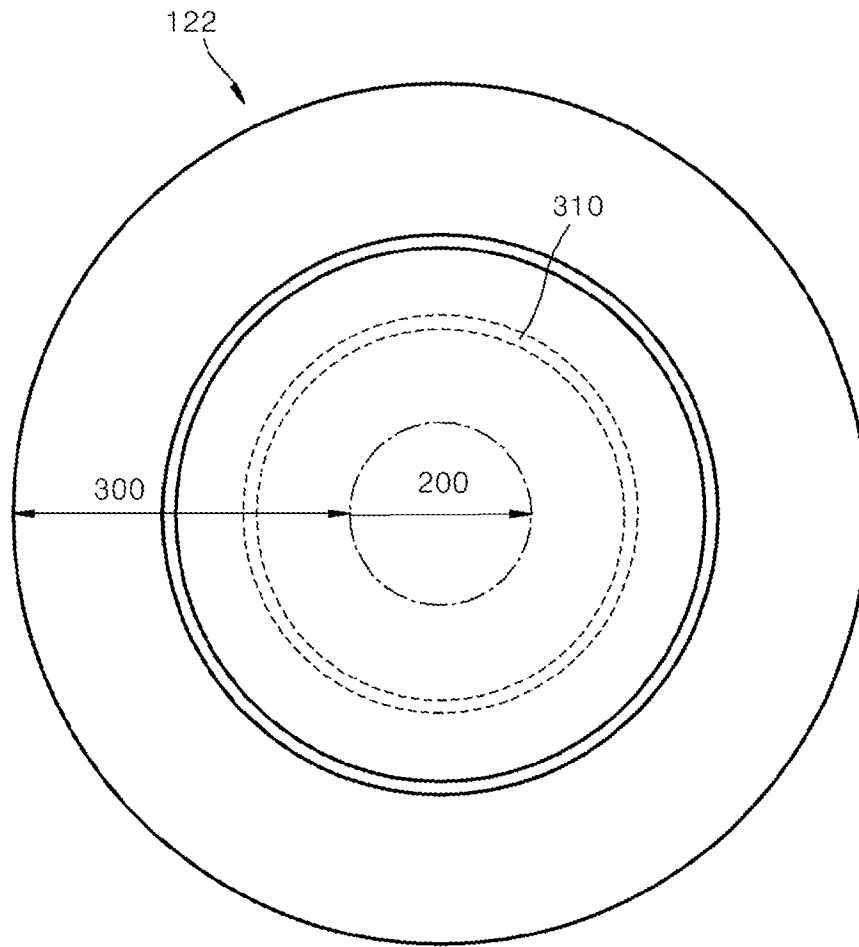


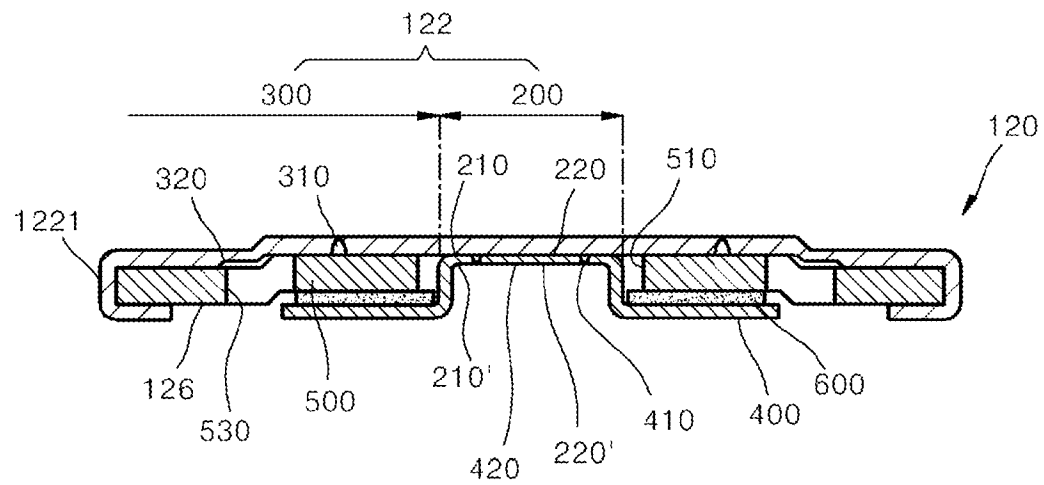
relieving portion adjacent to the breakable notch to space the current-carrying contact from the cap-down and forming a space where a rupture pressure is applied to the breakable junction.

- [Claim 10] The cap assembly according to claim 9, wherein the bottom width of the bending-relieving portion is larger than that of the breakable notch.
- [Claim 11] The cap assembly according to claim 1, wherein the cap-down comprises a through-hole that is spaced apart from and formed around the central hole and through which internal gases and liquids pass.
- [Claim 12] The cap assembly according to claim 1, wherein the insulating layer extends beyond the end of the sub-plate and reaches the end of the cap-down.
- [Claim 13] The cap assembly according to claim 12, wherein the insulating layer has a through-hole opposite to the through-hole of the cap-down.
- [Claim 14] The cap assembly according to claim 13, wherein spacer bridges are formed across the opposite through-hole.
- [Claim 15] The cap assembly according to claim 1, wherein a short-circuit preventing layer is formed at a position opposite to the end of the battery can through a gasket of the safety vent.
- [Claim 16] A secondary battery comprising the cap assembly according to any one of claims 1 to 15.

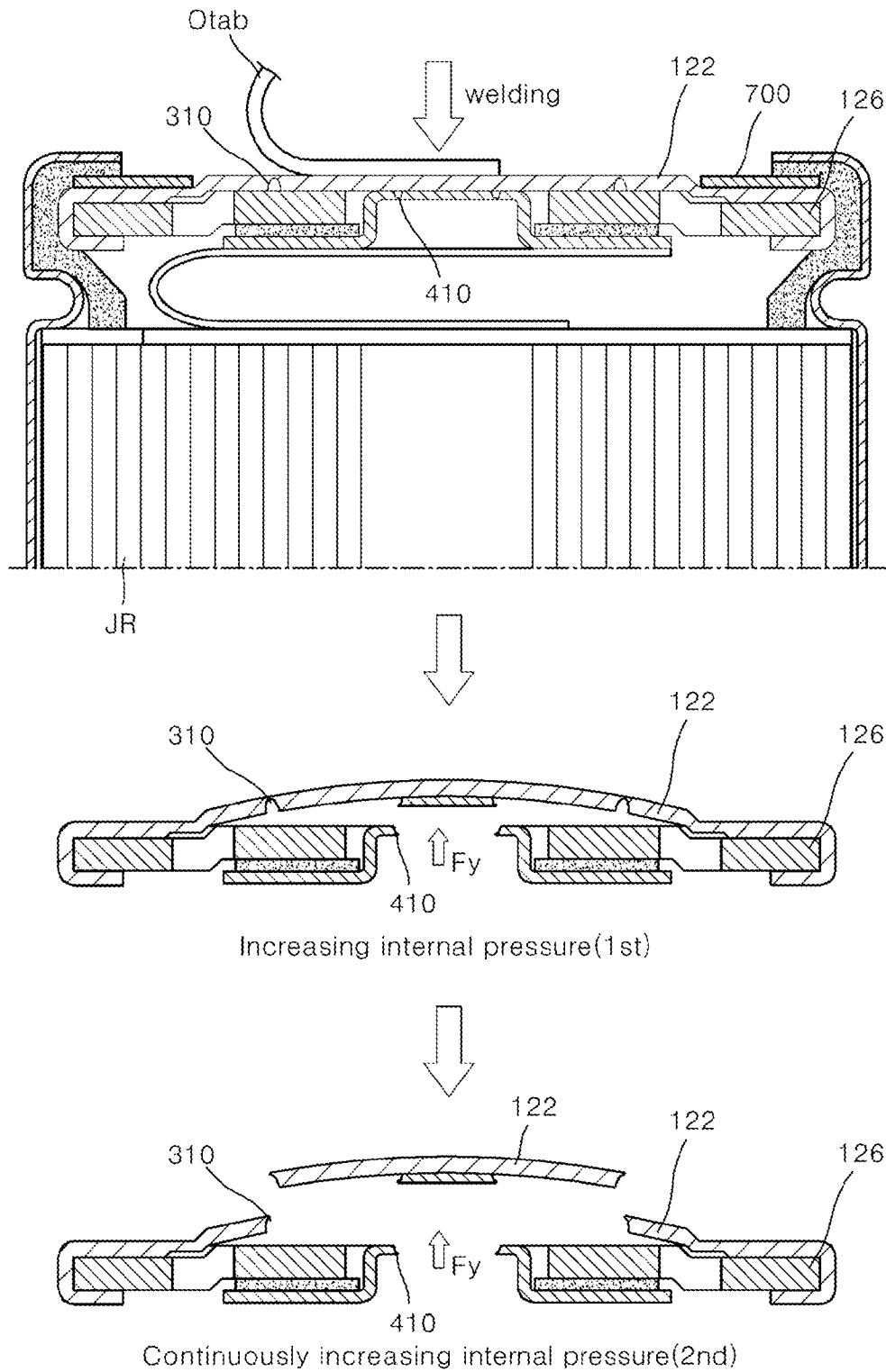
[Fig. 1]



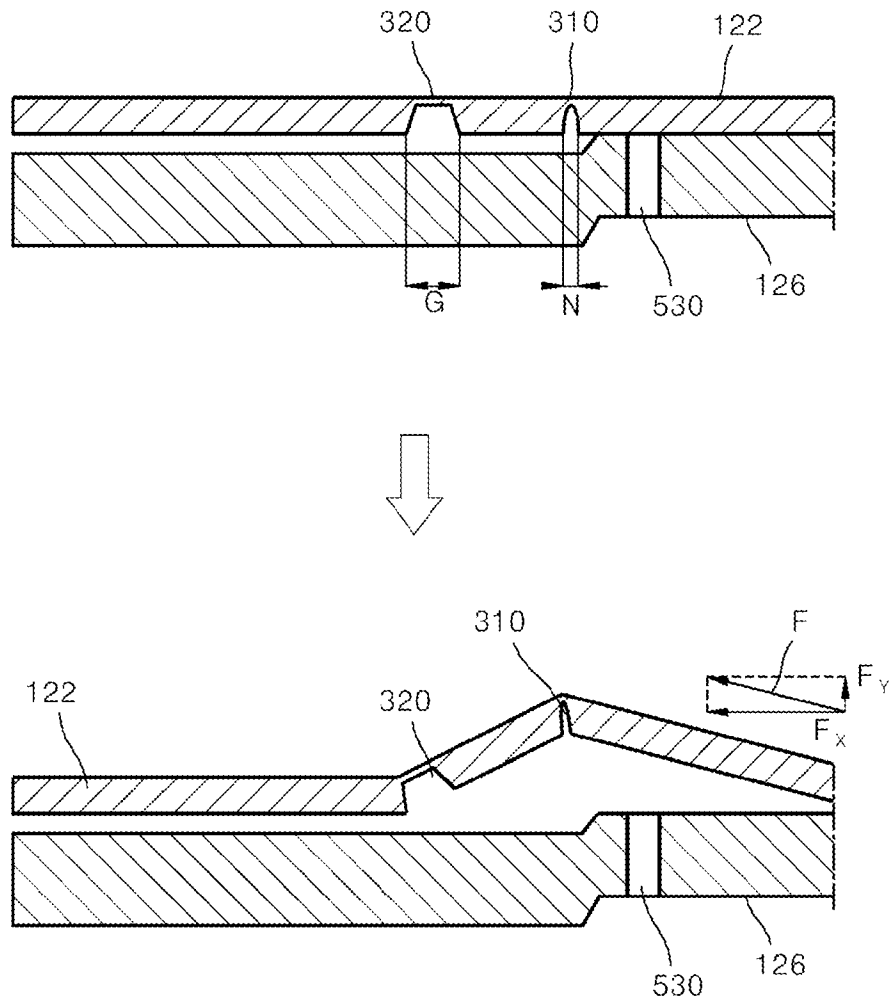
[Fig. 2]



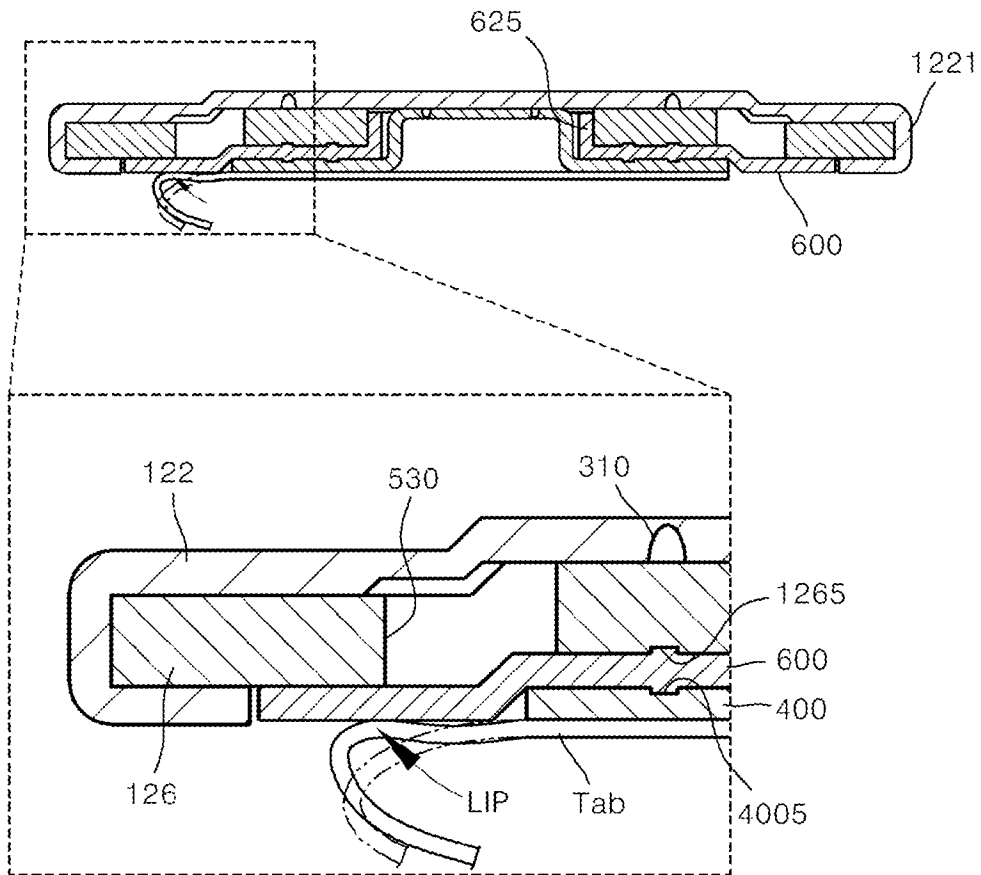
[Fig. 3]



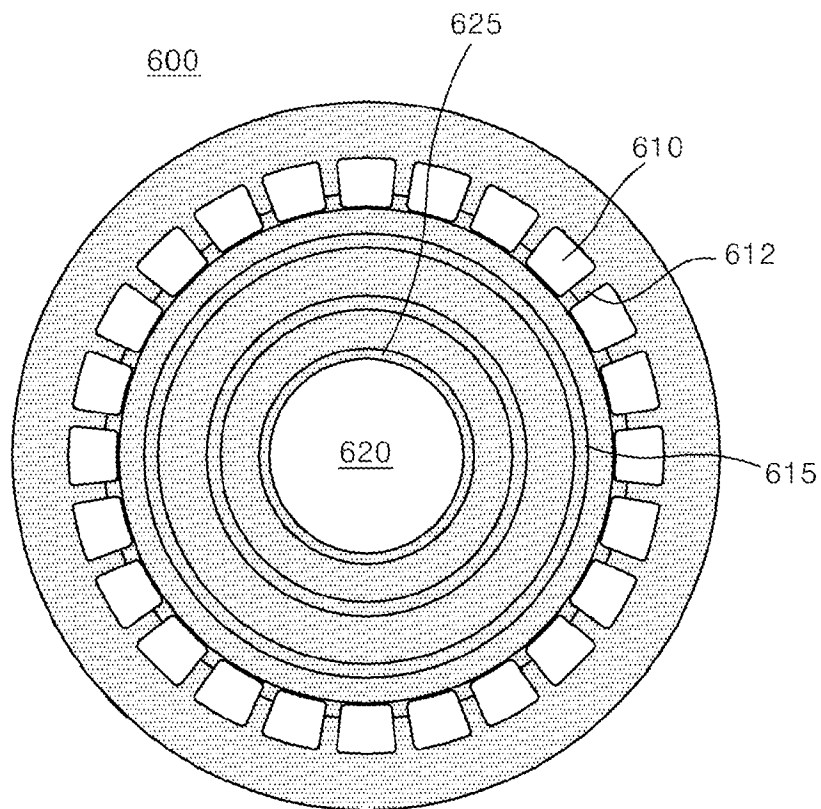
[Fig. 4]



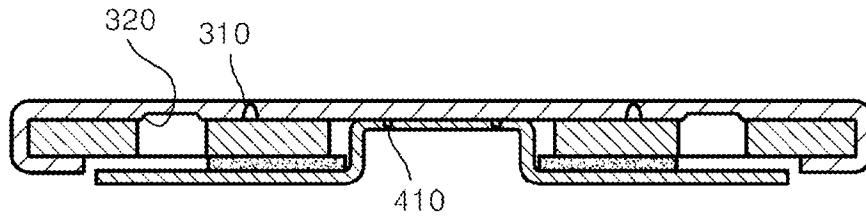
[Fig. 5]



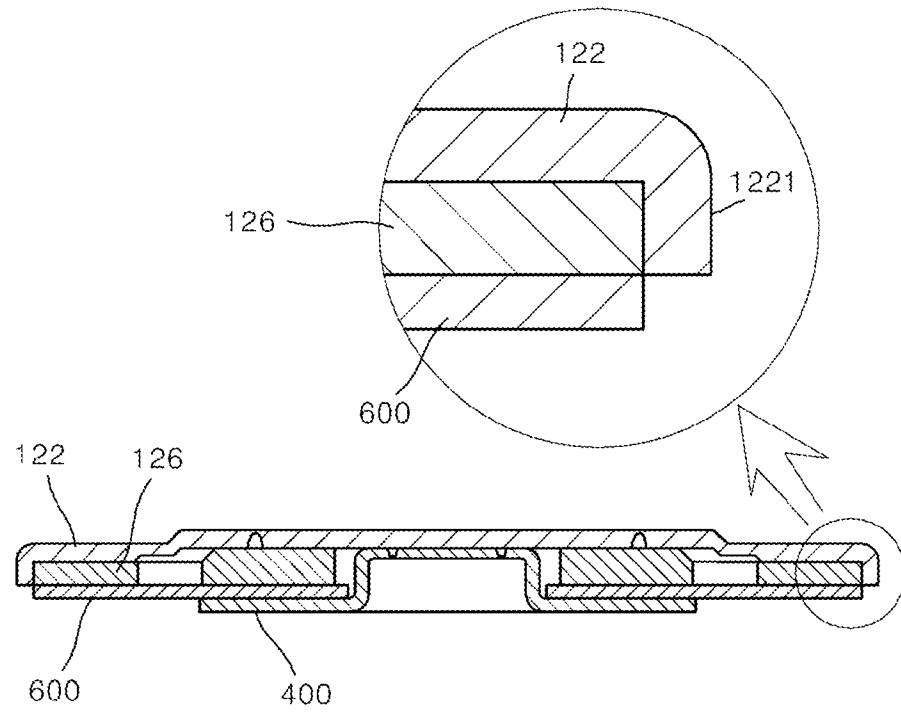
[Fig. 6]



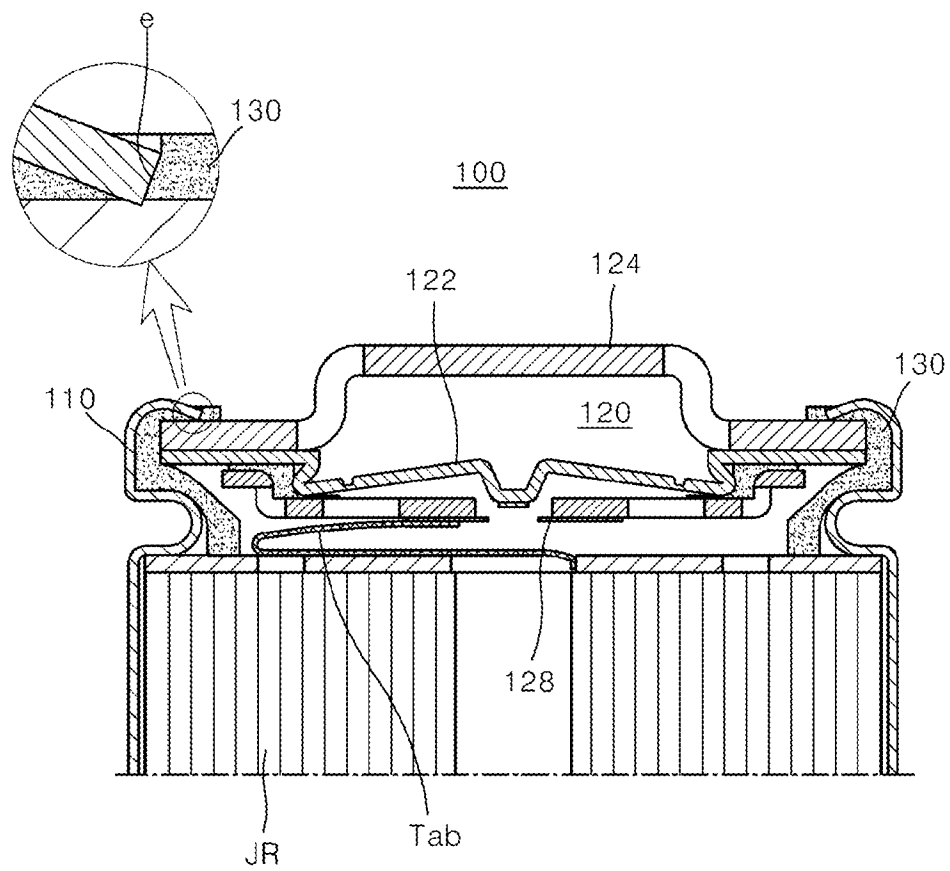
[Fig. 7]



[Fig. 8]



[Fig. 9]



INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2017/008736**A. CLASSIFICATION OF SUBJECT MATTER****H01M 2/04(2006.01)i, H01M 2/12(2006.01)i, H01M 10/04(2006.01)i**

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/04; H01M 2/12; H01M 2/08; H01M 2/34; H01M 2/20; H01M 2/26; H01M 2/10; H01M 2/02; H01M 10/04

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models

Japanese utility models and applications for utility models

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS(KIPO internal) & keywords: cap assembly, secondary battery, safety bent, shorting, notch

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KR 10-2016-0034794 A (SHIN HEUNG ENERGY & ELECTRONIC CO., LTD.) 30 March 2016 See figures 1-7; and claims 1-2.	1-16
A	US 2011-0206958 A1 (KIYAMA, A. et al.) 25 August 2011 See the entire document.	1-16
A	JP 2005-100927 A (MATSUSHITA ELECTRIC INDUSTRY CO., LTD.) 14 April 2005 See the entire document.	1-16
A	KR 10-2010-0062886 A (SAMSUNG SDI CO., LTD.) 10 June 2010 See the entire document.	1-16
A	KR 10-2006-0065378 A (LG CHEM, LTD.) 14 June 2006 See the entire document.	1-16



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

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INTERNATIONAL SEARCH REPORT

Information on patent family members

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PCT/KR2017/008736

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
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US 2011-0206958 A1	25/08/2011	JP 2010-113929 A	20/05/2010
		JP 4596289 B2	08/12/2010
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		US 8486546 B2	16/07/2013
KR 10-2006-0065378 A	14/06/2006	KR 10-0858414 B1	11/09/2008



IRISH PATENTS OFFICE

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CAP ASSEMBLY FOR RECHARGEABLE BATTERY

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Applicant(s): LG CHEMICAL LTD [KR] ± (주식회사 엘지화학)

Classification: - **international:** H01M2/04; H01M2/12
- **cooperative:** H01M50/152 (KR); H01M50/3425 (KR); H01M50/383 (KR); H01M2200/20 (KR); Y02E60/10 (EP, KR)

Application number: KR20160054122 20160502 Global Dossier

Priority number(s): KR20160054122 20160502

Also published as: KR102155706 (B1)

Abstract of KR20170124343 (A)

The present invention relates to a cap assembly for a secondary battery. According to the present invention, the cap assembly for a secondary battery comprises: a top cap located in an outer side of a battery wherein the top cap has a discharging hole formed therein; and a safe vent having a rupture notch ruptured in case of an internal pressure increase to discharge an internal gas to the discharging hole and having a bending notch formed on the rupture unit of which a location or a shape is changed due to ruptures wherein the bending notch is bent in case of ruptures of the rupture notch.



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(52) CPC특허분류

HOIM 2/1247 (2013.01)

HOLM 2/046 (2013.01)

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(74) 대리인

특허법인 태평양

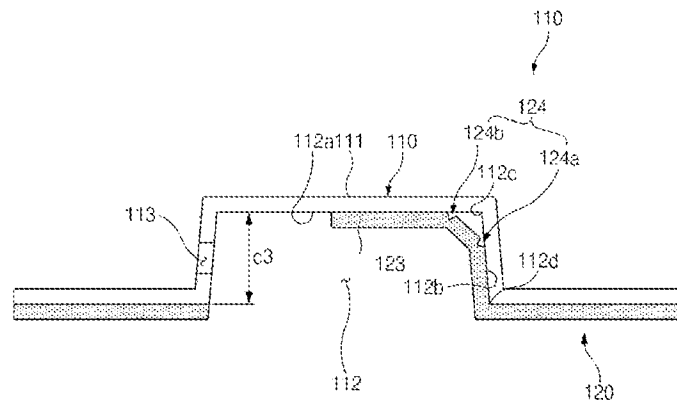
전체 청구항 수 : 총 26 항

(54) 발명의 명칭 이차전지용 캡 조립체

(00) 2 94

본 발명은 이차전지용 캡 조립체에 관한 것으로, 본 발명에 따른 이차전지용 캡 조립체는, 전지의 외각에 위치되어 배출구가 형성된 탑 캡 및 상기 배출구로 내부 가스의 배출이 가능하도록 내압 상승시 파열되는 파열 노치가 형성되고, 상기 파열 노치의 파열시 절곡되는 절곡 노치가 상기 파열로 인해 위치 또는 형태가 변하는 파열부 상에 형성되는 안전 배트를 포함한다.

44 E - 53



(52) CPC특허분류

H01M 2/1241 (2013.01)

H01M 2200/20 (2013.01)

Y02E 60/122 (2013.01)

명세서

청구범위

청구항 1

전지의 외각에 위치되어 배출구가 형성된 탭 캡; 및

상기 배출구로 내부 가스의 배출이 가능하도록 내압 상승시 파열되는 파열 노치가 형성되고, 상기 파열 노치의 파열시 절곡되는 절곡 노치가 상기 파열로 인해 위치 또는 형태가 변하는 파열부 상에 형성되는 안전 밴트를 포함하는 이차전지용 캡 조립체.

청구항 2

청구항 1에 있어서,

상기 탭 캡 보다 상기 전지의 내측에 위치된 상기 안전 밴트에서 상기 파열 노치의 파열시 상기 파열부가 상기 탭 캡과 충돌되며 상기 절곡 노치가 절곡되는 이차전지용 캡 조립체.

청구항 3

청구항 2에 있어서,

상기 파열 노치의 파열 전에는 상기 탭 캡의 상기 배출구는 상기 안전 밴트에 의해 모두 폐쇄되고,

상기 파열 노치의 파열 및 상기 절곡 노치의 절곡 후에는, 상기 배출구가 모두 개방되는 이차전지용 캡 조립체.

청구항 4

청구항 2에 있어서,

상기 파열 노치의 파열시 상기 파열부가 상기 탭 캡의 내측면과 대응되는 형태로 절곡되도록 상기 절곡 노치가 절곡되는 이차전지용 캡 조립체.

청구항 5

청구항 2에 있어서,

상기 파열 노치는 상기 안전 밴트의 평면도 상으로 일측이 개방된 원호 형태로 형성되는 이차전지용 캡 조립체.

청구항 6

청구항 5에 있어서,

상기 절곡 노치는 상기 평면도 상에서 상기 파열 노치의 원호 내측부에 형성되는 이차전지용 캡 조립체.

청구항 7

청구항 6에 있어서,

상기 절곡 노치는 상기 파열 노치의 개방 부위와 상기 파열 노치의 원호 중심 사이에 위치되는 이차전지용 캡 조립체.

청구항 8

청구항 6에 있어서,

상기 절곡 노치는 상기 평면도 상에서 직선 형태로 형성되는 이차전지용 캡 조립체.

청구항 9

청구항 6에 있어서,

상기 절곡 노치는 상기 평면도 상에 형성되는 상기 파열 노치의 원호 중심에서 상기 파열 노치의 개방 방향으로 연장된 선에 대해 수직으로 라인을 형성하는 이차전지용 캡 조립체.

청구항 10

청구항 5에 있어서,

상기 절곡 노치는 평면도 상에 형성되는 상기 파열 노치의 원호 중심에서 상기 파열 노치의 개방 부위로 향하는 방향에 대해 기울어진 방향으로 연장되며 형성되는 이차전지용 캡 조립체.

청구항 11

청구항 10에 있어서,

상기 절곡 노치의 양단은 상기 파열 노치의 원호 상에 위치되는 이차전지용 캡 조립체.

청구항 12

청구항 1 내지 청구항 11 중 어느 한 항에 있어서,

상기 파열 노치가 상기 절곡 노치 보다 먼저 파열되도록, 상기 파열 노치가 상기 절곡 노치 보다 크게 형성되는 이차전지용 캡 조립체.

청구항 13

청구항 12에 있어서,

상기 파열 노치의 깊이는 상기 절곡 노치의 깊이 보다 깊게 형성되는 이차전지용 캡 조립체.

청구항 14

청구항 12에 있어서,

상기 파열 노치의 폭은 상기 절곡 노치의 폭 보다 넓게 형성되는 이차전지용 캡 조립체.

청구항 15

청구항 1 내지 청구항 11 중 어느 한 항에 있어서,

상기 절곡 노치는 복수개로 형성되는 이차전지용 캡 조립체.

청구항 16

청구항 5 내지 청구항 10 중 어느 한 항에 있어서,

상기 탭 캡은 상기 안전 벨트의 상기 파열부가 수용되는 수용부를 포함하는 이차전지용 캡 조립체.

청구항 17

청구항 16에 있어서,

상기 배출구는 상기 수용부의 측면에 형성되어 상기 전지의 내부에서 발생하는 가스가 배출되게 하는 이차전지용 캡 조립체.

청구항 18

청구항 17에 있어서,

상기 탭 캡에서 수용부의 천장면과 측면 사이의 각도는 수직으로 형성되는 이차전지용 캡 조립체.

청구항 19

청구항 17에 있어서,

상기 탭 캡에서 수용부의 천장면과 측면 사이의 각도는 85~100° 로 형성되는 이차전지용 캡 조립체.

청구항 20

청구항 17에 있어서,

상기 안전 밴트에서 상기 파열 노치의 파열시 평면상으로 상기 파열 노치의 원호 상 노치가 형성되지 않은 부분인 비노치부가 상기 탭 캡 방향으로 절곡되도록 형성되는 이차전지용 캡 조립체.

청구항 21

청구항 20에 있어서,

상기 안전 밴트의 상기 비노치부는 상기 파열 노치의 파열시 상기 수용부의 가장자리에 의해 절곡되는 이차전지용 캡 조립체.

청구항 22

청구항 21에 있어서,

상기 절곡 노치는 상기 파열 노치의 파열시 상기 수용부의 천장 모서리부에 대응되는 위치에 위치되도록 형성되는 이차전지용 캡 조립체.

청구항 23

청구항 21에 있어서,

상기 절곡 노치는 제1 절곡 노치 및 제2 절곡 노치를 포함하고,

상기 제1 절곡 노치 및 상기 제2 절곡 노치는 상호 일정거리 이격되어 나란하게 형성되는 이차전지용 캡 조립체.

청구항 24

청구항 23에 있어서,

상기 비노치부에서 상기 제1 절곡 노치 까지의 거리는 상기 탭 캡의 상기 수용부 깊이 보다 작게 형성되고,

상기 비노치부에서 상기 제2 절곡 노치 까지의 거리는 상기 탭 캡의 수용부 깊이 보다 크게 형성되는 이차전지용 캡 조립체.

청구항 25

청구항 20에 있어서,

상기 수용부를 중심으로 4방향 중 어느 한 방향에 상기 안전 밴트의 상기 비노치부가 위치되고, 나머지 하나 이상의 방향에 상기 배출구가 위치되는 이차전지용 캡 조립체.

청구항 26

청구항 20에 있어서,

상기 수용부를 중심으로 4 방향 중 어느 한 방향에 상기 안전 밴트의 상기 비노치부가 위치되고, 나머지 3 방향에 상기 배출구가 위치되는 이차전지용 캡 조립체.

발명의 설명

기술분야

[0001] 본 발명은 이차전지용 캡 조립체에 관한 것이다.

배경기술

[0003] 이차 전지는 일차 전지와는 달리 재충전이 가능하고, 또 소형 및 대용량화 가능성으로 인해 근래에 많이 연구

개발되고 있다. 모바일 기기에 대한 기술 개발과 수요가 증가함에 따라 에너지원으로서의 이차 전지의 수요가 급격하게 증가하고 있다.

[0004] 특히, 리튬 이차 전지는 작동 전압이 3.6V로서, 전자 장비의 전원으로 많이 사용되는 니켈-카드뮴 전지 또는 니켈-수소 전지보다 약3배의 용량을 가지며, 단위 중량당 에너지 밀도가 높기 때문에 그 활용 정도가 급속도로 증가되는 추세에 있다.

[0005] 이러한 리튬 이차 전지는 주로 리튬계 산화물과 탄소재를 각각 양극 활물질과 음극 활물질로 사용한다. 또한, 리튬 이차 전지는 각형 전지, 원통형 전지, 파우치형 전지로 구분될 수 있다.

[0006] 리튬 이온 이차 전지는 양극/세퍼레이터/음극이 순차적으로 배치된 전극 조립체와, 전극 조립체를 전해액과 함께 밀봉 수납하는 외장재를 구비한다. 특히, 각형 또는 원통형 이차 전지의 외장재는 개방단이 형성된 케이스 및 케이스의 개방단에 밀봉 결합되는 캡 조립체를 구비한다.

선행기술문헌

특허문헌

[0008] (특허문헌 0001) 한국 공개특허 제10-2009-0105547호

발명의 내용

해결하려는 과제

[0009] 본 발명의 하나의 관점은 전지 내부의 가스 발생으로 전지의 내압 상승시 가스를 전지 외부로 용이하게 배출할 수 있는 이차전지용 캡 조립체를 제공하기 위한 것이다.

과제의 해결 수단

[0011] 본 발명의 일 실시예에 따른 이차전지용 캡 조립체는, 전지의 외각에 위치되어 배출구가 형성된 탑 캡 및 상기 배출구로 내부 가스의 배출이 가능하도록 내압 상승시 파열되는 파열 노치가 형성되고, 상기 파열 노치의 파열 시 절곡되는 절곡 노치가 상기 파열로 인해 위치 또는 형태가 변하는 파열부 상에 형성되는 안전 벤트를 포함할 수 있다.

발명의 효과

[0013] 본 발명에 따른 이차전지용 캡 조립체는 전지 내부의 가스 발생으로 전지의 내압 상승시 가스 배출이 용이한 형태로 탑 캡 및 안전 벤트가 형성되어 가스 배출의 용이성을 확보할 수 있다.

[0014] 특히, 본 발명에 따른 이차전지용 캡 조립체는 내압 상승으로 인해 파열되는 안전 벤트의 파열부가 가스 배출이 용이한 형태로 파열 및 절곡되어 가스 배출에 필요한 통로를 용이하게 확보할 수 있다.

도면의 간단한 설명

[0016] 도 1은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체가 설치되는 이차전지를 도시한 절개 사시도이다.

도 2는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체를 나타낸 단면도이다.

도 3은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 벤트가 파열된 후 상태를 나타낸 단면도이다.

도 4는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 벤트가 파열된 후 상태를 나타낸 개념도이다.

다.

도 5는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 밴트를 나타낸 평면도이다.

도 6는 도 5에서 A-A'선을 따라 절개한 단면이다.

발명을 실시하기 위한 구체적인 내용

- [0017] 본 발명의 목적, 특정한 장점들 및 신규한 특징들은 첨부된 도면들과 연관되어지는 이하의 상세한 설명과 바람직한 실시예들로부터 더욱 명백해질 것이다. 본 명세서에서 각 도면의 구성요소들에 참조번호를 부가함에 있어서, 동일한 구성 요소들에 한해서는 비록 다른 도면상에 표시되더라도 가능한 한 동일한 번호를 가지도록 하고 있음에 유의하여야 한다. 또한, 본 발명은 여러 가지 상이한 형태로 구현될 수 있으며 여기에서 설명하는 실시예에 한정되지 않는다. 그리고, 본 발명을 설명함에 있어서, 본 발명의 요지를 불필요하게 흐릴 수 있는 관련된 공지 기술에 대한 상세한 설명은 생략하도록 한다.
- [0019] 도 1은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체가 설치되는 이차전지를 도시한 절개 사시도이고, 도 2는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체를 나타낸 단면도이다.
- [0020] 도 1 및 도 2를 참고하면, 본 발명의 일 실시예에 따른 이차전지용 캡 조립체(100)는, 배출구(113)가 형성된 탑 캡(Top cap)(110) 및 이차 전지(10)의 내압 상승시 파열되는 파열 노치(Notch)(121) 및 절곡되는 절곡 노치(124)가 형성된 안전 밴트(bent)(120)를 포함한다.
- [0022] 도 3은 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 밴트가 파열된 후 상태를 나타낸 단면도이고, 도 4는 본 발명의 일 실시예에 따른 이차전지용 캡 조립체에서 안전 밴트가 파열된 후 상태를 나타낸 사시도이다.
- [0023] 이하에서, 도 1 내지 도 6을 참조하여, 본 발명의 일 실시예인 이차전지용 캡 조립체에 대해 보다 상세히 설명하기로 한다.
- [0024] 도 1 및 도 2를 참고하면, 탑 캡(110)은 이차 전지(10)의 측외각에 위치되어, 외부 단자(111)와, 수용부(112) 및 배출구(113)를 포함하여 형성될 수 있다.
- [0025] 외부 단자(111)는 탑 캡(110)의 상단에 돌출 형성되어 이차 전지(10)의 내부에 수용된 전극 적층체(12)와 전기적으로 연결되어 양극 단자 또는 음극 단자를 형성할 수 있다.
- [0026] 전극 적층체(12)는 케이스(11)에 수용되고, 케이스(11)의 내부에 전해액이 주입된다. 또한, 전극 적층체(12)는 충방전이 가능한 발전소자로서, 양극(12a)과 음극(12b) 및 분리막(12c)이 겹겹되어 적층된 구조를 형성한다. 여기서, 전극 적층체(12)의 양극(12a)은 탑 캡(110)의 외부 단자(111)와 전기적으로 연결되어 외부 단자(111)가 양극 단자의 기능을 할 수 있다.
- [0027] 분리막(12c)은 양극(12a)과 음극(12b)을 분리하여 전기적으로 절연시킨다. 이때, 양극(12a) 및 음극(12b)은 분리막(12c)과 함께 권취되어 젤리 롤(Jelly roll) 형으로 형성될 수 있다. 아울러, 분리막(12c)은 절연 재질로 이루어져 양극(12a)과 음극(12b) 사이를 전기적으로 절연한다. 여기서, 분리막(12c)은 예를 들어 미다공성을 가지는 폴리에틸렌, 폴리프로필렌 등 폴리올레핀계 수지막으로 형성될 수 있다.
- [0028] 한편, 전극 적층체(12)의 중심부에는 원통형의 센터 핀(13)이 구비될 수 있다. 센터 핀(13)은 소정의 강도를 부여하기 위해 금속 소재로 이루어질 수 있으며, 관재를 통괄해 절곡한 중공형의 원통형 구조로 이루어질 수 있다. 이러한 센터 핀(13)은 전극 적층체(12)를 고정 및 지지하는 작용과 충방전 및 작동시 내부 반응에 의해 발생하는 가스를 방출하는 통로로서 작용할 수 있다.
- [0029] 도 2 및 도 3을 참고하면, 수용부(112)는 외부 단자(111)의 하부에 안전 밴트(120)의 파열부(123)가 수용되도록 안전 밴트(120) 방면으로 개방된 형태로 형성될 수 있다.
- [0030] 또한, 탑 캡(110)에서 수용부(112)의 전장면(112a)과 측면(112b) 사이의 각도(α)는 85° ~ 100° 로 형성될 수 있다. 이때, 탑 캡(110)의 수용부(112)에서 전장면(112a)과 측면(112b) 사이의 각도는 예를 들어 수직으로 형성될 수 있다.

- [0031] 도 3 및 도 4를 참고하면, 배출구(113)는 수용부(112)의 측면(112b)에 형성되어 전지 내부의 가스를 배출할 수 있다. 또한, 배출구(113)는 수용부(112)를 중심으로 4 방향 중에서 어느 한 방향을 제외한 방향에 형성될 수 있다. 이때, 배출구(113)는 예를 들어 수용부(112)를 중심으로 4 방향 중에서 3 방향에 형성될 수 있다.
- [0033] 도 5는 본 발명의 일 실시예에 따른 이차전지용 랩 조립체에서 안전 밸트를 나타낸 평면도이고, 도 6은 도 5에서 A-A'선을 따라 절개한 단면이다.
- [0034] 도 3 및 도 5를 참고하면, 안전 밸트(120)는 랩 캡(110) 보다 전지의 내측에 위치되어, 파열 노치(121) 및 절곡 노치(124)가 형성될 수 있다.
- [0035] 또한, 안전 밸트(120)는 랩 캡(110)의 배출구(113)를 개방 및 폐쇄할 수 있다. 여기서, 랩 캡(110)의 배출구(113)와 전기 내부를 연결하는 통로 상에 안전 밸트(120)가 위치된다. 이에 따라, 파열 노치(121)의 파열 전에는 랩 캡(110)의 배출구(113)가 폐쇄되고, 전지의 내압 상승으로 안전 밸트(120)의 파열 노치(121)가 파열되면 랩 캡(110)의 배출구(113)가 개방될 수 있다.
- [0036] 파열 노치(121)는 전지 내압의 상승시 랩 캡(110) 방향으로 파열될 수 있다. 이때, 파열 노치(121)의 파열에 따라 위치 또는 형태가 변하는 파열부(123)는 랩 캡(110)의 수용부(112) 방향으로 파열되어 수용부(112)에 수용될 수 있다.
- [0037] 아울러, 파열 노치(121)는 평면도 상으로 일측이 개방된 원호 형태로 형성될 수 있다. 여기서, 파열 노치(121)의 개방 부위는 원호 중심으로 소정 각도로 형성될 수 있다. 이때, 파열 노치(121)의 개방 부위 각도(β)는 예를 들어 60° 이내로 형성될 수 있다.
- [0038] 절곡 노치(124)는 파열 노치(121)의 파열시 절곡되도록 파열 노치(121)의 파열에 따라 위치 또는 형태가 변하는 파열부(123) 상에 형성된다. 여기서, 파열부(123)는 파열시 랩 캡(110)과 충돌하게 되고, 랩 캡(110)과의 충돌력에 의해 절곡 노치(124)가 절곡되게 된다. 이때, 파열 노치(121)의 파열시 파열부(123)가 랩 캡(110)의 내측면과 대응되는 형태로 절곡되도록 절곡 노치(124)가 절곡될 수 있다. 따라서, 파열 노치(121)의 파열시 파열부(123)와 랩 캡(110)의 내측면 사이에 공간 발생을 줄여 랩 캡(110)의 배출구(113)가 모두 개방될 수 있다. 이에 따라, 전지의 내부 가스 배출에 필요한 통로를 최대한 확보할 수 있다. 특히, 랩 캡(110)에서 수용부(112)의 천장면(112a)과 측면(112b) 사이의 각도(α)가 $85^\circ\sim 100^\circ$ 또는 수직으로 형성되면, 파열부(123)가 랩 캡(110)의 수용부(112) 내측면에 대응되도록 절곡시 배출구(113)의 개방이 더욱 용이해짐에 따라 가스 배출의 용이성을 훨씬 더 효과적으로 확보할 수 있다. 여기서, 랩 캡(110)의 수용부(112) 내측면은 천장면(112a) 및 측면(112b)을 포함하여 구성될 수 있다.
- [0039] 또한, 절곡 노치(124)는 평면도 상으로 파열 노치(121)의 원호 내측부에 형성될 수 있다. 이때, 절곡 노치(124)는 파열 노치(121)의 개방 부위와 파열 노치(121)의 원호 중심(C) 사이에 위치될 수 있다. 그리고, 절곡 노치(124)는 평면도 상으로 직선 형태로 형성될 수 있다.
- [0040] 아울러, 절곡 노치(124)는 평면상으로 파열 노치(121)의 원호 중심에서 파열 노치(121)의 개방 부위로 향하는 방향에 대해 기울어진 방향으로 연장되며 형성될 수 있으며, 또한, 절곡 노치(124)는 평면상으로 파열 노치(121)의 원호 중심에서 파열 노치(121)의 개방 방향으로 연장되는 선(L)에 대해 수직으로 라인을 형성할 수 있다. 여기서, 절곡 노치(124)의 양단은 파열 노치(121)의 원호 상에 위치될 수 있다.
- [0041] 한편, 안전 밸트(120)에서 파열 노치(121)의 파열시 평면상으로 파열 노치(121)의 원호 상 노치가 형성되지 않은 부분인 비노치부(122)가 랩 캡(110) 방향으로 절곡되도록 형성될 수 있다. 여기서, 안전 밸트(120)의 비노치부(122)는 수용부(112)의 가장자리에 위치한 모서리(112d)에 대응되도록 위치되어, 파열 노치(121)의 파열시 수용부(112)의 가장자리에 위치한 모서리(112d)에 의해 절곡될 수 있다. 이때, 안전 밸트(120)에서 랩 캡(110)의 하면과 접촉되지 않는 파열부(123)가 비노치부(122)를 기준으로 랩 캡(110)의 수용부(112)에 수용되는 방향으로 절곡될 수 있다.
- [0042] 그리고, 절곡 노치(124)는 파열 노치(121)의 파열시 수용부(112)의 천장면(112a)과 측면(112b) 사이에 형성된 천장 모서리부(112c)에 대응되는 위치에 위치되도록 형성될 수 있다. 즉, 안전 밸트(120)에서 절곡되는 비노치부(122)에서 절곡 노치(124)까지의 거리는 수용부(112)의 깊이에 대응되도록 형성될 수 있다.
- [0043] 또한, 절곡 노치(124)는 복수개로 형성될 수 있다. 이때, 절곡 노치(124)는 제1 절곡 노치(124a) 및 제2 절곡 노치(124b)를 포함하고, 제1 절곡 노치(124a) 및 제2 절곡 노치(124b)는 상호 일정거리 이격되어 나란하게 형성

될 수 있다.

[0044] 여기서, 비노치부(122)에서 제1 절곡 노치(124a) 까지의 거리(c1)는 탑 캡(110)의 수용부(112) 깊이(c3) 보다 작게 형성되고, 비노치부(122)에서 제2 절곡 노치(124b)까지의 거리(c2)는 탑 캡(110)의 수용부(112) 깊이(c3) 보다 크게 형성될 수 있다.

[0045] 한편, 도 6을 참고하면, 파열 노치(121)가 절곡 노치(124) 보다 먼저 파열되도록, 파열 노치(121)가 절곡 노치(124) 보다 크게 형성되거나 깊게 형성될 수 있다. 여기서, 파열 노치(121)의 폭(a1)은 절곡 노치(124)의 폭(a2) 보다 넓게 형성될 수 있다. 또한, 파열 노치(121)의 길이(b1)는 절곡 노치(124)의 길이(b2) 보다 깊게 형성될 수 있다.

[0046] 한편, 도 4 및 도 5를 참고하면, 수용부(112)를 중심으로 4방향 중 어느 한 방향에 안전 벨트(120)의 비노치부(122)가 위치되고, 나머지 하나 이상의 방향에 상기 배출구(113)가 위치될 수 있다. 여기서, 수용부(112)를 중심으로 4 방향 중 어느 한 방향에 안전 벨트(120)의 비노치부(122)가 위치되고, 나머지 3 방향에 상기 배출구(113)가 위치 될 수 있다.

[0048] 이상 본 발명을 구체적인 실시예를 통하여 상세히 설명하였으나, 이는 본 발명을 구체적으로 설명하기 위한 것으로, 본 발명에 따른 이차전지용 캡 조립체는 이에 한정되지 않는다. 본 발명의 기술적 사상 내에서 당해 분야의 통상의 지식을 가진 자에 의해 다양한 실시가 가능하다고 할 것이다.

[0049] 또한, 발명의 구체적인 보호 범위는 첨부된 특허청구범위에 의하여 명확해질 것이다.

부호의 설명

- [0051]
- 10: 이차 전지
 - 11: 케이스
 - 12: 전극 적층체
 - 12a: 양극
 - 12b: 음극
 - 12c: 분리막
 - 13: 셀터면
 - 100: 캡 조립체
 - 110: 탑 캡
 - 111: 외부 단자
 - 112: 수용부
 - 112a: 전장면
 - 112b: 측면
 - 112c: 전장 모서리부
 - 112d: 모서리
 - 113: 배출구
 - 120: 안전 벨트
 - 121: 파열 노치
 - 122: 비노치부

123: 화열부

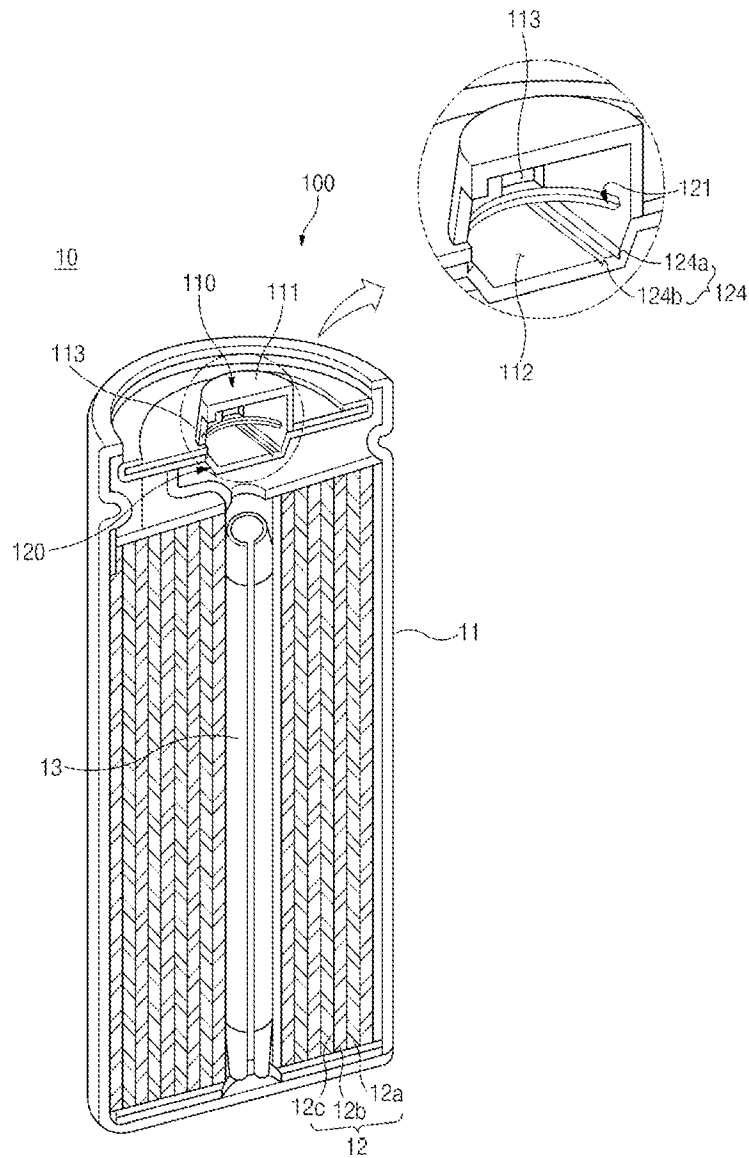
124: 절곡 노치

124a: 제1 절곡 노치

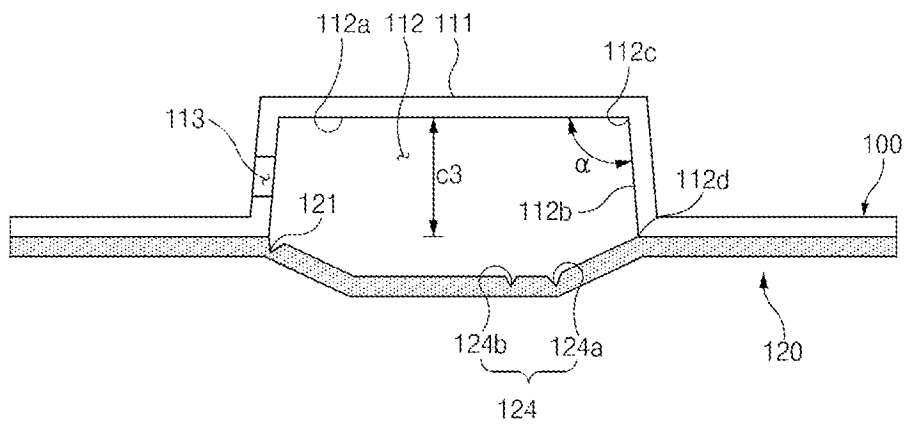
124b: 제2 절곡 노치

도면

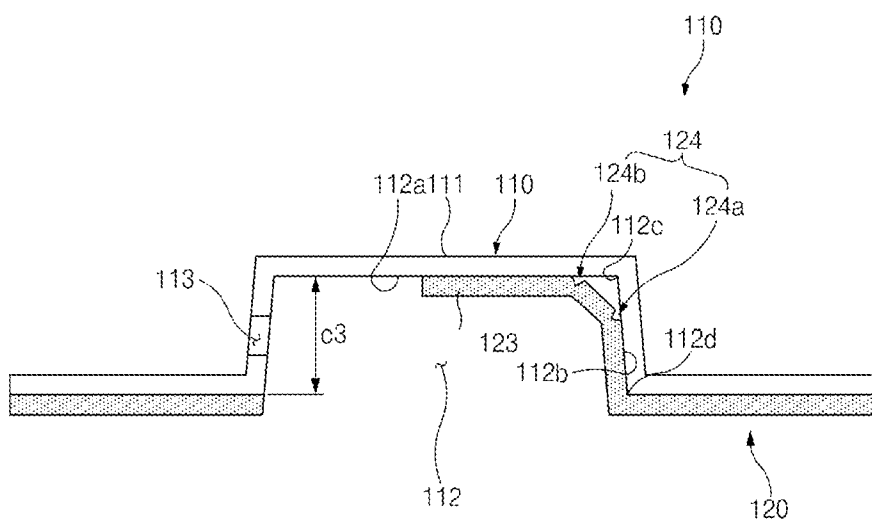
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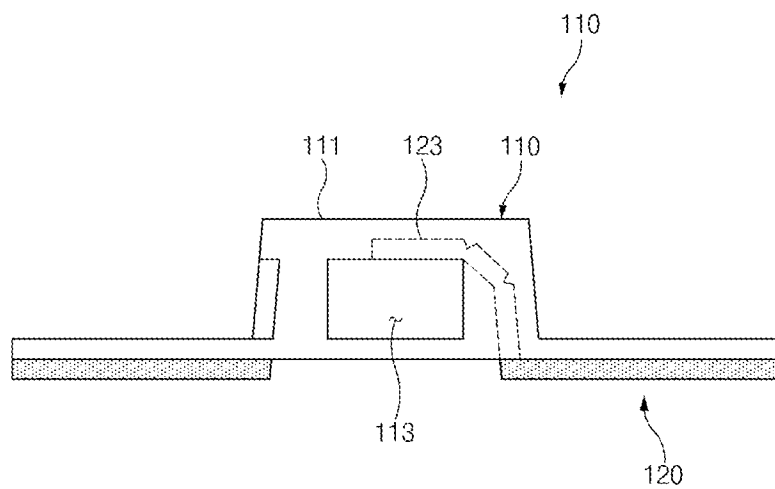
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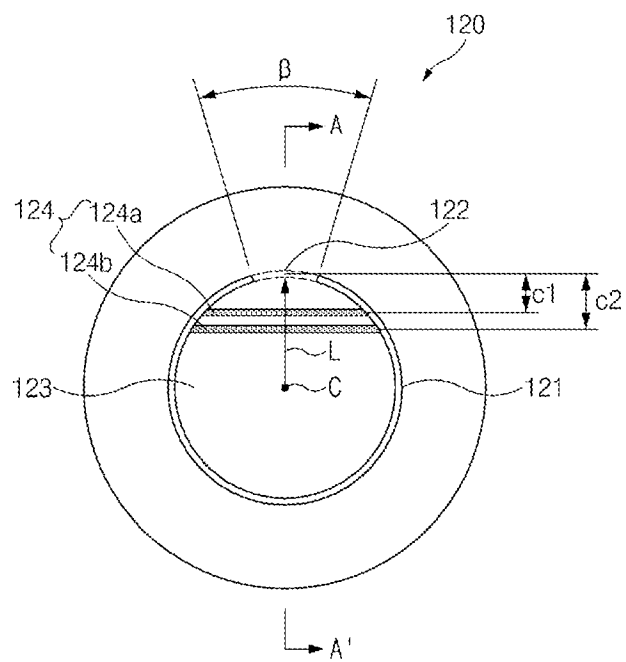
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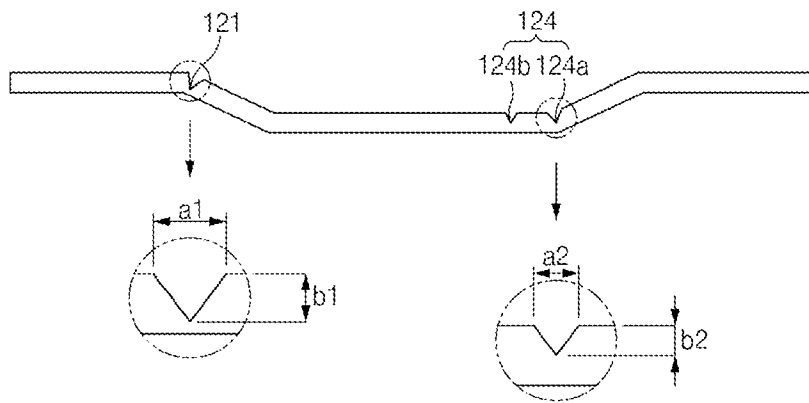
도면4



도면5



도면6





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CAP ASSEMBLY FOR RECHARGEABLE BATTERY

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Abstract of KR20170124343 (A)

The present invention relates to a cap assembly for a secondary battery. According to the present invention, the cap assembly for a secondary battery comprises: a top cap located in an outer side of a battery wherein the top cap has a discharging hole formed therein; and a safe vent having a rupture notch ruptured in case of an internal pressure increase to discharge an internal gas to the discharging hole and having a bending notch formed on the rupture unit of which a location or a shape is changed due to ruptures wherein the bending notch is bent in case of ruptures of the rupture notch.



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DESCRIPTION KR20170124343A

Cap assembly for secondary battery {CAP ASSEMBLY FOR RECHARGEABLE BATTERY}

[0001]

The present invention relates to a cap assembly for a secondary battery.

[0003]

Secondary batteries, unlike primary batteries, are rechargeable, and have been widely researched and developed in recent years due to their small size and large capacity. As technology development and demand for mobile devices increase, the demand for secondary batteries as an energy source is rapidly increasing.

[0004]

In particular, the lithium secondary battery has an operating voltage of 3.6V, which has about three times the capacity of a nickel-cadmium battery or a nickel-hydrogen battery, which are often used as power sources for electronic equipment, and has a high energy density per unit weight, so the degree of utilization is in a rapidly increasing trend.

[0005]

Such a lithium secondary battery mainly uses a lithium-based oxide and a carbon material as a positive electrode active material and a negative electrode active material, respectively.

In addition, the lithium secondary battery may be classified into a prismatic battery, a cylindrical battery, and a pouch-type battery.

[0006]

A lithium ion secondary battery includes an electrode assembly in which a positive electrode/a separator/a negative electrode is sequentially disposed, and a packaging material for sealingly housing the electrode assembly together with an electrolyte.

In particular, the casing of a prismatic or cylindrical secondary battery includes a case having an open end and a cap assembly sealingly coupled to the open end of the case.

[0008]

Korean Patent Publication No. 10-2009-0105547

[0009]

One aspect of the present invention is to provide a cap assembly for a secondary battery capable of easily discharging gas to the outside of the battery when the internal pressure of the battery is increased due to the generation of gas inside the battery.

[0011]

The cap assembly for a secondary battery according to an embodiment of the present invention includes a top cap positioned on the outer shell of the battery and having an outlet formed therein, and a rupture notch that ruptures when the internal pressure rises to allow the internal gas to be discharged to the outlet, and the rupture notch and a safety vent in which a bending notch that is bent upon rupture of the rupture is formed on the rupture portion whose position or shape is changed due to the rupture.

[0013]

In the cap assembly for a secondary battery according to the present invention, a top cap and a safety vent are formed in a form in which gas is easily discharged when the internal pressure of the battery is increased due to the generation of gas inside the battery, so that it is possible to secure the ease of gas discharge.

[0014]

In particular, in the cap assembly for a secondary battery according to the present invention, the rupture portion of the safety vent that ruptures due to an increase in internal pressure is ruptured and bent in a form for easy gas discharge, so that a passage necessary for gas discharge can be easily secured.

[0016]

1 is a cutaway perspective view illustrating a secondary battery in which a cap assembly for a secondary battery according to an embodiment of the present invention is installed.

2 is a cross-sectional view illustrating a cap assembly for a secondary battery according to an embodiment of the present invention.

3 is a cross-sectional view illustrating a state after a safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention.

4 is a conceptual diagram illustrating a state after a safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention.

5 is a plan view illustrating a safety vent in the cap assembly for a secondary battery according to an embodiment of the present invention.

FIG. 6 is a cross-sectional view taken along line A-A' in FIG. 5 .

[0017]

The objects, specific advantages and novel features of the present invention will become more apparent from the following detailed description and preferred embodiments taken in conjunction with the accompanying drawings.

In the present specification, in adding reference numbers to the components of each drawing, it should be noted that only the same components are given the same number as possible even though they are indicated on different drawings.

In addition, the present invention may be embodied in several different forms and is not limited to the embodiments described herein.

And, in describing the present invention, detailed descriptions of related known technologies that may unnecessarily obscure the gist of the present invention will be omitted.

[0019]

1 is a cutaway perspective view illustrating a secondary battery in which a cap assembly for a secondary battery according to an embodiment of the present invention is installed, and FIG. 2 is a cross-sectional view showing a cap assembly for a secondary battery according to an embodiment of the present invention.

[0020]

1 and 2, in the cap assembly 100 for a secondary battery according to an embodiment of the present invention, a top cap 110 having an outlet 113 is formed and the internal pressure of the secondary battery 10 is increased, and a safety vent 120 formed with a rupture notch 121 and a bending notch 124 to be bent.

[0022]

3 is a cross-sectional view illustrating a state after the safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention, and FIG. 4 is after the safety vent is ruptured in the cap assembly for a secondary battery according to an embodiment of the present invention. It is a perspective view showing the state.

[0023]

Hereinafter, a cap assembly for a secondary battery according to an embodiment of the present invention will be described in more detail with reference to FIGS. 1 to 6.

[0024]

1 and 2, the top cap 110 is positioned at the outermost portion of the secondary battery 10, and may be formed to include an external terminal 111, an accommodating part 112, and an outlet 113. have.

[0025]

The external terminal 111 may protrude from the top of the top cap 110 and may be electrically connected to the electrode stack 12 accommodated in the secondary battery 10 to form a positive terminal or a negative terminal.

[0026]

The electrode stack 12 is accommodated in the case 11, and an electrolyte is injected into the case 11.

In addition, the electrode stack 12 is a power generating element capable of charging and discharging, and the anode 12a, the cathode 12b, and the separator 12c are assembled to form a stacked structure.

Here, the positive electrode 12a of the electrode stack 12 is electrically connected to the external terminal 111 of the top cap 110 so that the external terminal 111 may function as a positive electrode terminal.

[0027]

The separator 12c separates and electrically insulates the anode 12a and the cathode 12b.

At this time, the positive electrode 12a and the negative electrode 12b may be wound together with the separator 12c to form a jelly roll type.

In addition, the separator 12c is made of an insulating material to electrically insulate between the anode 12a and the cathode 12b.

Here, the separator 12c may be formed of, for example, a polyolefin-based resin film such as polyethylene or polypropylene having microporosity.

[0028]

Meanwhile, a cylindrical center pin 13 may be provided at the center of the electrode stack 12.

The center pin 13 may be made of a metal material to impart a predetermined strength, and may have a hollow cylindrical structure obtained by bending a plate material in a round shape.

The center pin 13 may act as a passage for fixing and supporting the electrode stack 12 and for discharging gas generated by an internal reaction during charging, discharging, and operation.

[0029]

2 and 3, the receiving part 112 may be formed in an open shape toward the safety vent 120 so that the rupture part 123 of the safety vent 120 is accommodated in the lower part of the external terminal 111. can

[0030]

In addition, the angle α between the ceiling surface 112a and the side surface 112b of the receiving part 112 in the top cap 110 may be formed in a range of 85 to 100°.

In this case, the angle between the ceiling surface 112a and the side surface 112b in the receiving part 112 of the top cap 110 may be formed vertically, for example.

[0031]

Referring to FIGS. 3 and 4, the outlet 113 is formed on the side 112b of the receiving part 112 to discharge gas inside the battery.

In addition, the outlet 113 may be formed in a direction except for any one of the four directions with respect to the receiving portion 112.

In this case, the outlet 113 may be formed in, for example, 3 directions among 4 directions with respect to the receiving part 112.

[0033]

5 is a plan view illustrating a safety vent in the cap assembly for a secondary battery according to an embodiment of the present invention, and FIG. 6 is a cross-sectional view taken along line A-A' in FIG. 5.

[0034]

3 and 5, the safety vent 120 is located inside the battery rather than the top cap 110, so that a rupture notch 121 and a bending notch 124 may be formed.

[0035]

In addition, the safety vent 120 may open and close the outlet 113 of the top cap 110.

Here, the safety vent 120 is positioned on the passage connecting the outlet 113 of the top cap 110 and the inside of the battery.

Accordingly, before the rupture of the rupture notch 121, the outlet 113 of the top cap 110 is closed, and when the rupture notch 121 of the safety vent 120 ruptures due to an increase in internal pressure of the battery, the outlet 113 may be opened.

[0036]

The rupture notch 121 may rupture in the direction of the top cap 110 when the internal pressure of the battery increases.

At this time, the rupture part 123 whose position or shape is changed according to the rupture of the rupture notch 121 may be ruptured in the direction of the receiving part 112 of the top cap 110 to be accommodated in the receiving part 112.

[0037]

In addition, the rupture notch 121 may be formed in a circular arc shape with one side open in a plan view.

Here, the opening portion of the rupture notch 121 may be formed at a predetermined angle with respect to the center of the arc. In this case, the opening angle β of the rupture notch 121 may be formed within, for example, 60°.

[0038]

The bending notch 124 is formed on the tearing portion 123 whose position or shape changes according to the tearing of the tearing notch 121 so as to be bent upon the tearing of the tearing notch 121.

Here, the rupture part 123 collides with the top cap 110 when ruptured, and the bending notch 124 is bent by the collision force with the top cap 110.

In this case, the bending notch 124 may be bent so that the rupture part 123 is bent in a shape corresponding to the inner surface of the top cap 110 when the rupture notch 121 is ruptured.

Accordingly, when the rupture notch 121 is ruptured, space generation between the rupture part 123 and the inner surface of the top cap 110 is reduced, so that all of the outlets 113 of the top cap 110 can be opened.

Accordingly, a passage necessary for discharging the internal gas of the battery can be maximally secured.

In particular, when the angle α between the ceiling surface 112a and the side surface 112b of the receiving part 112 in the top cap 110 is 85 to 100° or is formed vertically, the rupture part 123 is formed in the top cap (As the opening of the outlet 113 becomes easier when bending to correspond to the inner surface of the receiving portion 112 of the 110), the ease of gas discharge can be secured much more effectively.

Here, the inner surface of the receiving part 112 of the top cap 110 may be configured to include a ceiling surface 112a and a side surface 112b.

[0039]

In addition, the bending notch 124 may be formed on the inner side of the arc of the tear notch 121 in a plan view.

In this case, the bending notch 124 may be located between the opening portion of the tear notch 121 and the arc center C of the tear notch 121.

In addition, the bending notch 124 may be formed in a straight shape in a plan view.

[0040]

In addition, the bending notch 124 may be formed to extend in a direction inclined with respect to the direction from the center of the arc of the tear notch 121 toward the opening portion of the tear notch 121 in plan view, and also the bending notch 124 () may form a line perpendicular to the line L extending in the opening direction of the tear notch 121 from the center of the circular arc of the tear notch 121 in plan view.

Here, both ends of the bending notch 124 may be positioned on the arc of the rupture notch 121.

[0041]

On the other hand, when the rupture notch 121 is ruptured in the safety vent 120, the non-notch portion 122, which is a portion where the notch is not formed on the arc of the rupture notch 121, is bent in the direction of the top cap 110 in a plan view. can

Here, the non-notch part 122 of the safety vent 120 is positioned to correspond to the edge 112d positioned at the edge of the receiving part 112, and is positioned at the edge of the receiving part 112 when the tear notch 121 is ruptured. It can be bent by the corner edge (112d).

At this time, in the safety vent 120, the rupture part 123 that does not come into contact with the lower surface of the top cap 110 is bent in a direction to be accommodated in the receiving part 112 of the top cap 110 with respect to the non-notch part 122. can be

[0042]

In addition, the bending notch 124 may be formed to be positioned at a position corresponding to the ceiling edge portion 112c formed between the ceiling surface 112a and the side surface 112b of the receiving part 112 when the tear notch 121 is ruptured, have.

That is, the distance from the non-notch part 122 bent in the safety vent 120 to the bent notch 124 may be formed to correspond to the depth of the receiving part 112 .

[0043]

In addition, the bending notches 124 may be formed in plurality.

At this time, the bending notch 124 includes a first bending notch 124a and a second bending notch 124b, and the first bending notch 124a and the second bending notch 124b are spaced apart from each other by a predetermined distance and arranged side by side. can be formed.

[0044]

Here, the distance c1 from the non-notch part 122 to the first bending notch 124a is formed to be smaller than the depth c3 of the receiving part 112 of the top cap 110 , and from the non-notch part 122 to the second The distance c2 to the bending notch 124b may be greater than the depth c3 of the receiving portion 112 of the top cap 110 .

[0045]

Meanwhile, referring to FIG. 6 , the tear notch 121 may be formed larger or deeper than the bending notch 124 so that the tear notch 121 ruptures before the bending notch 124 .

Here, the width a1 of the rupture notch 121 may be wider than the width a2 of the bending notch 124 .

In addition, the depth b1 of the rupture notch 121 may be formed to be deeper than the depth b2 of the bending notch 124 .

[0046]

Meanwhile, referring to FIGS. 4 and 5 , the non-notch part 122 of the safety vent 120 is positioned in any one direction of the four directions around the receiving part 112 , and the outlet 113 is located in the other one or more directions.) can be located.

Here, the non-notch part 122 of the safety vent 120 may be positioned in any one of four directions around the receiving part 112 , and the outlet 113 may be positioned in the other three directions.

[0048]

Although the present invention has been described in detail through specific embodiments, this is intended to describe the present invention in detail, and the cap assembly for a secondary battery according to the present invention is not limited thereto.

It will be said that various implementations are possible by those of ordinary skill in the art within the technical spirit of the present invention.

[0049]

In addition, the specific protection scope of the invention will become clear by the appended claims.

[0051]

10: Secondary Battery 11: Case 12: Electrode stack 12a: positive electrode 12b: negative electrode 12c: separator 13: center pin 100: cap assembly 110: top cap 111: external terminal 112: receiving part 112a: ceiling surface 112b: side 112c: ceiling edge Part 112d: edge 113: outlet 120: safety vent 121: tear notch 122: unnotched portion 123: burst 124: bend notch 124a: first bend notch 124b: second bend notch



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CLAIMS KR20170124343A

1.

a top cap positioned on the outer shell of the battery and formed with an outlet; and a safety vent in which a rupture notch that ruptures when the internal pressure rises is formed so as to allow the internal gas to be discharged to the outlet, and a bending notch that is bent when the rupture notch is ruptured is formed on the rupture part whose position or shape is changed due to the rupture. A cap assembly for a secondary battery comprising.

2.

The cap assembly for a secondary battery according to claim 1, wherein the rupture part collides with the top cap and the bent notch is bent when the rupture notch is ruptured in the safety vent located inside the battery than the top cap.

3.

The cap for a rechargeable battery of claim 2, wherein all of the outlets of the top cap are closed by the safety vent before the rupture of the rupture notch, and all of the outlets are opened after the rupture of the rupture notch and the bending of the bending notch assembly.

4.

The cap assembly for a rechargeable battery of claim 2, wherein the bending notch is bent so that the rupture part is bent in a shape corresponding to the inner surface of the top cap when the rupture notch is ruptured.

5.

The cap assembly for a secondary battery according to claim 2, wherein the rupture notch is formed in an arc shape with one side open in a plan view of the safety vent.

6.

The cap assembly for a secondary battery according to claim 5, wherein the bending notch is formed on an inner side of an arc of the rupture notch in the plan view.

7.

The cap assembly for a secondary battery according to claim 6, wherein the bending notch is positioned between an open portion of the rupture notch and an arc center of the rupture notch.

8.

The cap assembly for a secondary battery according to claim 6, wherein the bent notch is formed in a straight shape in the plan view.

9.

The cap assembly for a secondary battery according to claim 6, wherein the bent notch forms a line perpendicular to a line extending in an opening direction of the rupture notch from the center of an arc of the rupture notch formed in the plan view.

10.

The cap assembly for a secondary battery according to claim 5, wherein the bending notch extends in a direction inclined with respect to a direction toward an opening portion of the rupture notch from the center of a circular arc of the rupture notch formed in a plan view.

11.

The cap assembly for a secondary battery according to claim 10, wherein both ends of the bending notch are positioned on an arc of the rupture notch.

12.

The cap assembly for a secondary battery according to any one of claims 1 to 11, wherein the rupture notch is formed to be larger than the bending notch so that the rupture notch is ruptured before the bending notch.

13.

The cap assembly for a secondary battery of claim 12, wherein a depth of the rupture notch is greater than a depth of the bending notch.

14.

The cap assembly of claim 12, wherein a width of the rupture notch is wider than a width of the bending notch.

15.

The cap assembly for a secondary battery according to any one of claims 1 to 11, wherein the bent notches are formed in plurality.

16.

The cap assembly for a secondary battery according to any one of claims 5 to 10, wherein the top cap includes a receiving part in which the rupture part of the safety vent is accommodated.

17.

The cap assembly for a secondary battery according to claim 16, wherein the outlet is formed on a side surface of the accommodating part to discharge gas generated inside the battery.

18.

The cap assembly for a secondary battery according to claim 17, wherein an angle between a ceiling surface and a side surface of the receiving part in the top cap is formed vertically.

19.

The cap assembly for a secondary battery according to claim 17, wherein an angle between the ceiling surface and the side surface of the receiving part in the top cap is 85 to 100°.

20.

The cap assembly for a secondary battery according to claim 17, wherein when the tear notch is ruptured in the safety vent, a non-notch portion, which is a portion on an arc of the tear notch in which a notch is not formed, is formed to be bent in the direction of the top cap.

21.

The cap assembly for a secondary battery according to claim 20, wherein the non-notch portion of the safety vent is bent by an edge of the receiving portion when the tear notch is ruptured.

22.

The cap assembly for a secondary battery according to claim 21, wherein the bending notch is formed to be positioned at a position corresponding to a ceiling edge of the receiving part when the rupture notch is ruptured.

23.

The cap assembly for a secondary battery of claim 21, wherein the bending notch includes a first bending notch and a second bending notch, and the first bending notch and the second bending notch are spaced apart from each other by a predetermined distance and formed in parallel.

24.

The method according to claim 23, The distance from the non-notch portion to the first bending notch is formed to be smaller than the depth of the receiving portion of the top cap, and the distance from the non-notch portion to the second bending notch is greater than the depth of the receiving portion of the top cap. A cap assembly for a secondary battery that is formed large.

25.

The cap assembly for a secondary battery according to claim 20, wherein the non-notch portion of the safety vent is positioned in any one direction among four directions with respect to the receiving part, and the outlet is positioned in one or more other directions.

26.

The cap assembly for a secondary battery according to claim 20, wherein the non-notch part of the safety vent is positioned in any one of four directions around the receiving part and the outlet is positioned in the other three directions.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17259395
	Filing Date		2021-07-16
	First Named Inventor	Geon Woo Min	
	Art Unit	OPAP	
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGCHEM 3.3F-3059	

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	OPAP
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	OPAP
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2022-03-04
Name/Print	Jordan T. Riviello	Registration Number	72,515

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Dated: March 4, 2022

Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: July 16, 2021

Art Unit: OPAP

For: Secondary Battery

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Sir:

It is respectfully requested that the reference listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of the reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited reference is legally available prior art or that the same is pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-3059, from which the undersigned is authorized to draw.

Dated: March 4, 2022

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant(s)

LD-458A

Electronic Acknowledgement Receipt

EFS ID:	45149846
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Stephen M. Lund/Kathleen Reynolds
Filer Authorized By:	Stephen M. Lund
Attorney Docket Number:	LGCHEM 3.3F-3059
Receipt Date:	04-MAR-2022
Filing Date:	16-JUL-2021
Time Stamp:	15:41:19
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Foreign Reference	LGCHEM3430_- _JP2013-502035A.pdf	777926	no	29
			25fbac7597f8c1eb8689096068427e3bcb3 d7d50		

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2	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM3430_-_Information_Disclosure_Statement_Fillable_PDF.pdf	1263203 e3139975834a34c1dc17fe8711c705833091eb1d	no	4
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Information:					
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3	Transmittal Letter	IDS_-_Information_Disclosure_Statement_-_Reference_Enclosed_LD-458.pdf	21375 3154c5fcf5f1938b6a0b11de850db4c2704a7d47	no	1
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Bibliographic data: JP2013502035 (A) — 2013-01-17

CYLINDRICAL SECONDARY BATTERY OF IMPROVED SAFETY

Inventor(s):

Applicant(s):

Classification: - international: H01M10/0567; H01M2/02; H01M2/04; H01M2/06;
H01M2/08; H01M2/12; H01M2/26; H01M2/34
- cooperative: H01M10/0431 (EP, US); H01M10/0587 (KR);
H01M50/3425 (EP, US); H01M50/578 (EP, US);
H01M10/0525 (KR); Y02E60/10 (EP); Y02P70/50 (EP)

Application number: JP20120524651 20100813 Global Dossier

Priority number(s): KR20090075156 20090814 ; WO2010KR05338 20100813

Also published as: CN102473883 (A) CN102473883 (B) EP2466668 (A2)
EP2466668 (A4) EP2466668 (B1) JP5577405 (B2)
KR101058386 (B1) KR20110017835 (A) US2012114979 (A1)
US9112234 (B2) WO2011019237 (A2) WO2011019237 (A3) less

Abstract not available for JP2013502035 (A)

Abstract of corresponding document: US2012114979 (A1)

Provided is a cylindrical battery in which an electrode assembly fabricated by rolling a cathode/separator/anode and an electrolyte are provided in a cylindrical can, wherein a cap assembly mounted on the opening top of the cylindrical can comprises: a safety vent provided with a predetermined notch, to allow breakage due to high-pressure gas of the battery, a current interruptive device to interrupt current, welded to the bottom of the safety vent, and a gasket for the current interruptive device to surround the periphery of the current interruptive device, wherein the current interruptive device comprises two or more through holes to allow exhaustion of gas, wherein the through holes have a size of 20 to 50% with respect to the total area of the current interruptive device.

(19)

(12) 公表特許公報 (A)

JP 2013-502035 A 2013.1.17

(11)

特表2013-502035

(P 2 0 1 3 - 5 0 2 0 3 5 A)

(43)

25 1 17 (2013.1.17)

(51)Int.Cl.

<i>H O 1 M</i>	<i>2/34</i>	<i>(2006. 01)</i>	<i>2/34</i>
<i>H O 1 M</i>	<i>10/0567</i>	<i>(2010. 01)</i>	<i>10/00</i>
<i>H O 1 M</i>	<i>2/02</i>	<i>(2006. 01)</i>	<i>2/02</i>
<i>H O 1 M</i>	<i>2/04</i>	<i>(2006. 01)</i>	<i>2/04</i>
<i>H O 1 M</i>	<i>2/06</i>	<i>(2006. 01)</i>	<i>2/06</i>

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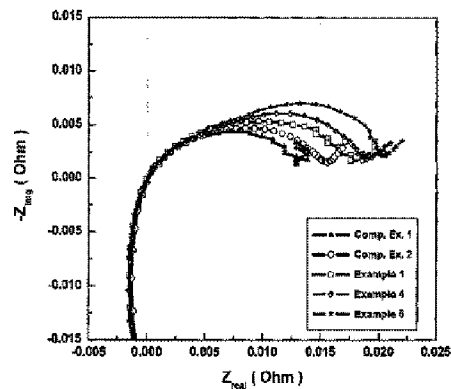
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(87)	WO2011/019237
(87)	23 2 17 (2011.2.17)
(31)	10-2009-0075156
(32)	21 8 14 (2009.8.14)
(33)	(KR)

(71)	500239823
(74)	100110364
(74)	100122161
(72)	

(54)

(57)

[Fig. 16]



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coefficient

positive temperature

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I LiClO₄ LiBF₄ LiB₁₀Cl₁₀ LiPF₆ LiCF₃SO₃ LiCF₃CO₂ LiAsF₆ LiSbF₆ LiAlCl₄
 CH₃SO₃Li CF₃SO₃Li (CF₃SO₂)₂NLi

30

N -

hexaphosphoric triamide

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beading member

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the anode collector

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beading

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frequency

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<表 1> (単位：電流 (A))

	1	2	平均値
実施例 1	46.8	50	48.4
実施例 4	44.5	45.8	45.2
実施例 5	40.9	43.2	42.0
実施例 6	45.3	45.6	45.4
実施例 7	43.9	42.1	43.0
比較例 1	51.2	53.6	52.4
比較例 2	51	52	51.5

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<表 2>

	開始温度 (°C)	熱容量(J/g)			
		1ピーク	2ピーク	3ピーク	合計
実施例 1	206±1	71±4	47±4	-	118±6
実施例 4	202±1	55±3	45±5	-	100±6
実施例 5	204±1	50±4	40±4	-	90±4
実施例 6	203±1	53±2	50±4	-	103±4
実施例 7	205±1	48±5	43±5	-	91±6
比較例 1	198±1	223±6			223±6
比較例 2	204±1	98±11	56±5	49±6	203±6

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30

specific heat capacity

40

nail test

50

<表 3>

	貫通孔の面積比	安全ベントの切断領域	爆発	電池缶の変形
実施例 1	30%	第一ノッチ	×	×
実施例 2	30%	第二ノッチ	×	△
実施例 3	30%	内側のノッチ (増加した直径)	×	△
比較例 3	18%	第一ノッチ	○	○

10

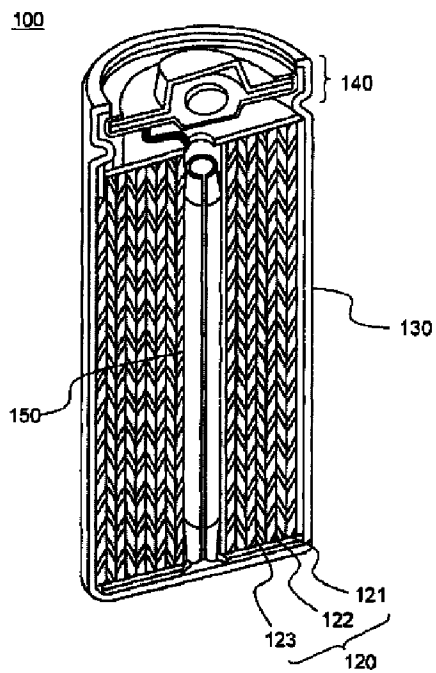
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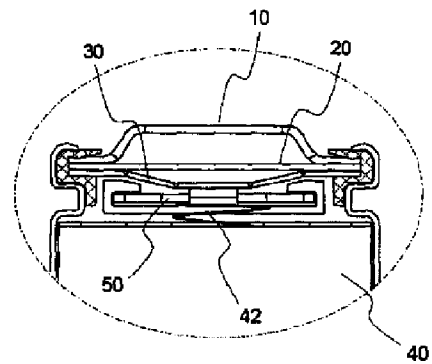
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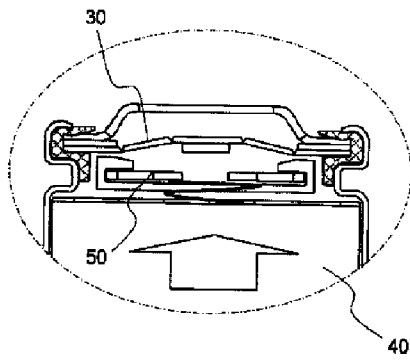
[Fig. 1]



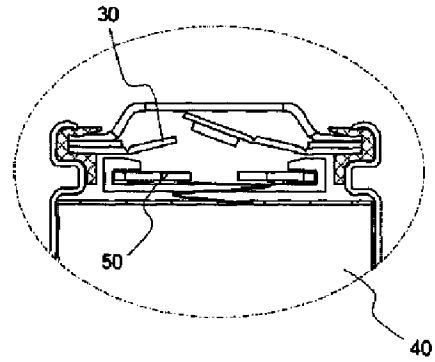
[Fig. 2]



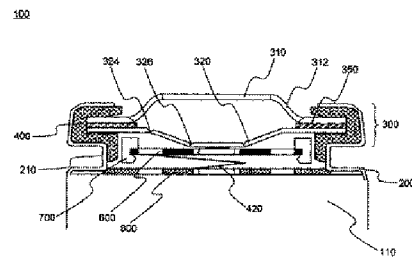
[Fig. 3]



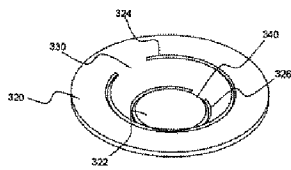
[Fig. 4]



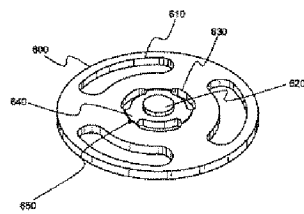
[Fig. 5]



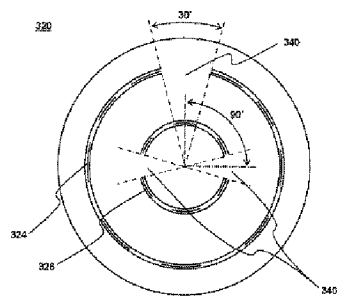
[Fig. 6]



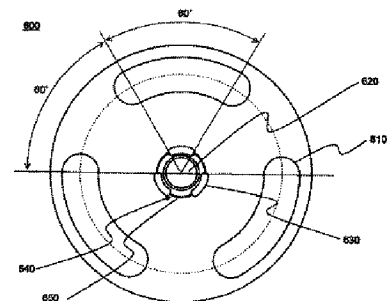
[Fig. 8]



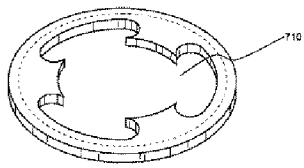
[Fig. 7]



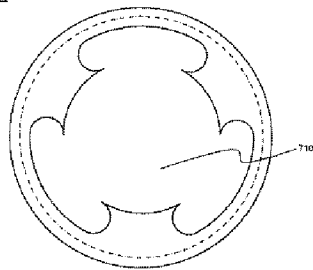
[Fig. 9]



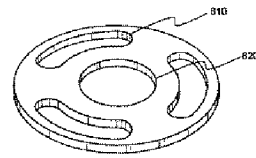
[Fig. 10]
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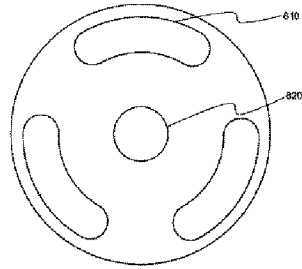
[Fig. 11]
700



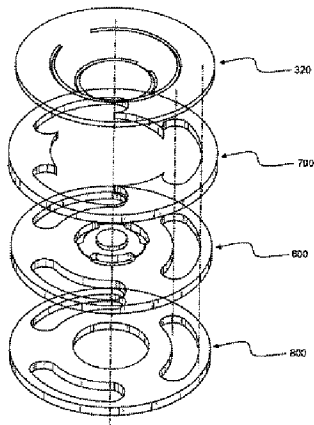
[Fig. 12]
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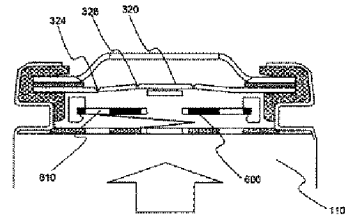
[Fig. 13]
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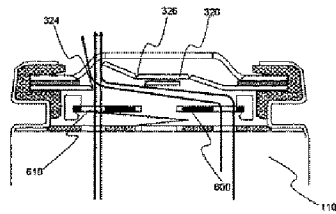
[Fig. 14]



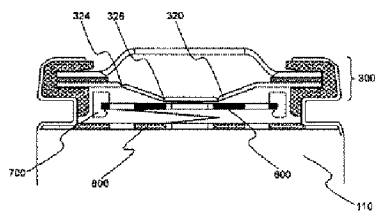
[Fig. 16]

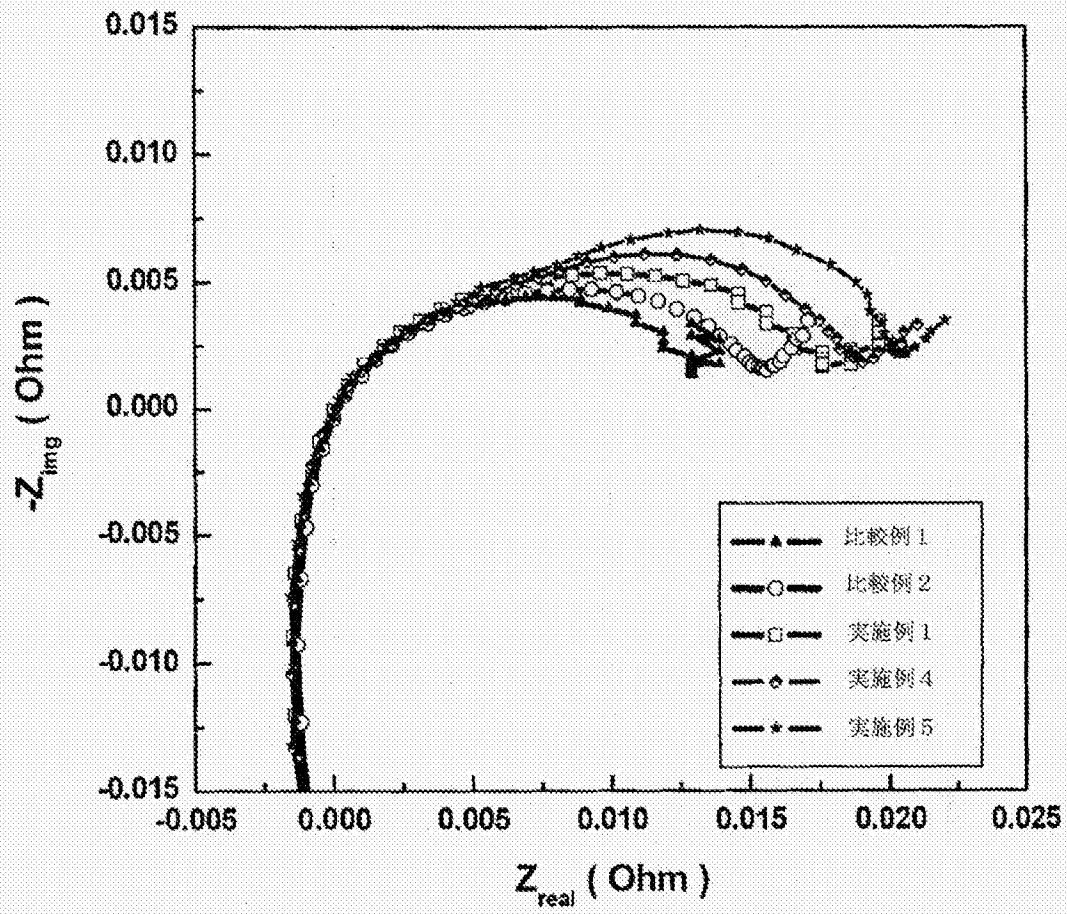


[Fig. 17]



[Fig. 15]





INTERNATIONAL SEARCH REPORT

International application No.

PCT/KR2010/005338**A. CLASSIFICATION OF SUBJECT MATTER****H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/08(2006.01)i**

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/12; H01M 10/058; H01M 10/0525; H01M 2/26; H01M 2/10; H01M 10/36; H01M 10/50

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean Utility models and applications for Utility models: IPC as above

Japanese Utility models and applications for Utility models: IPC as above

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS (KIPO internal) & Keywords: cap assembly, safety belt, current interrupt, gasket, through-hole

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KR 10-0855181 B1 (LG CHEM. LTD.) 29 August 2008 See abstract, claims 1-6 and figures 1-6b	1-22
A	KR 10-0619631 B1 (LG CHEM. LTD.) 12 September 2006 See abstract, claims 1-6 and figures 1-4c	1-22
A	KR 10-2004-0037579 A (SAMSUNG SDI CO., LTD.) 07 May 2004 See abstract, claims 1-23 and figures 1-4	1-22
A	JP 2008-262832 A (MATSUSHITA ELECTRIC IND CO LTD) 30 October 2008 See abstract, claims 1-8 and figure 1	1-22

☐ Further documents are listed in the continuation of Box C.☒ See patent family annex.

* Special categories of cited documents:

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"G" document member of the same patent family

Date of the actual completion of the international search

08 MARCH 2011 (08.03.2011)

Date of mailing of the international search report

08 MARCH 2011 (08.03.2011)

Name and mailing address of the ISA/KR


 Korean Intellectual Property Office
 Government Complex-Daejeon, 139 Seonsa-ro, Daejeon 302-701,
 Republic of Korea

Facsimile No. 82-42-472-7140

Authorized officer

Telephone No.

INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.

PCT/KR2010/005338

Patent document cited in search report	Publication date	Patent family member	Publication date
KR 10-0855181 B1	29.08.2008	NONE	
KR 10-0619631 B1	12.09.2006	NONE	
KR 10-2004-0037579 A	07.05.2004	NONE	
JP 2008-262832 A	30.10.2008	CN 101542820 A	23.09.2009
		JP 2008-262832 A	30.10.2008
		KR 20090028729A	19.03.2009
		US 2009-311585 A1	17.12.2009
		WO 2008-136177 A1	13.11.2008

국제조사보고서

국제출원번호
PCT/KR2010/005338

A. 발명이 속하는 기술분류(국제특허분류(IPC)) H01M 2/12(2006.01)i, H01M 2/34(2006.01)i, H01M 2/08(2006.01)i		
B. 조사된 분야 조사된 최소문헌(국제특허분류상 기재) H01M 2/12; H01M 10/058; H01M 10/0525; H01M 2/26; H01M 2/10; H01M 10/36; H01M 10/50 조사된 기술분야에 속하는 최소문헌 이외의 문헌 한국등록실용신안공보 및 한국공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 일본등록실용신안공보 및 일본공개실용신안공보: 조사된 최소문헌란에 기재된 IPC 국제조사에 이용된 전산 데이터베이스(데이터베이스의 명칭 및 검색어(해당하는 경우)) eCOMPASS(특허청 내부 검색시스템) & 키워드: 캡 어셈블리, 안전펜트, 전류차단, 가스켓, 판봉구,		
C. 관련 문헌		
카테고리*	인용문헌명 및 관련 구절(해당하는 경우)의 기재	관련 청구항
A	KR 10-0855181 B1 (주식회사 엘지화학) 2008.08.29 요약, 청구항 1-6항 및 도면 1-6a참조	1-22
A	KR 10-0619631 B1 (주식회사 엘지화학) 2006.09.12 요약, 청구항 1-6항 및 도면 1-4c참조	1-22
A	KR 10-2004-0037579 A (삼성세디아이 주식회사) 2004.05.07 요약, 청구항 1-23항 및 도면 1-4참조	1-22
A	JP 2008-262832 A (MATSUSHITA ELECTRIC IND CO LTD) 2008.10.30 요약, 청구항 1-8항 및 도면 1참조	1-22
☐ 추가 문헌이 C(계속)에 기재되어 있습니다. ☒ 대응특허에 관한 별지를 참조하십시오.		
* 인용된 문헌의 특별 카테고리: "A" 특별히 관련이 없는 것으로 보이는 일반적인 기술수준을 정의한 문헌 "T" 국제출원일 또는 우선일 후에 공개된 문헌으로, 출원과 상충하지 않으며 발명의 기초가 되는 원리나 이론을 이해하기 위해 인용된 문헌 "B" 국제출원일보다 빠른 출원일 또는 우선일을 가져나 국제출원일 이후에 공개된 선출원 또는 특허 문헌 "X" 특별한 관련이 있는 문헌. 해당 문헌 하나만으로 청구된 발명의 신규성 또는 진보성이 없는 것으로 본다. "L" 우선권 주장에 의문을 제기하는 문헌 또는 다른 인용문헌의 공개일 또는 다른 특별한 이유(이유를 명시)를 밝히기 위하여 인용된 문헌 "Y" 특별한 관련이 있는 문헌. 해당 문헌이 하나 이상의 다른 문헌과 조합하는 경우로 그 조합이 당업자에게 자명한 경우 청구된 발명은 진보성이 없는 것으로 본다. "O" 구조, 개시, 사용, 전시 또는 기타 수단을 언급하고 있는 문헌 "&" 동일한 대응특허문헌에 속하는 문헌 "P" 우선일 이후에 공개되었으나 국제출원일 이전에 공개된 문헌		
국제조사의 실제 완료일 2011년 03월 08일 (08.03.2011)		국제조사보고서 발송일 2011년 03월 08일 (08.03.2011)
ISA/KR의 명칭 및 우편주소  대한민국 특허청 (302-701) 대전광역시 서구 선사로 139, 정부대전청사 팩스 번호 82-42-472-7140		심사관 이창희 전화번호 82-42-481-5644 

서식 PCT/ISA/210 (두 번째 용지) (2009년 7월)

국제조사보고서
대응특허에 관한 정보

국제출원번호

PCT/KR2010/005338

국제조사보고서에서 인용된 특허문헌	공개일	대응특허문헌	공개일
KR 10-0855181 B1	2008.08.29	없음	
KR 10-0619631 B1	2006.09.12	없음	
KR 10-2004-0037579 A	2004.06.07	없음	
JP 2008-262832 A	2008.10.30	CN 101542820 A	2009.09.23
		JP 2008-262832 A	2008.10.30
		KR20090028729A	2009.03.19
		US 2009-311585 A1	2009.12.17
		WO 2008-136177 A1	2008.11.13

서식 PCT/ISA/210 (대응특허 추가용지) (2009년 7월)

(51)Int.Cl.

H O 1 M	2/08	(2006. 01)	2/08
H O 1 M	2/12	(2006. 01)	2/12
H O 1 M	2/26	(2006. 01)	2/26

(81) AP(BW,GH,GM,KE,LR,LS,MW,MZ,NA,SD,SL,SZ,TZ,UG,ZM,ZW),EA(AM,AZ,BY,KG,KZ,MD,RU,TJ,TM),EP(AL,AT,BE,BG,CH,CY,CZ,DE,DK,EE,ES,FI,FR,GB,GR,HR,HU,IE,IS,IT,LT,LU,LV,MC,MK,MT,NL,NO,PL,PT,RO,SE,SI,SK,SM,TR),OA(BF,BJ,CF,CG,CI,CM,GA,GN,GQ,GW,ML,MR,NE,SN,TD,TG),AE,AG,AL,AM,AO,AT,AU,AZ,BA,BB,BG,BH,BR,BW,BY,BZ,CA,CH,CL,CN,CO,CR,CU,CZ,DE,DK,DM,DO,DZ,EC,EE,EG,ES,FI,GB,GD,GE,GH,GM,GT,HN,HR,HU,ID,IL,IN,IS,JP,KE,KG,KM,KN,KP,KZ,LA,LC,LK,LR,LS,LT,LU,LY,MA,MD,ME,MG,MK,MN,MW,MX,MY,MZ,NA,NG,NI,NO,NZ,OM,PE,PG,PH,PL,PT,RO,RS,RU,SC,SD,SE,SG,SK,SL,SM,ST,SV,SY,TH,TJ,TM,TN,TR,TT,TZ,UA,UG,US,UZ,VC,VN,ZA,ZM,ZW

(72)

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(72)

(72)

() 5H011 AA13 BB03 CC01 CC06 DD15 EE04 GG02 HH02 JJ11 KK01
 KK02 KK04
 5H012 AA01 BB02 CC01 DD02 DD17 EE04 FF01 GG01 JJ01 JJ10
 5H029 AJ12 AK03 AL07 AM05 AM07 BJ02 BJ14 BJ27 CJ08 CJ13
 DJ02 DJ08 EJ11 HJ01 HJ02 HJ07 HJ12 HJ14
 5H043 AA04 AA11 BA19 CA03 CA12 EA06 EA15 EA16 EA29 EA36
 EA39 GA07 GA13 HA04E HA06E HA10E HA11E JA01E JA02E JA04E
 JA06E JA12E JA13E KA08E KA09E KA19 KA44E LA02E LA11E LA14
 LA15 LA21E LA22E LA23E LA35E LA41

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17259395
	Filing Date		2021-07-16
	First Named Inventor	Geon Woo Min	
	Art Unit		
	Examiner Name	Not Yet Assigned	
	Attorney Docket Number	LGCHEM 3.3F-3059	

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	1	20130216871	A1	2013-08-22	LEE et al.	

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	1	2572931	CA	A1	2008-06-21	Wu, Donald P. H		
	2	102473883	CN	A	2012-05-23	Lg Chemical Ltd	U.S. Publication No. 20120114979 A1 is an English language counterpart, which has been previously cited.	×
	3	103155225	CN	A	2013-06-12	Lg Chemical Ltd	U.S. Publication No. 20130216871 A1 is an English language counterpart.	☒

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

If you wish to add additional Foreign Patent Document citation information please click the Add button

NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	Search Report dated June 22, 2022 from the Office Action for Chinese Application No. 201980046222.8 issued June 30, 2022, 3 pages.	×

If you wish to add additional non-patent literature document citation information please click the Add button

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2022-09-14
Name/Print	Jordan T. Riviello	Registration Number	72,515

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: September 14, 2022
Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: July 16, 2021

Art Unit: Not Yet Assigned

For: Secondary Battery

Examiner: Not Yet Assigned

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Dear Commissioner:

It is respectfully requested that the references listed on the enclosed form be made of record and considered with respect to the above-referenced U.S. patent application. A copy of each reference which is not a U.S. patent or patent application is enclosed. Submission of the present Information Disclosure Statement should not be taken as an admission that the cited references are legally available prior art or that the same are pertinent or material.

Applicant believes no fee is due with this response. However, if a fee is due, please charge our Deposit Account No. 12-1095, under Order No. LGCHEM 3.3F-3059, from which the undersigned is authorized to draw.

Dated: September 14, 2022

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER, DAVID, LITTENBERG, KRUMHOLZ &
MENTLIK, LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant(s)

LD-458

Electronic Acknowledgement Receipt

EFS ID:	46613849
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Kevin M. Kocun/Sharon McGee
Filer Authorized By:	Kevin M. Kocun
Attorney Docket Number:	LGCHEM 3.3F-3059
Receipt Date:	14-SEP-2022
Filing Date:	16-JUL-2021
Time Stamp:	12:17:51
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Foreign Reference	LGCHEM3430_-_CA2572931A1.pdf	2493836 4438af5cdf1cf288032417168d78defeed4b7e9a	no	25

Warnings:

Information:					
2	Foreign Reference	LGCHEM3430_-_CN102473883.pdf	3664450	no	27
			3473d76c2fa8a718fe9e28bb884f5a4dede7f53c		
Warnings:					
Information:					
3	Foreign Reference	LGCHEM3430_-_CN103155225A.pdf	1860766	no	12
			c58cdc3bcb9e811794967833e3de58712dbc6cb27		
Warnings:					
Information:					
4	Non Patent Literature	LGCHEM3430_-_CN_SR_201980046222.pdf	225102	no	3
			87541125274911ff4aad7f350fe795dad5c878a		
Warnings:					
Information:					
5	Non Patent Literature	LGCHEM3430_-_CN_SR_2019800462228_-_English_Translation.pdf	236114	no	2
			228298b3a303e2fb5d689e05a69e0a755e65f147		
Warnings:					
Information:					
6	Information Disclosure Statement (IDS) Form (SB08)	LGCHEM3430_-_Information_Disclosure_Statement_Fillable_PDF.pdf	1263477	no	4
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Warnings:					
Information:					
Total Files Size (in bytes):			9765126		

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New International Application Filed with the USPTO as a Receiving Office

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Espacenet

Bibliographic data: CA2572931 (A1) — 2008-06-21

SAFETY STRUCTURE FOR A PLASTIC BATTERY CASE

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Applicant(s): WU DONALD P H [TW] ± (WU, DONALD P. H)

Classification: - **international:** H01M2/02
- **cooperative:** H01M10/0413 (EP); H01M10/0431 (EP);
H01M10/0463 (EP); H01M10/052 (EP); H01M10/0587
(EP); H01M50/121 (EP); H01M50/15 (EP);
H01M50/152 (EP); H01M50/3425 (EP); H01M50/528
(EP); Y02E60/10 (EP)

Application number: CA20062572931 20061221

Priority number(s): CA20062572931 20061221

Abstract of CA2572931 (A1)

A safety structure for a plastic battery case, the battery case is a plastic unitary structure, in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces the interior of the battery case, so as to facilitate the collection of pressure, and in the bottom of the groove is formed a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the closest grooves will bulge instantly, and the bulged working portion will not be ruptured right away. The working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure safely.



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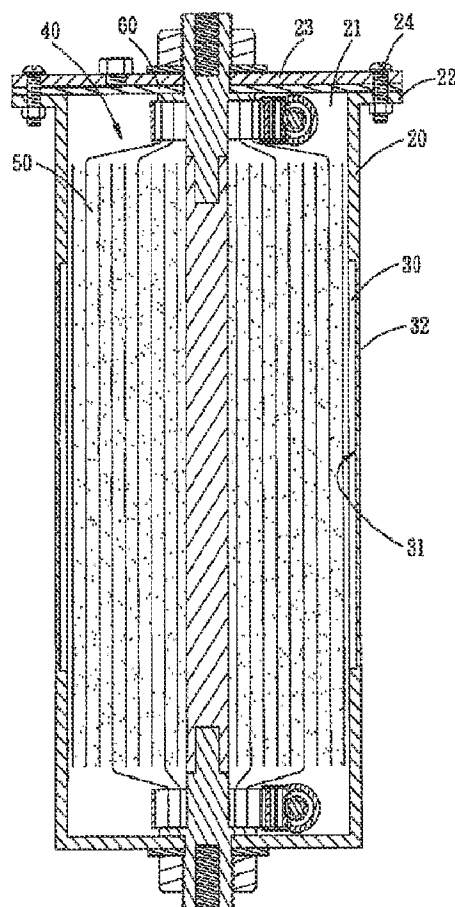
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(54) Titre : STRUCTURE DE SECURITE POUR BOITIER DE PILE EN PLASTIQUE

(54) Title: SAFETY STRUCTURE FOR A PLASTIC BATTERY CASE



(57) Abrégé/Abstract:

A safety structure for a plastic battery case, the battery case is a plastic unitary structure, in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces the interior of the battery case, so as to facilitate the

(57) **Abrégé(suite)/Abstract(continued):**

collection of pressure, and in the bottom of the groove is formed a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the closest grooves will bulge instantly, and the bulged working portion will not be ruptured right away. The working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure safely.

ABSTRACT OF THE DISCLOSURE

A safety structure for a plastic battery case, the battery case is a plastic unitary structure, in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces the interior of the battery case, so as to
5 facilitate the collection of pressure, and in the bottom of the groove is formed a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the closest grooves will bulge instantly, and the bulged working portion will not be ruptured right away. The working portion of the groove will bulge outward and become thinner and
10 consequently form a slit to release the pressure safely.

SAFETY STRUCTURE FOR A PLASTIC BATTERY CASE

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a safety structure for a plastic battery case, and
5 more particularly to a safety design for a plastic battery case for safely releasing the
internal pressure thereof.

Description of the Prior Art

Due to the great break-through in material technique, lithium secondary battery,
as a high capacity power supply (such as LiFePO_4 type lithium secondary battery),
10 has been used on power devices, such as electric bicycle, electric wheelchair, and etc.
The power storage and supply capacity of such a high capacity non-aqueous
electrolyte type lithium secondary battery are greater than that of a conventional
lithium battery. The battery case of the abovementioned lithium secondary battery is
usually a metal case made of aluminum. Since the battery case is filled with electrode
15 board and non-aqueous electrolyte, and in order to ensure a long term and stable
sealing performance, the open end of the battery case should be sealed with a sealing
plate by laser welding.

If the abovementioned lithium secondary battery is overcharged and
consequently the interior current value exceeds the normal value, or the battery is
20 misused and cause short circuit, the non-aqueous electrolyte in the battery case will be
fully decomposed and produce a great deal of gas. When the gas pressure inside the

battery case exceeds the maximum pressure limit, the battery case will be ruptured and gas will be gushed out of the battery case instantly, causing damage to the powered device the battery and causing electrolyte contamination.

Therefore, the aluminum sealing plate of the conventional non-aqueous
5 electrolyte lithium secondary battery is usually defined with an air hole sealed with a thin aluminum safety valve. When the internal gas pressure of the battery case exceeds a predetermined value, the gas will break through the aluminum valve, thus releasing the pressure from the battery case.

It is to be noted that the safety valve is located on the surface of the sealing
10 plate, considering the battery as a whole, the safety valve is located comparatively far away from mid portion of the electrode board, and since the air hole is limited by the size of the sealing plate, the inner diameter of the air hole will not be large. The gas will take a long time to break through the safety valve. The normal condition is that, when the internal pressure of the battery case increases, the mid portion of the battery
15 case will be inflated and ruptured instantly, however, the safety valve doesn't work. Obviously, such a conventional safety valve is unable to solve the safety problem of the battery case.

Therefore, as shown in Fig. 1, a safety mechanism for an aluminum rectangular battery case was made and was disclosed in US Pat. No. 6,964,690, in the elongated
20 side surface of the battery case 10 is formed a plurality of cutting grooves 11. Between a groove bottom surface of the cutting groove 11 and an inner surface of the

battery case 10 is provided a thin-walled easily-rupturable portion designed to rupture at an instant when an internal pressure of the battery case reaches a predetermined value, thus releasing the gas pressure from the battery case.

For such a safety mechanism design of cutting the cutting grooves in the metal
5 battery case, when the internal pressure of the battery case increases, the easily-rupturable portion will be ruptured. It is to be noted that a great deal of gas pressure impacts the small easily-rupturable portion in a very short time, the resultant ruptured grooves in the battery case will be very large, and as a result, the gas and the non-aqueous electrolyte will be gushed out of the battery case via the ruptured
10 grooves, and the electrode board inside the battery case will be exposed out of the ruptured portion. This is a very large threat to a powered device.

The safety structure of the present invention is such that, when the internal pressure of the battery case increases, the strong impact force caused by the unusual gas pressure increase is buffered and then is released through small slits.

15

SUMMARY OF THE INVENTION

The primary objective of the present invention is to provide a plastic integrally formed battery case, in the inner surface of the battery case is formed a plurality of grooves whose open side face the interior of the battery case. When the internal pressure of the battery case is too high, the closest grooves will bulge instantly, and
20 the bulged working portion will not be ruptured right away. When the thickness of the working portion is thinned to zero, a small slit will appear. At this moment, the gas

can be released from the slit, thus preventing a great deal of gas from gushing out of the battery case in an instant and contaminating the powered device.

The secondary objective of the present invention is to provide a plastic integrally formed battery case, in the inner surface of the battery case is formed a plurality of grooves whose open side face the interior of the battery case, the open side of the groove is opposite the direction of the gas pressure, thus facilitating the collection of pressure, and making it easier for the thinnest working portion to bulge outward.

BRIEF DESCRIPTION OF THE DRAWINGS

10 Fig. 1 is a perspective view of showing a conventional safety structure for an aluminum rectangular battery disclosed in US Pat. No. 6,964,690;

Fig. 2 is an exploded view of a circular plastic battery case and a core assembly in accordance with the present invention;

15 Fig. 3 is a longitudinal cross sectional view in accordance with the present invention of showing a circular plastic battery case;

Fig. 4 is a transverse cross sectional view in accordance with the present invention of showing a circular plastic battery case;

Fig. 5 is a longitudinal cross sectional view in accordance with the present invention of showing that the working portion in Fig. 4 is being bulged;

20 Fig. 6 is a longitudinal cross sectional view in accordance with the present invention of showing that the working portion in Fig. 4 is bulged and ruptured to

release the gas;

Fig. 7 is an exploded view of a circular plastic battery case with longitudinal and transverse grooves and a core assembly in accordance with the present invention;

Fig. 8 is a longitudinal cross sectional view in accordance with the present invention of showing that the longitudinal and transverse working portions in Fig. 7 is bulged and ruptured to release the gas;

Fig. 9 is an exploded view of a rectangular plastic battery case and a core assembly in accordance with the present invention; and

Fig. 10 is a longitudinal cross sectional view in accordance with the present invention of showing that the working portion in Fig. 9 is bulged and ruptured to release the gas.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention will be more clear from the following description when viewed together with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiment in accordance with the present invention.

Referring to Figs. 2, 3 and 4, a safety structure for a plastic battery case in accordance with a preferred embodiment of the present invention comprises at least one groove formed in a battery case, wherein:

The battery case 20 is a circular plastic unitary structure in which is defined a receiving space 21 with an open end for accommodation of a core assembly 40, a leakage-proof washer 60, and non-aqueous electrolyte 50. A connecting flange 22 is

formed around the peripheral edge of open end of the receiving space, a cover 23 is fixed to the connecting flange 22 by screws 24 to seal the open end, thus creating an airtight receiving space 21.

In the inner surface of the receiving space 21 of the battery case 20 is formed at least one groove 30 whose open side 31 faces the interior of the battery case 20, and between the bottom of the groove 30 and the outer surface of the battery case 20 is a working portion 32 which is the thinnest portion of the wall of the battery case 20.

When unusual factors (including: overcharge, short circuit of the electrode plate, overload) occur and consequently the non-aqueous electrolyte is decomposed into high pressure gas, the working portion can substantially bulge outward and then become thinner and consequently form a slit to release the gas from the slit.

The battery case of the present invention is made of plastic material, in the inner surface of the battery case is formed at least one groove, and when the internal pressure of the battery case exceeds a predetermined value, the working portion of the groove will bulge outward and become thinner and consequently form a slit for releasing the pressure. And such a pressure-releasing process can prevent the battery from explosion, and ensure that the internal pressure can be released safely. The operation of the embodiment is explained as follows:

Referring to Figs. 2, 3 and 4 again, the battery case 20 is a unitary structure made of plastic material, in the inner surface of the battery case 20 is formed a plurality of grooves 30, and between the bottom of the grooves 30 and the outer wall

of the battery case 20 is a working portion 32 which is the thinnest portion of the wall of the battery case 20. The number of the grooves 30 is based on the size of the battery case 20. For example, a 20 cm diameter circular battery case 20 can be axially formed in its inner surface with four grooves 30, and the grooves 30 are equidistantly
5 arranged.

Referring then to Figs. 4, 5 and 6, when unusual factors occur and consequently the non-aqueous electrolyte is decomposed into high pressure gas 70, the gas pressure will make the battery case 20 bulge outward. It is to be noted that the location where the unusual factors take place may be located at one of the inner sides of the battery
10 case 20 only, and the pressure on this inner side of the battery case 20 is the greatest. The working portion 32 of the grooves 30 located closest to the high pressure point will bulge instantly. The battery case 20 is made of plastic material, and the plastic surface of the battery case 20 has certain flexibility, so that the bulged working portion 32 will not be ruptured right away. The rupture time is prolonged, and then a
15 small slit 33 will appear when the thickness of the working portion 32 is thinned to zero. At this moment, the gas 70 will be released from the battery via the slit 33.

It has to be mentioned that the battery case of the present invention is made of plastic material, and as compared with the conventional metal battery case, the plastic battery case makes it easier for the battery to pass the free-fall test.

20 It is to be noted that, as shown in Fig. 4, each of the grooves 30 is V-shaped in cross section, and the open side 31 of the grooves 30 faces the interior of the battery

case 20. When the unusual gas pressure is applied to the inner surface of the battery case 20, the open side 31 of the grooves 30 is opposite the direction of the gas pressure for facilitating the collection of pressure, and making it easier for the thinnest working portion 32 to bulge toward the radial direction of the battery and consequently form a slit 33. And thus the pressure can be released safely. Therefore, it is very important that the open side 31 of the grooves 30 opens toward the interior of the battery case 20.

In addition, the location of the grooves 30 is the position where the gas 70 inside the battery case 20 is to be released. With the structural design, the grooves 30 can be arranged at a predetermined position, so that the position where the gas 70 to be released can be anticipated, thus preventing the powered device from being contaminated and damaged. In addition to the longitudinal direction as shown in Fig. 7, the direction in which the grooves 30 are arranged can also be arranged in transverse direction. For example, as shown in Fig. 8, the battery case can be formed with longitudinal grooves 30 and transverse grooves 30a, and it has the same pressure releasing effect.

In addition to being used on circular battery case, the safety structure design of the present invention is also suitable for use in a rectangular battery case. With reference to Fig. 9, the grooves 30 and the working portion 32 are formed in the rectangular battery case. And as shown in Fig. 10, which is an illustrative view of showing that the working portion 32 of the rectangular battery 20a is bulged and

ruptured.

It can be seen from the above description that the battery case of the present invention is formed in its inner surface thereof with at least one groove, and since the plastic battery case is flexible, when the internal pressure of the battery case exceeds a
5 predetermined value, the working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure. In other words, the safety pressure release is achieved by the flexible inflation of the plastic material, so that the strong impact force caused by the unusual gas pressure increase is buffered and then is released.

10 While we have shown and described various embodiments in accordance with the present invention, it is clear to those skilled in the art that further embodiments may be made without departing from the scope of the present invention.

WHAT IS CLAIMED IS:

1. A safety structure for a plastic battery case, the battery case being a plastic unitary structure in which being defined a receiving space with an open end for accommodation of electrode plate and electrolyte, the open end being sealed with a cover, thus creating an airtight receiving space; the safety structure is characterized in that:

in an inner surface of the receiving space of the battery case is formed at least one groove whose open side faces an interior of the battery case, and between a bottom of the groove and an outer surface of the battery case is a working portion which is the thinnest portion of the battery case, when internal pressure of the battery case exceeds a predetermined value, the working portion of the groove will bulge outward and become thinner and consequently form a slit to release the pressure.

2. The safety structure for a plastic battery case as claimed in claim 1, wherein the groove is V-shaped in cross section, and the open side of the groove faces the interior of the battery case.

3. The safety structure for a plastic battery case as claimed in claim 1, wherein the groove is arranged in a longitudinal direction.

4. The safety structure for a plastic battery case as claimed in claim 1, wherein the groove is arranged both in longitudinal and transverse directions.

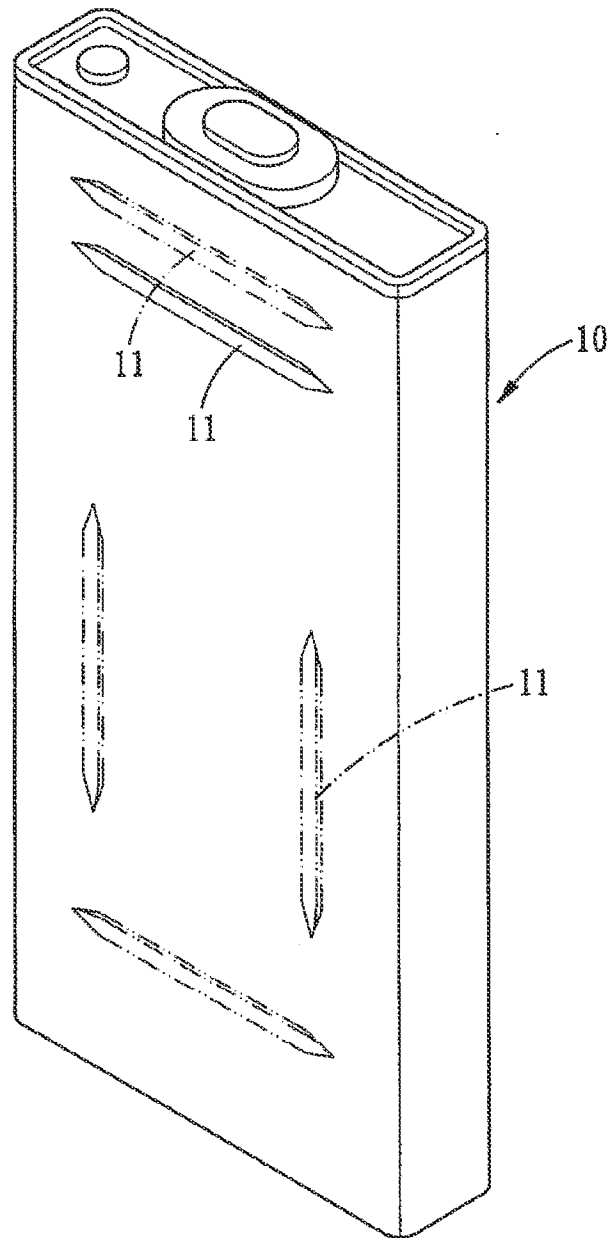
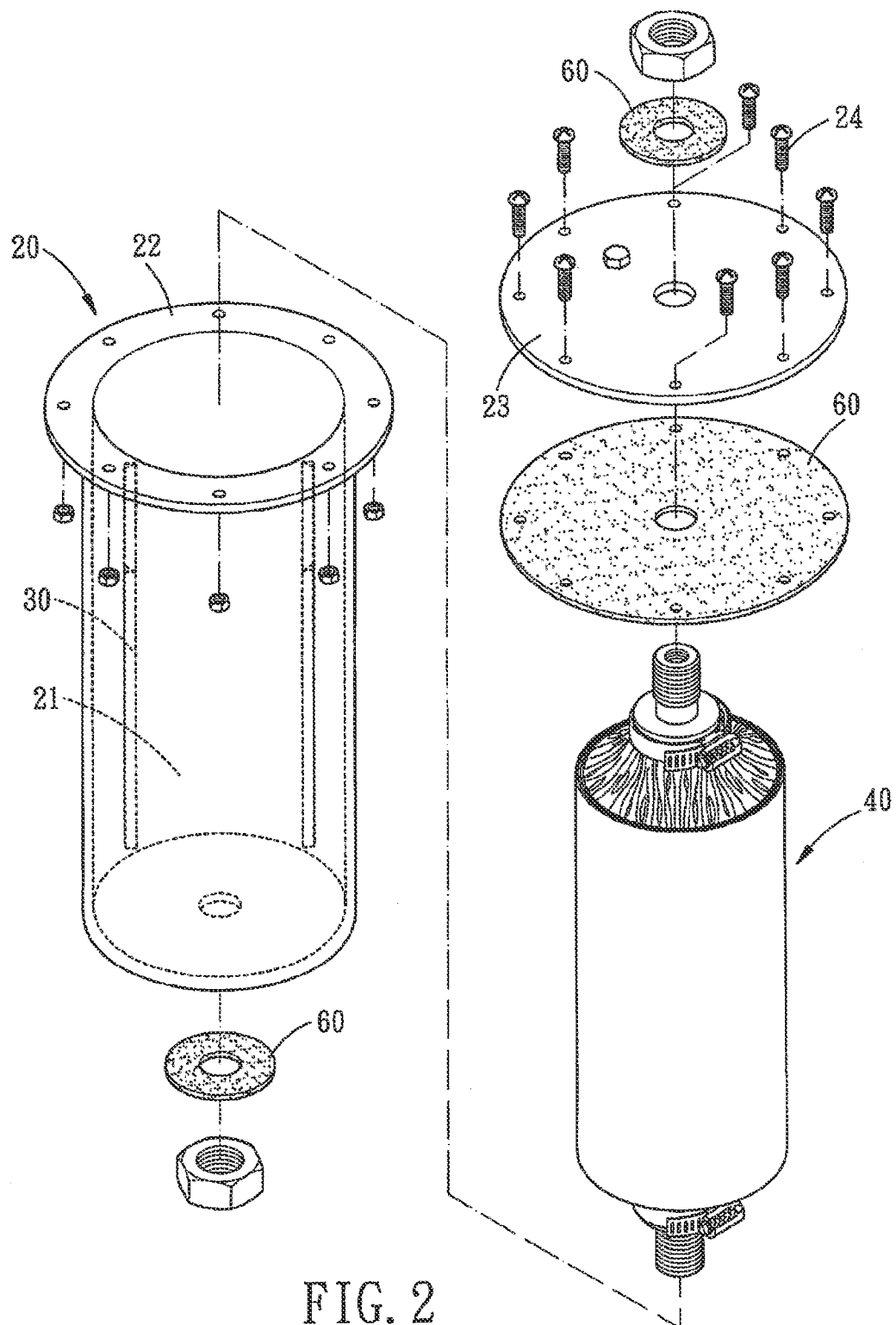


FIG. 1
PRIOR ART



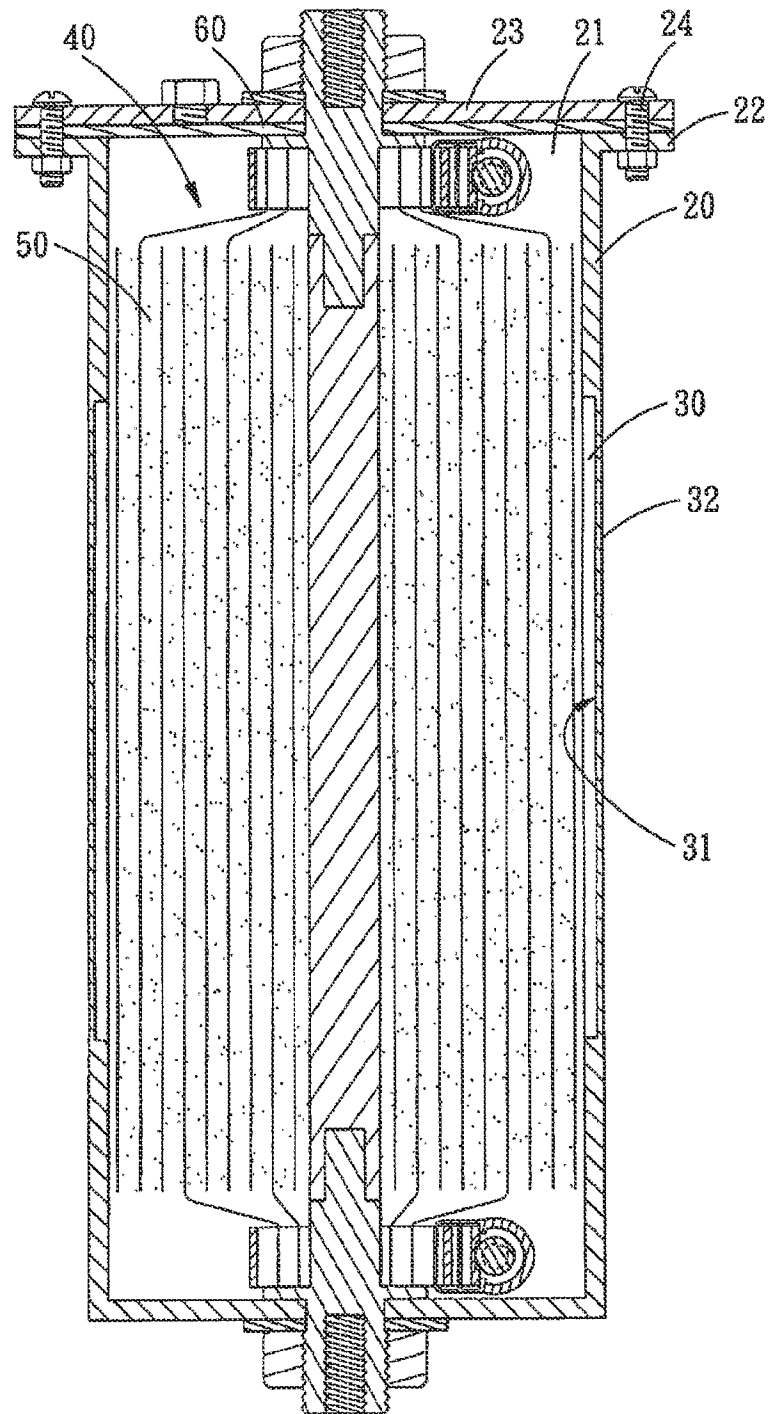


FIG. 3

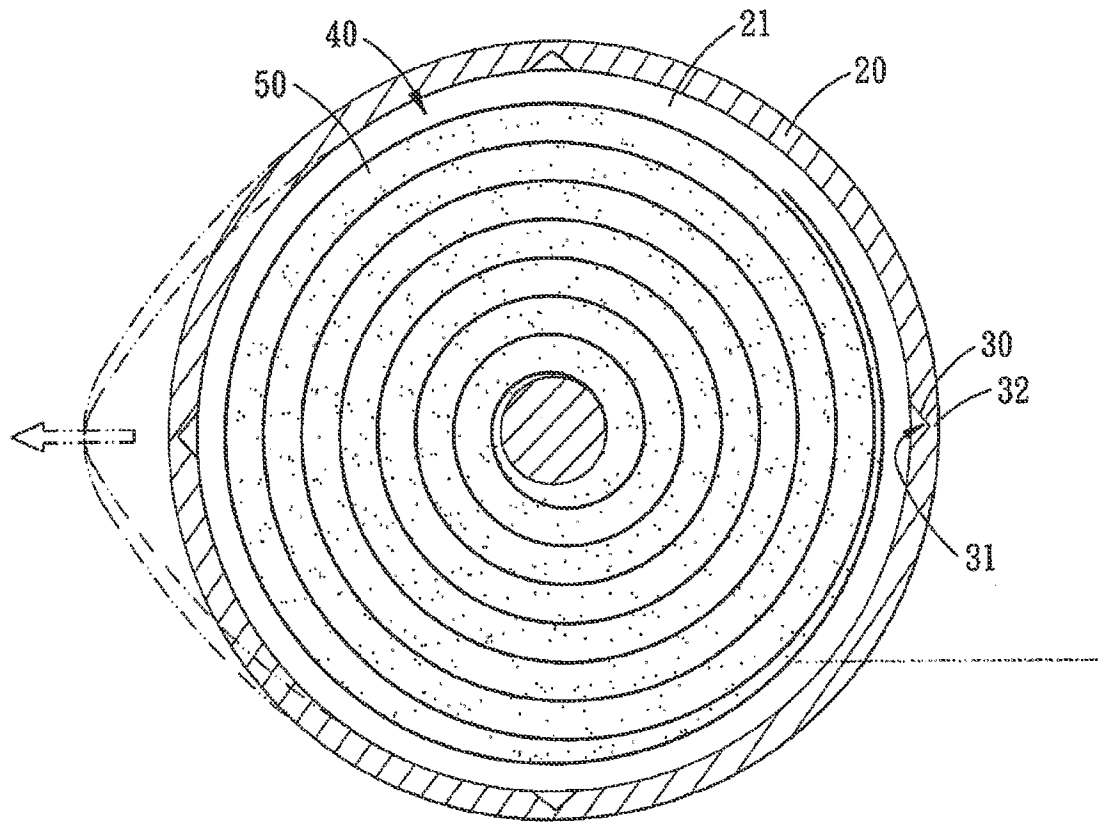


FIG. 4

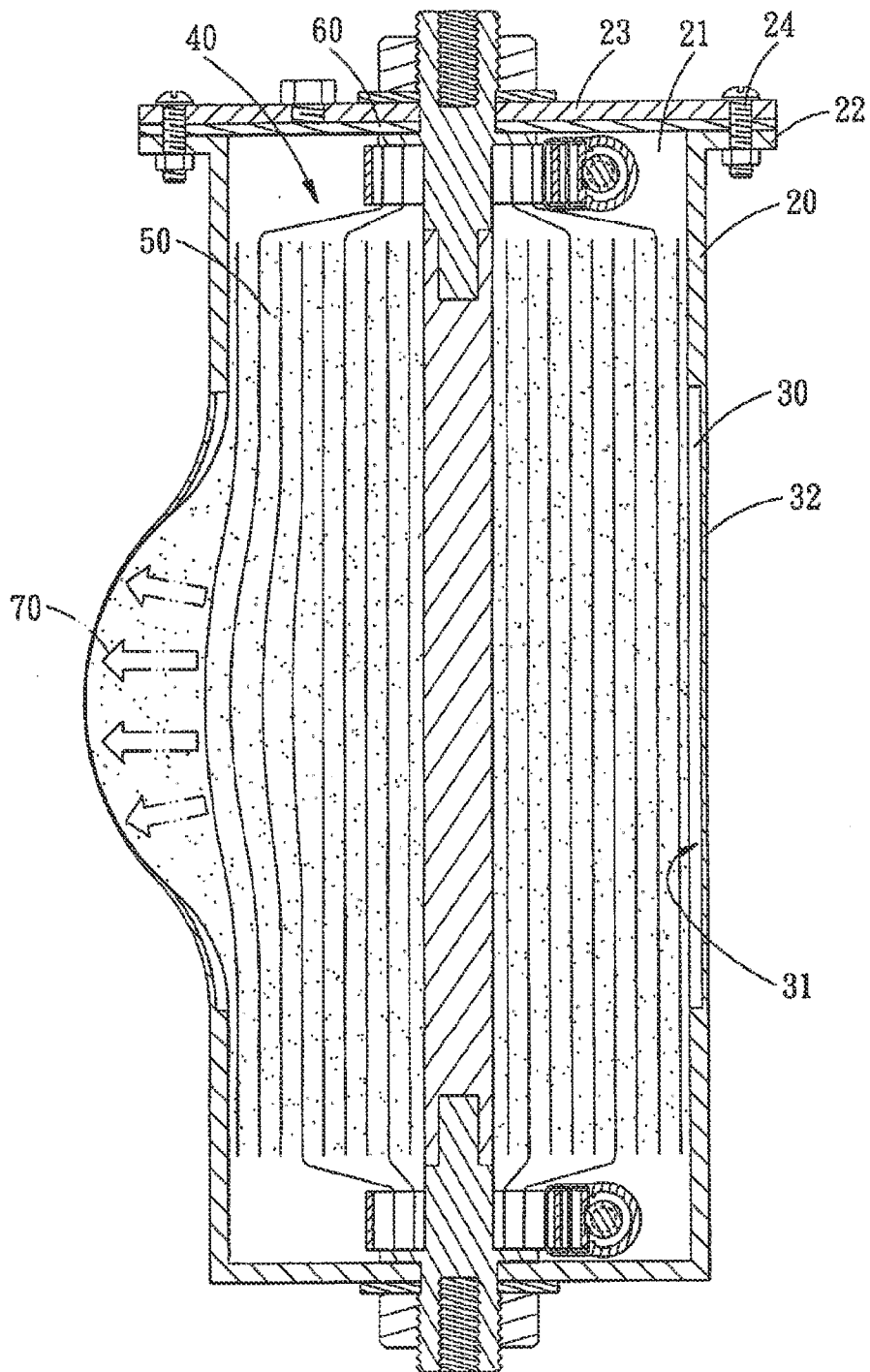


FIG. 5

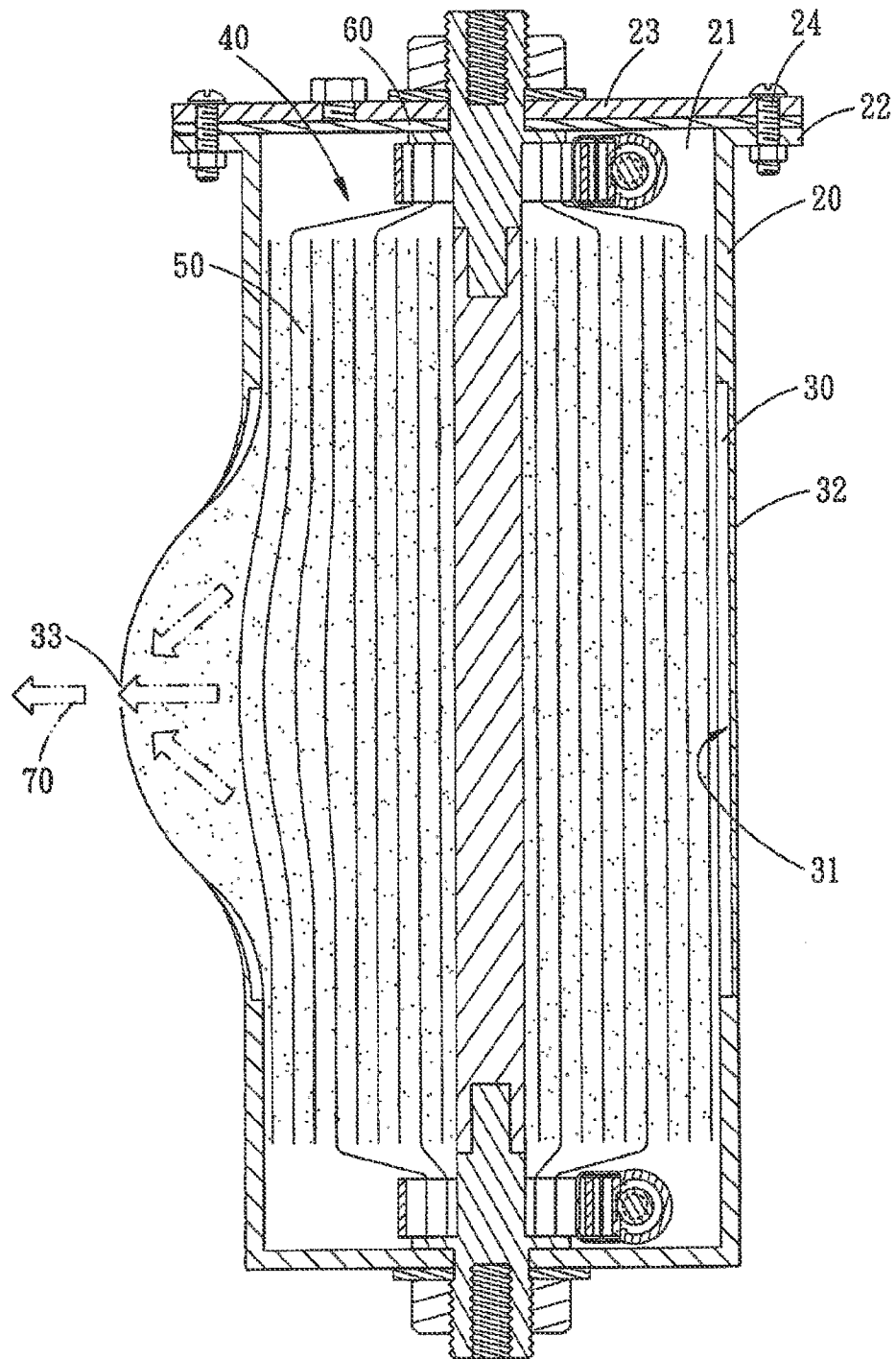


FIG. 6

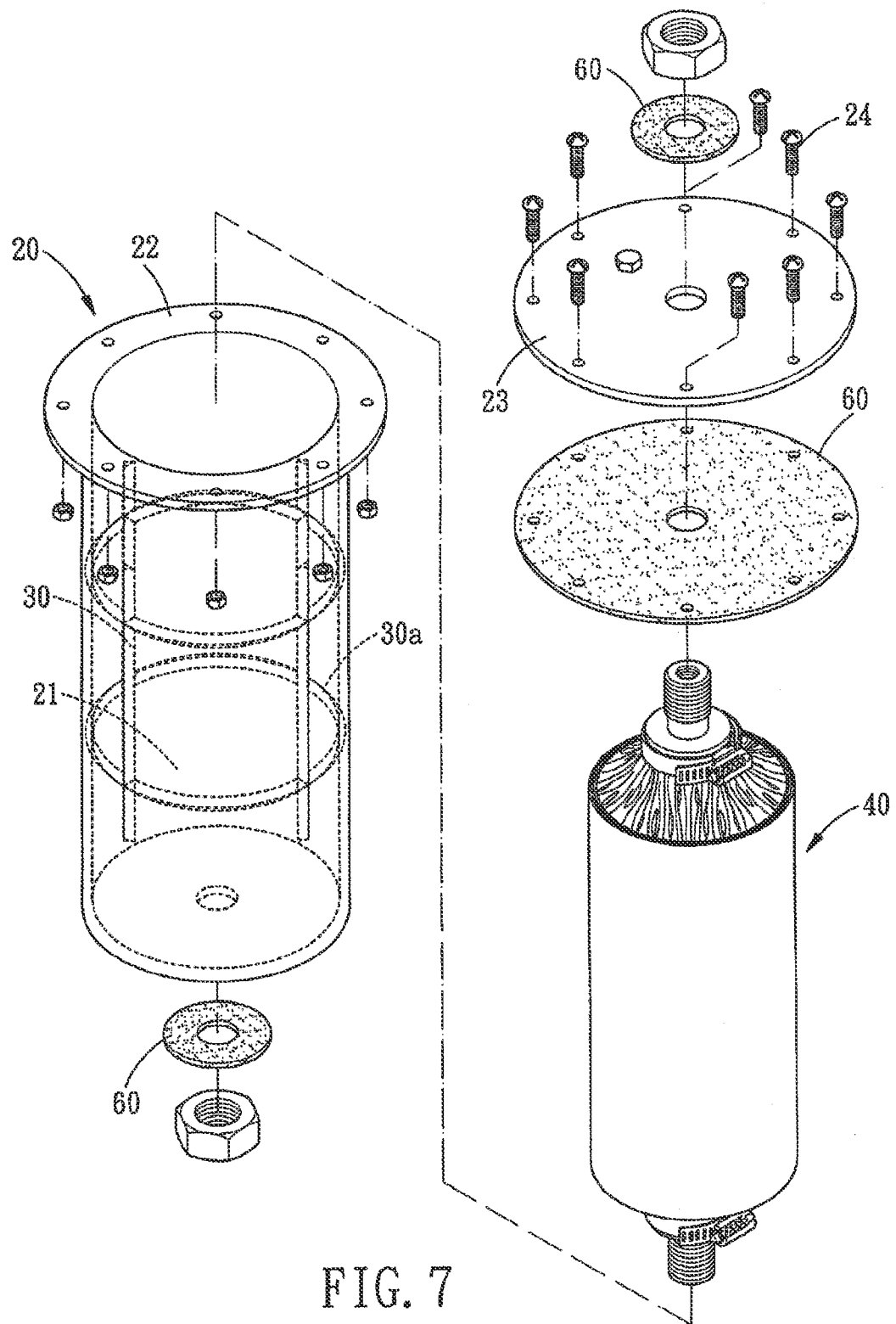


FIG. 7

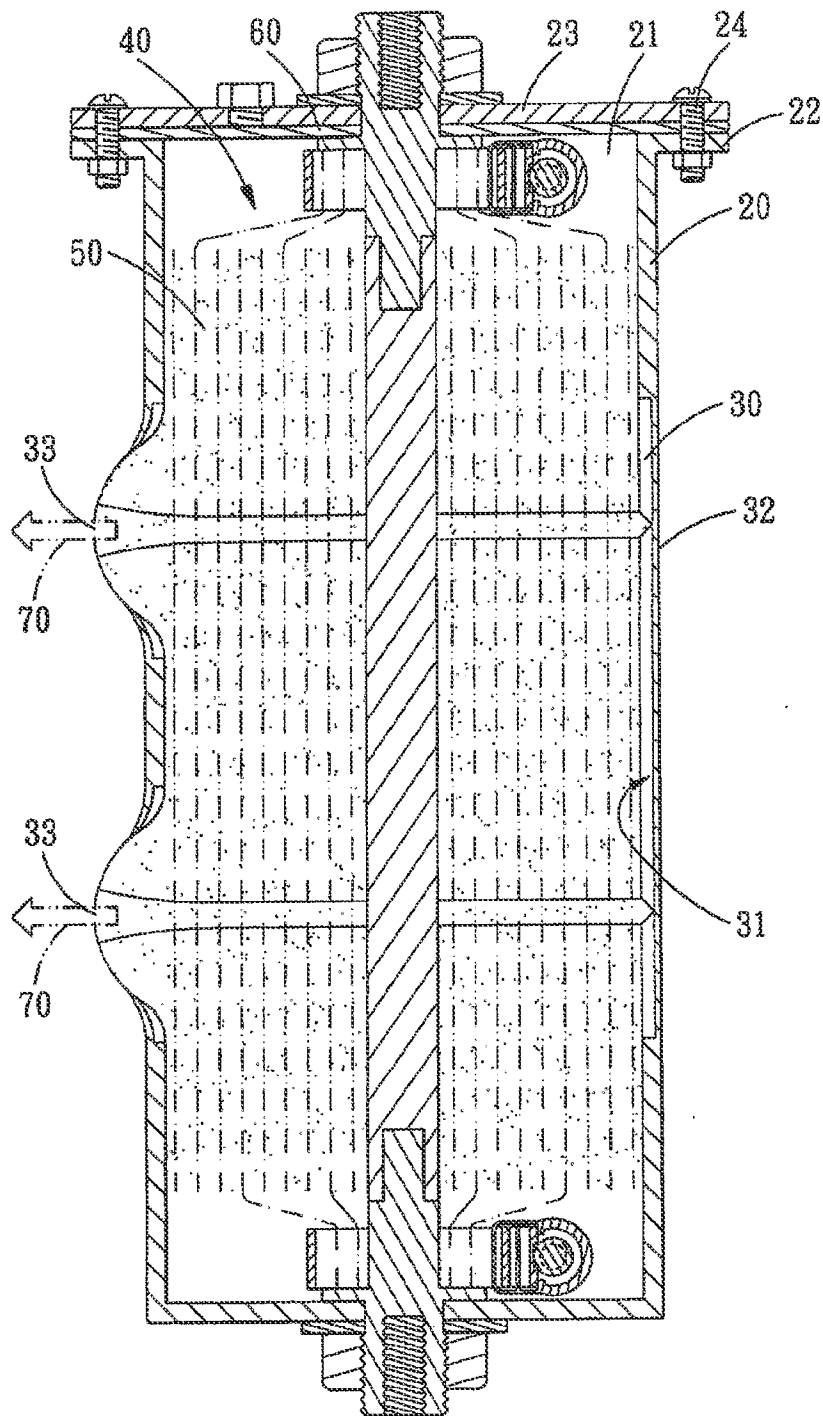


FIG. 8

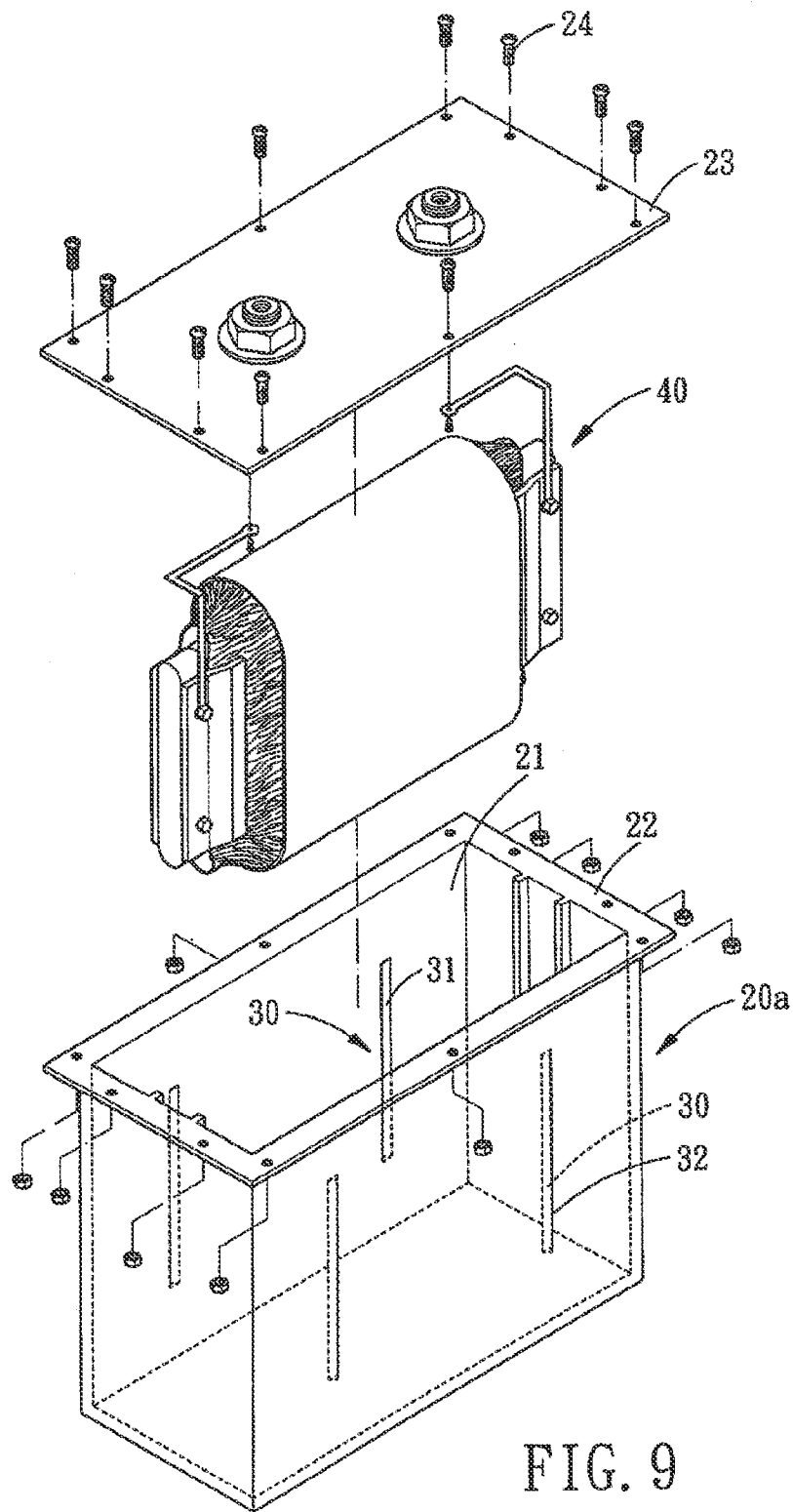


FIG. 9

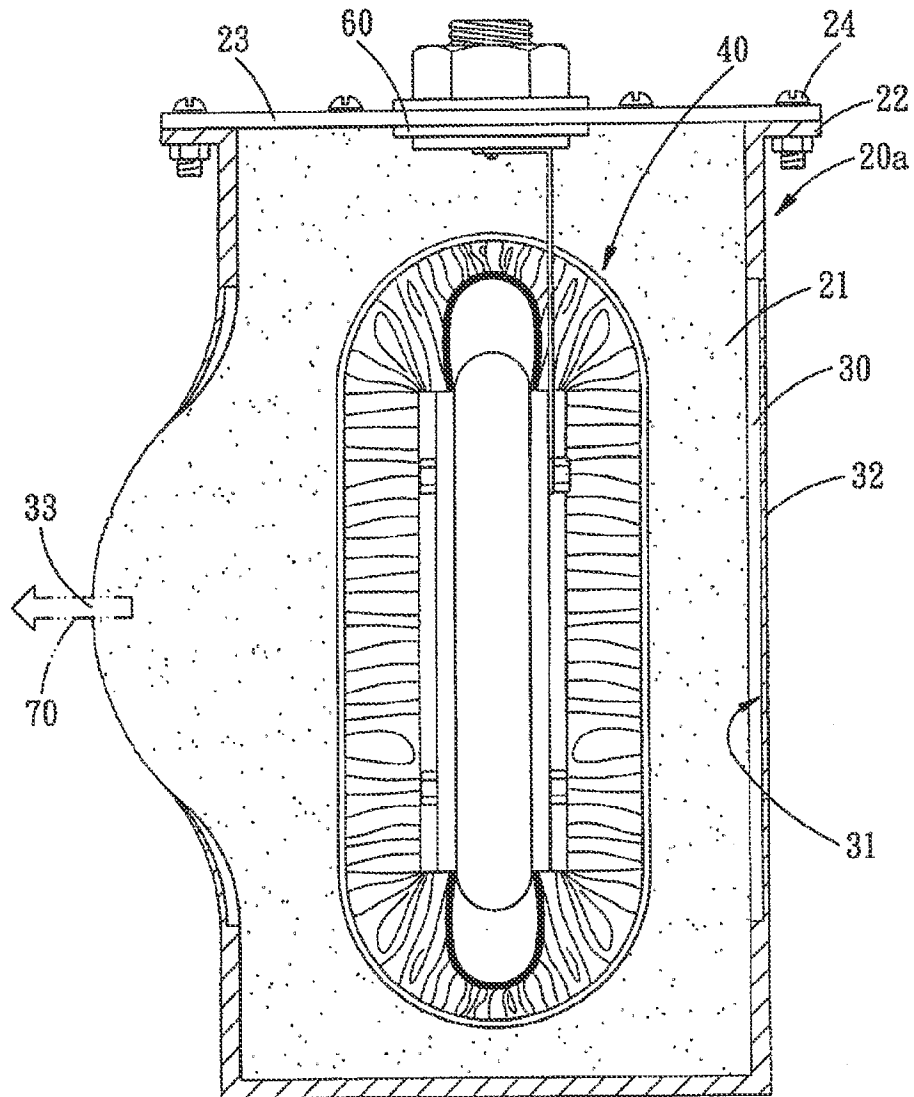
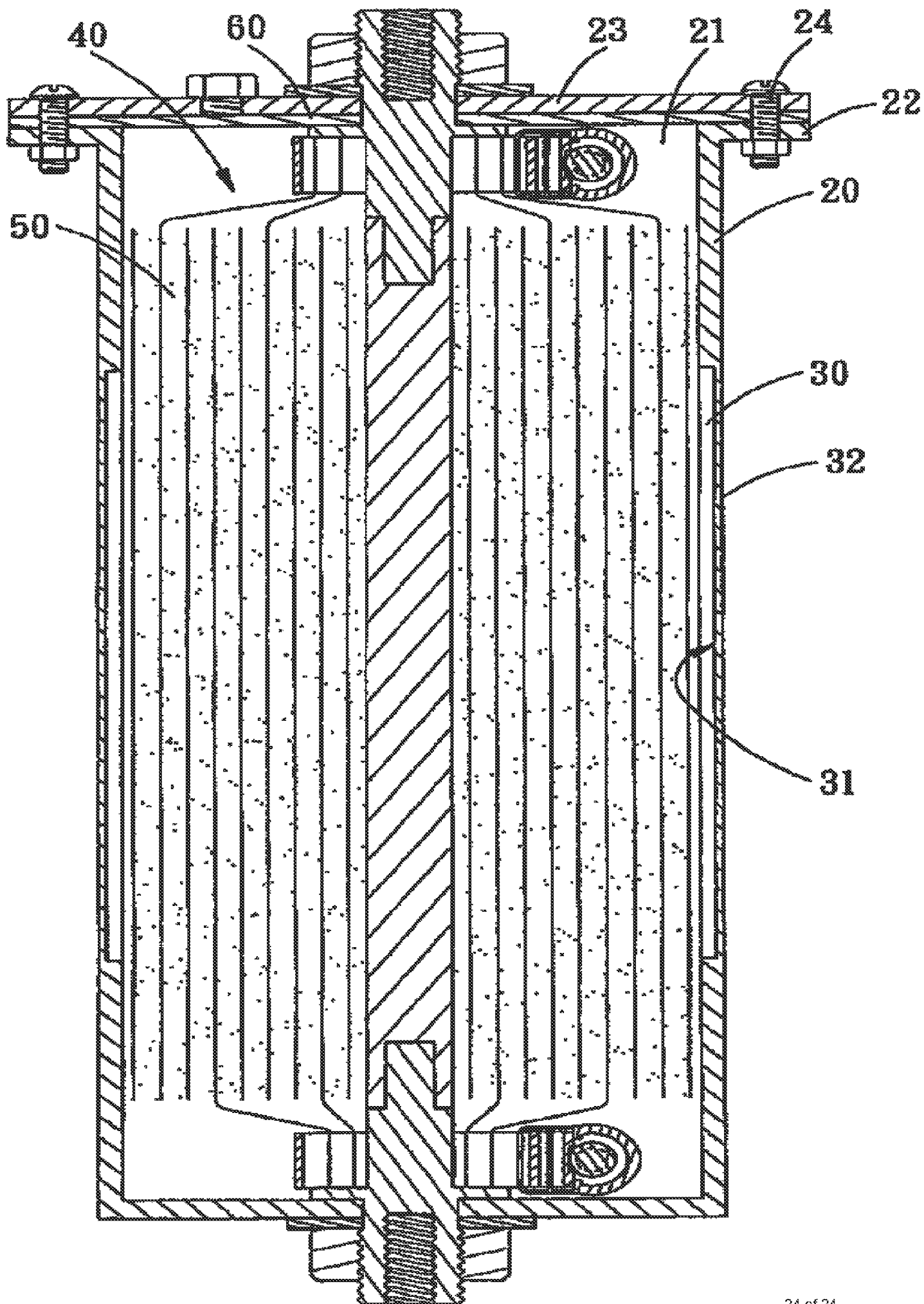


FIG. 10





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Cylindrical rechargeable battery with improved stability

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- **cooperative:** H01M10/0431 (EP, US); H01M10/0587 (KR);
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H01M10/0525 (KR); Y02E60/10 (EP); Y02P70/50 (EP)

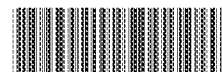
Application number: CN20108034848 20100813 Global Dossier

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EP2466668 (B1) JP2013502035 (A) JP5577405 (B2)
KR101058386 (B1) KR20110017835 (A) US2012114979 (A1)
US9112234 (B2) WO2011019237 (A2) WO2011019237 (A3) less

Abstract of CN102473883 (A)

Provided is a cylindrical rechargeable battery in which an electrode assembly produced by the winding of an anode/separating film/cathode and an electrolytic solution are housed inside a cylindrical can, wherein a cap assembly mounted on the open upper part of the cylindrical can comprises: a safety vent formed with a predetermined notch so as to be ruptured by high-pressure gas in the battery; a current-blocking member which is coupled by welding to the lower edge of the safety vent and blocks current when the internal pressure of the battery rises; and a gasket for the current-blocking member surrounding the outer circumferential surface of the current-blocking member, and wherein the current-blocking member comprises two or more through holes for discharging gas, and the through holes are formed with a size such that their overall surface area is from 20 to 50% of the overall surface area of the current-blocking member.



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代理人 陈海涛 穆德骏

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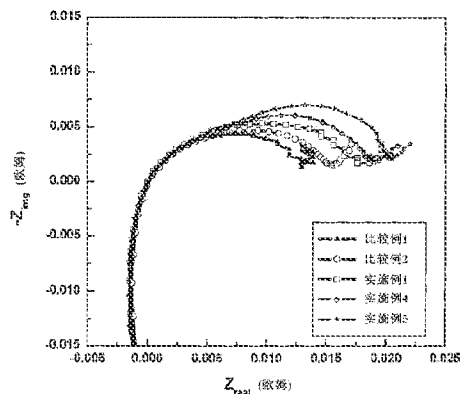
权利要求书 2 页 说明书 13 页 附图 10 页

(54) 发明名称

具有改善的安全性的圆筒形二次电池

(57) 摘要

本发明提供了一种其中通过卷绕正极 / 隔膜 / 负极而制造的凝胶卷型电极组件以及电解质设置在圆筒形罐中的圆筒形二次电池, 其中安装在所述圆筒形罐的开放顶部上的盖组件包含: 安全排气口, 所述安全排气口具有预定凹口以允许由于电池的高压气体而造成的破裂; 用于中断电流的电流中断装置, 所述电流中断装置焊接至安全排气口的底部; 和用于所述电流中断装置的衬垫, 所述衬垫用于围绕所述电流中断装置的周边, 其中所述电流中断装置包含两个以上允许排放气体的通孔, 其中所述通孔具有相对于所述电流中断装置的总面积为 20% 至 50% 的尺寸。



1. 一种圆筒形电池,其中通过卷绕正极/隔膜/负极而制造的电极组件(凝胶卷)和电解质设置在圆筒形罐中,

其中安装在所述圆筒形罐的开放顶部上的盖组件包含:

安全排气口,所述安全排气口具有预定凹口以允许由于所述电池的高压气体而造成的破裂;

用于中断电流的电流中断装置,所述电流中断装置焊接至所述安全排气口的底部;和用于所述电流中断装置的衬垫,所述衬垫用于围绕所述电流中断装置的周边,

其中所述电流中断装置包含两个以上允许排放气体的通孔,其中所述通孔具有相对于所述电流中断装置的总面积为 20%至 50%的尺寸。

2. 根据权利要求 1 所述的圆筒形电池,其中所述电流中断装置具有如下结构:在其中心处具有向上突出且焊接至所述安全排气口的凹陷部的底部的突出部,并且连接至所述凝胶卷的正极的正极引线通过所述突出部之外的区域的所述底部发生电连接;和桥,所述桥具有三至五个通孔和用于连接相邻通孔的凹口,所述通孔基于所述突出部的中心同心地形成。

3. 根据权利要求 1 所述的圆筒形电池,其中所述通孔具有相对于所述电流中断装置的总面积为 30%至 40%的面积。

4. 根据权利要求 1 所述的圆筒形电池,其中所述通孔在基于所述电流中断装置的中心径向对称的位置处以圆弧形式形成。

5. 根据权利要求 2 所述的圆筒形电池,其中所述三个通孔被布置成使得它们基于所述电流中断装置的中心以圆弧形式彼此间隔预定距离,所述圆弧具有与 50 度至 80 度的角度相对应的长度。

6. 根据权利要求 2 所述的圆筒形电池,其中所述通孔被布置成使得它们基于所述电流中断装置的中心以圆弧形式彼此间隔预定距离,所述圆弧具有与 40 度至 55 度的角度相对应的长度。

7. 根据权利要求 1 所述的圆筒形电池,其中所述安全排气口具有向下凹陷的中心,并且构成所述凹陷部的上弯曲部和下弯曲部分别具有第一凹口和第二凹口,

其中设置在上部区域上的所述第一凹口具有其中一侧敞开的开口曲线。

8. 根据权利要求 7 所述的圆筒形电池,其中所述第二凹口具有其中彼此面对的两端都敞开的开口弯曲结构。

9. 根据权利要求 7 所述的圆筒形电池,其中所述第一凹口具有与假想线的尺寸相等的尺寸,所述假想线用于连接所述通孔的中心线。

10. 根据权利要求 7 所述的圆筒形电池,其中相对于所述安全排气口的中心,所述第一凹口的开口区域具有与 20 度至 50 度的角度相对应的圆弧长度,从而在所述第一凹口被切开时防止所述凹陷部被完全切开。

11. 根据权利要求 8 所述的圆筒形电池,其中在所述第二凹口中敞开的所述两端形成于不与所述第一凹口的开口区域相对应的区域中。

12. 根据权利要求 1 所述的圆筒形电池,其中所述盖组件具有电流中断装置、用于电流中断装置的衬垫、安全排气口、PTC 元件和其中形成一个或多个气体出口的顶盖的层压结构,其中在所述层压结构的外围上安装有衬垫。

13. 根据权利要求 1 所述的圆筒形电池,其中所述衬垫在其中心处具有开口,并且所述开口与所述电流中断装置的所述通孔连通。

14. 根据权利要求 13 所述的圆筒形电池,其中所述电流中断装置的所述通孔、用于电流中断装置的衬垫的所述中心开口和所述安全排气口的所述凹口以使得它们在直线上彼此连通的方式形成。

15. 根据权利要求 1 所述的圆筒形电池,其中所述圆筒形电池包含含添加剂(电解质添加剂)的电解质以提高热稳定性。

16. 根据权利要求 15 所述的圆筒形电池,其中所述电解质添加剂在 150℃ 以上的温度下随着温度增加而引起连续的电解质分解,从而产生气体。

17. 根据权利要求 16 所述的圆筒形电池,其中所述电解质添加剂是在 150℃ 至 250℃ 下引起一次电解质分解和在 250℃ 至 300℃ 下引起二次电解质分解的材料。

18. 根据权利要求 16 所述的圆筒形电池,其中所述电解质添加剂为氟代碳酸亚乙酯(FEC)。

19. 根据权利要求 15 所述的圆筒形电池,其中基于电解质的总重量,所述电解质添加剂的含量为 11 重量%至 40 重量%。

20. 根据权利要求 15 所述的圆筒形电池,其中基于电解质的总重量,所述电解质添加剂的含量为 15 重量%至 35 重量%。

21. 一种安装在电池中的圆筒形罐的开放顶部上的盖组件,在所述电池中电极组件和电解质设置在圆筒形罐中,

其中,基于电解质的重量,电池的电解质包含 11 重量%至 40 重量%的添加剂,从而在 150℃ 以上的温度下随着温度增加通过连续的电解质分解而产生气体,

其中,在凝胶卷的顶部上安装有板型绝缘构件,所述板型绝缘构件在其中心处具有允许排放气体以及连接电极端子的穿孔开口,并具有布置在所述开口周围的一个或多个通孔,所述板型绝缘构件的所述通孔具有相对于所述板型绝缘构件的总面积为 20%至 50% 的尺寸。

22. 根据权利要求 21 所述的盖组件,其中所述板型绝缘构件的所述通孔的位置使得所述通孔在直线上与用于电流中断装置的衬垫的中心开口和所述电流中断装置的通孔连通。

具有改善的安全性的圆筒形二次电池

技术领域

[0001] 本发明涉及圆筒形二次电池。更具体而言,本发明涉及其中通过卷绕正极/隔膜/负极而制造的电极组件和电解质设置在圆筒形罐中的圆筒形电池,其中安装在圆筒形罐的开放顶部上的盖组件可包含:安全排气口,所述安全排气口具有预定凹口以允许因电池的高压气体而造成的破裂;电流中断装置,所述电流中断装置焊接至安全排气口的底部并用于中断电流;和用于所述电流中断装置的衬垫,所述衬垫用于围绕所述电流中断装置的周边,其中所述电流中断装置包含两个以上允许排放气体的通孔,其中所述通孔具有相对于所述电流中断装置的总面积为 20%至 50%的尺寸。

背景技术

[0002] 技术发展和对移动设备增加的需求已经导致对作为能源的二次电池的需求迅速增加。在这些二次电池中,正在进行与具有高能量密度和放电电压的锂二次电池相关的大量研究,并且这种锂二次电池可商购获得并被广泛使用。

[0003] 根据电池壳体的形状,二次电池分为其中电极组件安装在圆筒形或角形金属容器内的圆筒形电池和角形电池以及其中电极组件安装在由铝层压片制成的袋型壳体內的袋型电池。其中,圆筒形电池具有容量和结构稳定性相对高的优点。

[0004] 安装在电池壳体中的电极组件是用于产生电的可再充电且可放电的装置,并且具有正极/隔膜/负极的层压结构。电极组件分为:凝胶卷型(jelly-roll type)电极组件,所述电极组件通过将隔膜夹在负极和涂布了活性材料的具有长片形状的正极之间,然后进行卷绕而制成;和堆叠型电极组件,所述电极组件通过依次层压具有预定尺寸的多个正极和具有预定尺寸的多个负极,使得将隔膜夹在每个正极和每个负极之间而制成。其中,凝胶卷型电极组件具有易于制造且单位重量的能量密度高的优点。

[0005] 在这一点上,图 1 是示意性地示出一般圆筒形电池的横截面透视图。

[0006] 参考图 1,通过将凝胶卷型电极组件 120 放置在圆筒形壳体 130 中;将电解质注入到圆筒形壳体 130 内,并且将具有电极端子(例如,正极端子,未示出)的顶盖 140 连接到壳体 130 的开放顶部,制造了圆筒形二次电池 100。

[0007] 电极组件 120 具有如下结构,其中正极 121、负极 122 和夹在两者之间的隔膜 123 被卷绕,并且圆筒形中心销 150 插入在电极组件 120 的卷中心(凝胶卷中心)。中心销 150 通常由金属材料制成以提供预定强度并具有其中片材料被圆形弯曲的中空圆筒形结构。中心销 150 固定和支撑电极组件并充当通道,从而使得可排放由于在充电/放电时的内部反应而产生的气体。

[0008] 同时,锂二次电池具有低稳定性的缺点。例如,在其中将电池过充电至约 4.5V 以上的情况中,正极活性材料发生分解,在负极上生长锂枝晶,并且电解质发生分解。这些过程涉及热反应和分解反应并且因此迅速进行多个副反应。最终,可能发生电池的燃烧和爆炸。

[0009] 因此,为了解决这些问题,一般的圆筒形二次电池具有电流中断装置(CID)和安

全排气口以在电池异常运行时中断电流并减小设置在电极组件和顶盖之间的空间中的内部压力。

[0010] 将参考图 2 至 4 对该机构进行描述。

[0011] 参考附图,顶盖 10 形成突出形式的正极端子并具有穿孔排气口。在顶盖 10 下方依次设置有:正温度系数 (PTC) 元件 20,所述正温度系数 (PTC) 元件 20 在电池的内部温度增大时大大增加电池的电阻并从而中断电流;安全排气口 30,所述安全排气口 30 在正常状态下向下突出,但在电池的内部压力增大时突出并同时发生破裂,从而导致排放气体;和连接板 50,所述连接板 50 的顶部一侧连接至安全排气口 30 且其底部的另一侧连接至电极组件 40 的正极。

[0012] 因此,在正常运行条件下,电极组件 40 的正极依次通过引线 42、连接板 50、安全排气口 30 和 PTC 元件 20 连接至顶盖 10 以施加电。

[0013] 然而,如图 3 中所示,当由于例如过充电而从电极组件 40 产生气体并且内部压力增大时,安全排气口 30 的形状反转,由此向上突出。此时,安全排气口 30 与连接板 50 分离,并从而中断电流。因此,确保了安全性,以便防止进一步过充电。然而,如图 4 中所示,当内部压力继续增大时,安全排气口 30 发生破裂,并且压力气体穿过破裂的区域,然后经顶盖 10 的排气口排出,从而防止电池爆炸。

[0014] 这种运行过程取决于在电极组件中产生的气体的量和气体到安全排气口的输送效率。例如,虽然产生大量气体,但在其中气体未有效输送至安全排气口的情况中,不能进行期望的安全运行过程。此外,除非在短时间内产生大量气体,并且气体到达安全排气口并诱发预定的运行过程,否则电池的内部压力将迅速增大,从而引起爆炸。

[0015] 此外,由电解质经由高温加热的分解而引起气体的产生。当从电池产生的热量在短时间内迅速增加时,可能发生热失控。热失控现象发生在电池处于连续加电状态时。当该现象进一步加速时,电池可能燃烧或爆炸的风险大大增加,从而不利地引起严重的安全问题。

[0016] 因此,对于开发如下圆筒形二次电池具有日益增加的需求,所述圆筒形二次电池能够在电池异常运行状态下更稳定地诱导电池的内部气体并将所述气体迅速排放至电池外部。

发明内容

[0017] 【技术问题】

[0018] 因此,完成了本发明以解决上述问题以及其他尚待解决的技术问题。

[0019] 本发明的一个目的是提供一种二次电池,所述二次电池使得所产生的气体可以经由具有特定结构的盖组件而有效地通过安全构件,并且使得预定的安全过程能够进行,从而最终提高电池安全性。

[0020] 本发明的另一个目的是提供一种二次电池,所述二次电池随着温度的增加而诱发电解质的连续分解反应,并且抑制电池内部压力的迅速增加,从而使得安全构件能够有效运行。

[0021] 本发明的又一个目的是提供一种二次电池,所述二次电池诱发电解质的分解反应,同时使产生的热最小化并抑制热失控的发生。

【0022】 【技术解决方案】

[0023] 因此,根据本发明的圆筒形二次电池是其中通过卷绕正极/隔膜/负极而制造的凝胶卷型电极组件以及电解质设置在圆筒形罐中的圆筒形二次电池,其中安装在所述圆筒形罐的开放顶部上的盖组件包含:安全排气口,所述安全排气口具有预定凹口以允许由于电池的高压气体而造成的破裂;电流中断装置,所述电流中断装置焊接至安全排气口的底部并用于中断电流;和用于所述电流中断装置的衬垫,所述衬垫用于围绕所述电流中断装置的周边,其中所述电流中断装置包含两个以上允许排放气体的通孔,其中所述通孔具有相对于所述电流中断装置的总面积为 20%至 50%的尺寸。

[0024] 因此,圆筒形电池使得产生的气体可以通过具有特定结构的盖组件而直接作用于安全构件,诱发期望的断电并且通过使安全排气口破裂而使得能够将气体迅速排放至外部,从而确保电池的安全性。

[0025] 电流中断装置是当电池的内部压力增大时发生破裂并从而中断电流的构件,并且可具有如下结构:在其中心处具有向上突出且焊接至安全排气口的凹陷部的底部的突出部,并且连接至所述凝胶卷的正极的正极引线通过所述突出部之外的区域的底部发生电连接。

[0026] 照这样,由于将电流中断装置连接至安全构件,所以在电流中断装置和正极之间的连接区根据安全排气口的形状反转而发生破裂,从而导致电流中断。在具体实例中,可在突出部的周边表面处形成凹口,所述凹口根据安全排气口的形状反转而发生破裂,并且所述突出部可容易地与电流中断装置和正极之间的连接区分离,同时突出部仍然连接至安全排气口。

[0027] 此外,电流中断装置具有两个以上通孔以使得气体可向上移动,从而导致电池的内部气体增加,并由此使安全排气口反转。

[0028] 然而,本发明人确认了,在设置在电流中断装置处的通孔的面积小于 20%的情况下,当在电池中产生高压时,电池的燃烧或爆炸率增加。认为这是由于以下事实所造成的:由于穿过通孔的气体的量少而导致有效断电失败,或者由于气体的向上移动通道发生阻塞或变化而导致中断电流需要花费长时间。同时,从难以确保元件的预定强度和扭转(torsion)并且在加工时发生部件的扭转的观点来看,通孔具有过大的尺寸是不优选的。

[0029] 因此,在本发明的圆筒形电池中,电流中断装置包含两个以上允许排放气体的通孔,并且所述通孔具有相对于电流中断装置的总面积为 20%至 50%、更优选 30%至 40%的面积,从而提高向上移动(排放)的气体的量并诱导气体的快速向上移动(排放)。

[0030] 优选地,在基于电流中断装置的中心径向对称的位置处以圆弧形形成通孔。在其中在这种径向对称位置处形成通孔的情况下,有利之处在于,可以防止气体排放集中在一个方向上并且可以确保电流中断装置的强度。

[0031] 此时,通孔的数量没有特别限制,并且优选为 2 至 4 个。

[0032] 在一个优选实施方式中,可以设置三个通孔,使得它们基于电流中断装置的中心以圆弧形彼此间隔预定距离,所述圆弧具有与 50 至 80 度的角度相对应的长度。

[0033] 同时,所述突出部具有桥,所述桥具有三个通孔和用于连接相邻通孔的凹口,所述通孔基于所述突出部的中心而同心地形成。可以对通孔进行设置,使得它们基于电流中断装置的中心以圆弧形彼此间隔预定距离,所述圆弧在与通孔相对应的位置处且具有与 40

至 55 度的角度相对应的长度。

[0034] 安全排气口是在电池的內部压力增大时排放气体从而确保电池安全性的元件。例如,当在电池内部产生气体并且内部压力由此增大至临界水平以上时,从破裂区域排放的气体可通过设置在顶盖上的一个或多个气体出口排放。

[0035] 优选地,安全排气口具有向下凹陷的中心。在这种结构中,当电池的內部压力增大时,向下的凹陷部反转成向上的凹陷部,与电流中断装置分离且中断电连接。

[0036] 考虑到具有这种结构的安全排气口的运行过程,在电池内部产生的气体穿过通孔并对安全排气口施加向上的压力,安全排气口的向下凹陷部在预定压力(简称为“第一临界压力”)以上的压力下发生反转,附着至凹陷部的突出部与电流中断装置分离并且中断从电流中断装置到安全排气口的电施加。

[0037] 凹陷部包含例如上弯曲部和下弯曲部,并且所述上弯曲部优选具有第一凹口。在常规方法中,提出了其中仅有下弯曲部被切开的结构。然而,下弯曲部具有狭窄的区域并因此不适合有效地排放气体。另一方面,根据本发明,在上弯曲部中设置有用于破裂的凹口,从而增加切口面积并使得能够将大量气体迅速排放至外部。

[0038] 当压力继续增大并达到预定水平(简称为“第二临界压力”)时,尽管安全排气口中断电流,但是第一凹口可被切开。为此,受压的内部气体通过设置在顶盖上的气体出口而平稳地排放至外部。

[0039] 同时,为了防止整个上弯曲部与安全排气口分离,第一凹口可具有其中一侧敞开的开口曲线。即,一侧的开口充当用于将安全排气口的凹陷部连接至其剩余区域的桥。

[0040] 第一凹口的尺寸没有特别限制,并且可以优选地对应于假想线的尺寸,所述假想线用于连接通孔的中心线以使得气体可以从电流中断装置向上移动而对第一凹口直接施加压力。

[0041] 此外,第一凹口的开口区域具有如上所述当第一凹口被切开时防止凹陷部被完全切开的尺寸,并且相对于安全排气口的中心,优选具有与 20 度至 50 度、更优选 25 至 40 度的角度相对应的圆弧长度。

[0042] 在一个优选实施方式中,在下弯曲部中形成第二凹口,但具有其中彼此面对的两端敞开的开口弯曲结构,以便在内部压力增大时使得能够仅将上弯曲部切开。其中两端敞开的这种结构防止下弯曲部被切开,并且通过切开后弯曲部而更有效地引起打开。

[0043] 优选地,在不与第一凹口的开口区域相对应的区域中形成在第二凹口中开口的两端。在一个更优选的实施方式中,以约 ± 90 度的角度形成在第二凹口中开口的两端。

[0044] 在一个优选实施方式中,将第一凹口形成为大于第二凹口的深度,从而使得第一凹口在第二凹口之前被切开。

[0045] 安全排气口没有特别限制,并且可以由具有 0.15 至 0.4mm 厚度的铝板材料制成以提供强度,从而使得在内部压力增大至临界水平以上时可以将凹口切开。

[0046] 根据本发明的圆筒形电池的盖组件还可包含诸如正温度系数(PTC)元件的构件,从而在电池的內部温度增大时大大增加电池的电阻并由此中断电流。

[0047] 盖组件具有电流中断装置、电流中断装置用衬垫、安全排气口、PTC 元件、以及其中形成一个或多个气体出口的顶盖的层压结构。可以将衬垫进一步添加至层压结构的外围。可以将 PTC 元件夹在顶盖和安全排气口之间,使得其附着至设置在两者之间的区域。

[0048] 此外,电流中断装置用衬垫围绕电流中断装置的周边。在一个优选实施方式中,衬垫在其中心处具有开口。受压气体优选以直线移动,以便使得内部气体可有效地向上移动。因此,中心开口具有不遮挡电流中断装置的通孔的形状。

[0049] 在常规技术中,由于衬垫遮挡电流中断装置的通孔的一部分,所以穿过通孔的受压气体被衬垫干扰,结果,排气量减小并且通道发生变化,从而使得难以诱发安全排气口的即时运行。然而,如上所述,其中衬垫不遮挡通孔的结构可确保电池的稳定性,因为高压气体完全直接到达安全排气口。

[0050] 在一个优选实施方式中,以使得它们在直线上彼此连通的方式形成电流中断装置的通孔、电流中断装置用衬垫的中心开口和安全排气口的凹口。

[0051] 因此,尽管在电池中产生的高压气体经由中心开口移动穿过电流中断装置的通孔而到达安全排气口的凹口,但是气体通道可保持直线,由此诱发安全排气口的即时运行并确保电池稳定性。

[0052] 在一个优选实施方式中,圆筒形电池可包含含添加剂(电解质添加剂)以提高热稳定性的电解质。

[0053] 电解质添加剂可随着温度增加而通过电解质的连续分解反应产生气体,并且例如可在 150℃ 以上的温度下发生反应。

[0054] 因此,如在下面的实验例中所证明的,根据本发明的圆筒形电池随着温度增加而诱发电解质的连续分解反应,并由此抑制热失控和电池内部压力的迅速增加。这种电解质连续分解反应的诱发防止了在将大量气体输送至安全元件时发生一种瓶颈现象。因此,电解质添加剂发挥与特定盖组件结构配合的协同效应。

[0055] 在一个优选实施方式中,电解质添加剂是在 150℃ 至 250℃ 下引起一次电解质分解和在 250℃ 至 300℃ 下引起二次电解质分解的材料。例如,电解质添加剂在一次电解质分解反应中引起约 2/3 的总分解反应,然后在高于一次电解质分解温度的温度下的二次电解质分解反应中引起约 1/3 的总分解反应。

[0056] 因此,气体随着温度的增加而连续地产生,从而使得电流中断装置和安全排气口可以运行。为此,能够解决与在短时间内产生大量气体相关的问题,并且如上所述,尽管连续产生气体,但所产生的气体经由特有的盖组件结构而直接起作用,从而诱发期望的断电。通过安全排气口的破裂可以将气体迅速排放至电池的外部,并且能最终有效地防止电池的燃烧和爆炸。

[0057] 电解质添加剂的优选实例是氟代碳酸亚乙酯(FEC),但本发明不限于此。

[0058] 本发明的发明人通过实验证明了,如上所述,电解质添加剂随着温度的增加而诱发连续的电解质分解,并且在正极的表面上形成保护膜而增加正极界面处的电阻;并且这种电阻的增加减少了短路时施加的电量,从而导致改善的电池稳定性。

[0059] 基于电解质的总重量,电解质添加剂优选以 11 重量%至 40 重量%、更优选 15 重量%至 35 重量%的量存在。

[0060] 当电解质添加剂的含量过小时,不能实现高温下的连续气体产生和在短路时的电流降低效果,另一方面,当电解质添加剂的含量过大时,不利地减小了电池容量。

[0061] 优选地,将电解质添加剂用于诸如在 4.3V 至 4.35V 的充电/放电电压下的高压电池。在这种情况下,电解质添加剂的含量优选为 20 重量%至 35 重量%。

[0062] 电解质通常包含在非水有机溶剂中的锂盐,这在本领域中是已知的。

[0063] 作为可用于本发明的非水电解液,例如可以提及的有非质子有机溶剂,如 N- 甲基 -2- 吡咯烷酮、碳酸亚丙酯、碳酸亚乙酯、碳酸亚丁酯、碳酸二甲酯、碳酸二乙酯、 γ -丁内酯、1,2-二甲氧基乙烷、四羟基 Franc、2- 甲基四氢呋喃、二甲基亚砜、1,3- 二氧戊环、甲酰胺、二甲基甲酰胺、二氧戊环、乙腈、硝基甲烷、甲酸甲酯、乙酸甲酯、磷酸三酯、三甲氧基甲烷、二氧戊环衍生物、环丁砜、甲基环丁砜、1,3- 二甲基 -2- 咪唑啉酮、碳酸亚丙酯衍生物、四氢呋喃衍生物、醚、丙酸甲酯以及丙酸乙酯。其中,优选的是由线性碳酸酯化合物、环状碳酸酯化合物或其组合构成的碳酸酯基溶剂。

[0064] 锂盐是易溶于上述非水电解质的材料,且可以包括例如 LiCl、LiBr、LiI、LiClO₄、LiBF₄、LiB₁₀Cl₁₀、LiPF₆、LiCF₃SO₃、LiCF₃CO₂、LiAsF₆、LiSbF₆、LiAlCl₄、CH₃SO₃Li、CF₃SO₃Li、(CF₃SO₂)₂NLi、氯硼烷锂、低级脂族羧酸锂、四苯基硼酸锂和酰亚胺。

[0065] 另外,为了改善充放电特性和阻燃性,可以向非水电解质内添加例如吡啶、亚磷酸三乙酯、三乙醇胺、环醚、乙二胺、正甘醇二甲醚、六磷酸三酰胺(hexaphosphoric triamide)、硝基苯衍生物、硫、酞亚胺染料、N- 取代的咪唑烷酮、N, N- 取代的咪唑烷、乙二醇二烷基醚、铵盐、吡咯、2- 甲氧基乙醇、三氯化铝等。如果需要,为了赋予不燃性,非水电解质还可包含含卤素的溶剂如四氯化碳和三氟乙烯。此外,为了改善高温贮藏特性,非水电解质可另外包含二氧化碳气体和丙烯磺酸内酯(PRS)。

[0066] 本发明还提供了安装在电池中的圆筒形罐的开放顶部上的盖组件,在所述电池中电极组件和电解质设置在圆筒形罐中。具体地,基于电解质的重量,电池的电解质包含 11 重量%至 40 重量%的添加剂以在 150℃ 以上的温度下随着温度的增加而通过连续的电解质分解产生气体。对于盖组件,可在凝胶卷的顶部上安装板型绝缘构件,所述构件在中心处具有穿孔开口以使得可排放气体并连接电极端子,并且具有设置在开口周围的一个或多个通孔。

[0067] 在一个优选实施方式中,板型绝缘构件的通孔可具有这样的位置,使得其在直线上与电流中断装置用衬垫的中心开口和电流中断装置的通孔连通。板型绝缘构件的通孔具有相对于板型绝缘构件的总面积为 20%至 50%的尺寸。

[0068] 可以使用任何绝缘构件而没有任何限制,只要其由绝缘材料制成即可,并且例如可以是聚丙烯树脂。其中,特别优选的是高强度聚丙烯树脂,其能够在恒定的厚度条件下发挥优异的机械强度并且在施加强度的外部冲击时防止凝胶卷移动和分离。

[0069] 【有利效果】

[0070] 如从上文显而易见的,本发明提供了一种圆筒形电池,其包含具有多个具有相对大面积的通孔的电流中断装置,从而有效地排放内部高压气体。此外,在形成在安全排气口中发生破裂的凹口以对应于通孔的情况中,当产生高压气体时,高压气体可以直线排放,且可以诱发快速断电,气体排放可迅速进行,并且可以显著地提高电池安全性。

[0071] 此外,在向电解质中添加特定电解质添加剂的情况中,添加剂随着温度的增加而引起连续的电解质分解,从而稳定地产生气体并减小短路电流,由此大大降低由发热而引起的燃烧或爆炸的可能性。

附图说明

[0072] 根据结合附图的以下详细描述,将更清楚地理解本发明的上述和其他目的、特征及其它优点,在附图中:

[0073] 图 1 是示意性地示出一般圆筒形电池的横截面透视图;

[0074] 图 2 至 4 是示出其中通过常规圆筒形电池中的安全元件的运行而中断电流并排放高压气体的一系列过程的横截面图;

[0075] 图 5 是示出根据本发明一个实施方式的圆筒形电池的示意性截面图;

[0076] 图 6 和 7 示出了用于图 5 的盖组件结构的安全排气口的透视图和平面图;

[0077] 图 8 和 9 是示出用于图 5 的圆筒形电池的电流中断装置的透视图和平面图;

[0078] 图 10 和 11 是示出用于图 5 的圆筒形电池的电流中断装置用衬垫的透视图和平面图;

[0079] 图 12 和 13 是示出用于图 5 的圆筒形电池的板型绝缘构件的透视图和平面图;

[0080] 图 14 是示出用于层压根据本发明一个实施方式的板型绝缘构件、电流中断装置用衬垫和电流中断装置的过程的示意图;

[0081] 图 15 至 17 是示出其中通过图 5 的圆筒形电池中的安全元件的运行而中断电流并排放高压气体的一系列过程的横截面图;以及

[0082] 图 18 是示出在实施例 1、4 和 5 以及比较例 1 和 2 中制造的电池的正极界面电阻结果的图。

具体实施方式

[0083] 现在,将参考附图更详细地描述本发明,并且不应将其理解为限制本发明的范围和主旨。

[0084] 图 5 示出了根据本发明一个实施方式的圆筒形电池的盖组件结构。图 6 和 7 是用于图 5 的盖组件结构的安全排气口的透视图和平面图。

[0085] 参考图 5,通过下列方式来制造圆筒形电池 100:将凝胶卷 110 插入到圆筒形罐 200 中;将电解质注入到其中;将板型绝缘构件 800 安装在凝胶卷 110 的顶部上;并且将盖组件 300 安装在圆筒形罐 200 的开放顶部上。

[0086] 电解质包含添加剂以使热产生最小化并诱发电解质的连续分解反应。

[0087] 盖组件 300 具有如下结构,其中顶盖 310、PTC 元件 350 和用于减小内部压力的安全排气口 320 附着至用于气密性的衬垫 400 的内部,所述衬垫 400 安装在圆筒形罐 200 的上部卷边构件 210 上。

[0088] 电流中断装置 600 通过焊接连接至安全排气口 320 的底部,所述电流中断装置 600 在电池内部压力增大时发生破裂并从而中断电流,且用于所述电流中断装置的衬垫 700 围绕电流中断装置的周边。

[0089] PTC 元件 350 夹在顶盖 310 和安全排气口 320 之间,使得其附着至设置在其间的电流电路,并且在电池内部温度增大时大大增加电池电阻,从而中断电流。顶盖 310 具有向上突出的中心,由此通过连接至外部电路而充当正极端子。沿着突出部的周边形成用于排放在圆筒形罐 200 中存在的受压气体的多个气体出口 312。

[0090] 将参考图 6 和 7 更详细地描述具有这种结构的盖组件 300 中的安全排气口 320 的结构。

[0091] 安全排气口 320 具有导电薄膜结构,所述结构具有在其中心处的向下凹陷的凹陷部 322、在其上弯曲部处的第一凹口 324 和在其下弯曲部处的第二凹口 326。

[0092] 如示出安全排气口 320 的平面图的图 7 中所示,设置在上弯曲部上的第一凹口 324 具有其中一侧敞开的开口曲线。该开口区域为非凹口区域 330,这防止了整个第一凹口的分开,尽管将其中由高压气体形成凹口的区域切开。

[0093] 此外,第二凹口 326 形成在下弯曲部中并具有其中彼此面对的两端敞开的开口弯曲结构。两个非凹口区域 340 设置在第二凹口 326 的两个开口端上。非凹口区域 340 防止通过切开面将第二凹口 326 分开。为了更精确地防止第二凹口 326 分开,非凹口区域 340 优选设置在不与第一凹口 324 中形成的非凹口区域 330 相对应的区域中,更优选地,如图中所示,基于第一凹口 324 的非凹口区域 330 的中心线,以约 ± 90 度的角度形成非凹口区域 340,并且可以以彼此面对的方式设置两个非凹口区域 340。

[0094] 图 8 和 9 示出了用于图 5 的圆筒形电池的电流中断装置 600。

[0095] 参考图 8 和 9 以及图 5,在电流中断装置 600 的中心处形成向上突出的突出部 620,所述突出部 620 通过焊接而附着至安全排气口 320 的凹陷部底部 (322,参考图 6),并且将连接至凝胶卷 110 的正极的正极引线 420 电连接至电流中断装置 600 底部上的突出部 620 之外的区域。

[0096] 突出部 620 具有桥 640,所述桥 640 具有三个通孔 630 和用于连接相邻通孔 630 的凹口 650,所述通孔基于突出部 620 的中心同心地形成。

[0097] 此外,在电流中断装置 600 的周边上,在基于中心轴径向对称的位置处,以圆弧形式形成用于气体排放的三个通孔 610。通孔 610 具有相对于电流中断装置 600 的总面积为 30% 的尺寸,并且增加了在电池内部存在的高压气体的排放量,从而能发挥可靠的电流中断。通孔 610 以约 120 的角度彼此隔开且具有相同的形状和尺寸,并且其间的距离基本均匀。基于这种结构,电流中断装置 600 使气体排放最大化并保持高机械强度。

[0098] 此时,优选地,用于连接通孔 610 的中心线的假想线 (参见虚线) 的尺寸基本等于安全排气口 (320,参见图 6 和 7) 的第一凹口 324 的尺寸。因此,当产生预定水平以上的内部压力时,穿过通孔 610 的高压气体经由直线通道压制第一凹口 324,并且促进切开第一凹口 324,从而使得能够快速排放气体。

[0099] 图 10 和 11 示出了围绕电流中断装置 600 的电流中断装置用衬垫 700。为了便于说明,简单地示出电流中断装置用衬垫以仅表示开口的形状。

[0100] 参考图 10 和 11 以及图 5,电流中断装置用衬垫 700 在其中心处具有开口 710,并且中心开口 710 具有不遮挡电流中断装置 600 的通孔 610 的形状。即,电流中断装置用衬垫 700 的形状基本对应于其中在电流中断装置 600 中未形成开口的区域的周边的形状,由此围绕电流中断装置 600 的周边且不遮挡通孔 610,从而确保通孔 610 起到排气口的作用。

[0101] 同时,图 12 和 13 示出了安装在根据图 5 的对称电池 100 中的凝胶卷 110 的顶部上的板型绝缘构件 800 的实例。参考这些附图以及图 2,板型绝缘构件 800 在其中心处具有穿孔开口 820 以使得可排放气体并连接电极端子,并且具有设置在开口 820 周围的一个或多个通孔 810。板型绝缘构件 800 的通孔 810 可以以与电流中断装置 600 的通孔 610 相对应的形状、尺寸和位置形成,并且具有相对于板型绝缘构件 800 的总面积为 30% 的尺寸。

[0102] 图 14 示意性地示出了其中在圆筒形罐上依次安装有板型绝缘构件 800、电流中断

装置用衬垫 700、电流中断装置 600 和安全排气口 320 的层压体。

[0103] 参考图 14 以及以上所示的图 2 至 6, 形成板型绝缘构件 800 的通孔 810、电流中断装置用衬垫 700 的中心开口 710、电流中断装置 600 的通孔 610 和安全排气口 320 的第一凹口 324, 使得它们在直线上彼此连通, 以便使得可以将电池内部的受压气体有效地排放到外部。

[0104] 图 15 至 17 示意性地示出了其中在电池的内部压力异常增大的情况中排放气体的过程。

[0105] 参考这些附图, 当电池的内部压力异常增大时, 受压气体穿过通孔 630 以及电流中断装置 600 的通孔 610, 并对安全排气口 320 施加向上的压力。

[0106] 这种压力导致安全排气口的向下凹陷部 322 升高, 而同时导致焊接至向下凹陷部 322 的突出部 620 在第一临界压力以上的压力下与电流中断装置 600 分离, 从而中断从电流中断装置 600 到安全排气口 320 的电施加, 如图 16 中所示。

[0107] 尽管具有这种电流阻断, 但是如果压力继续增大并随后达到第二临界压力以上, 则安全排气口 320 的第一凹口 324 发生破裂, 并且同时, 通过气体出口 312 将内部受压气体排放到外部, 如图 17 中所示。

[0108] 与其中将具有小直径的第二凹口切开的情况相比, 根据本发明的其中将具有大直径的第一凹口 324 切开的情况显示更大的气体排放量, 从而诱导高压气体的快速排放。

[0109] 下面将参考以下实施例更详细地描述本发明。提供这些实施例仅用于说明本发明, 且不应被理解为限制本发明的主旨和范围。

[0110] [实施例 1]

[0111] 以下列方式制造圆筒形二次电池。

[0112] 1-1 正极和负极的制造

[0113] 将包含 LiCoO_2 的正极活性材料涂布到铝集电器上, 并且将正极抽头 (tap) 附着至集电器的端部, 使得正极抽头向上突出, 从而制造正极。将包含人造石墨的负极活性材料涂布到铜集电器上, 并且将负极抽头附着至负极集电器的端部, 使得负极抽头向上突出, 从而制造负极。

[0114] 1-2. 电解质的制备

[0115] 通过基于电解质的总重量, 将 11 重量% 的作为添加剂的氟代碳酸亚乙酯添加至非水溶剂中来制备用于锂二次电池的电解质, 所述非水溶剂包含 1.0M LiPF_6 作为电解质并由 30/5/65 (体积比) 的碳酸亚乙酯 (EC) / 碳酸亚丙酯 (PC) / 碳酸二甲酯 (DMC) 组成。

[0116] 1-3 使用中心销卷绕层压体

[0117] 在 1-1 部分中获得的其中隔膜夹在正极和负极之间的片型层压体中插入中心销, 然后卷绕至 $350\text{g}/\text{cm}^2$ 的应力以制造凝胶卷。

[0118] 将由此制造的凝胶卷置于连接至引线的圆筒形壳体中, 并且将 1-2 部分中制备的电解质插入到其中。

[0119] 1-4 圆筒形二次电池的制造

[0120] 在具有 16mm 的外径和 0.3mm 的厚度的铝板材料上的具有 30 度角度的圆弧之外的区域中, 形成了具有 8.0mm 的直径和 0.06mm 的厚度的第一凹口, 如图 7 中所示, 并且制造了具有凹陷部的安全排气口, 所述凹陷部向下突出至 0.65mm 的深度。

[0121] 此外,在具有 11mm 的外径和 0.5mm 的厚度的铝板材料上径向地形成三个通孔,使得通孔的总面积相对于板材料的总面积为约 30%,如图 9 中所示。然后,在其中心处形成具有 1.53mm 的直径和 0.20mm 的高度的突出部,在距离中心突出部的中心为 1.5mm 的距离处打出具有 0.6mm 的宽度和 2.61mm 的周长的三个通孔,并且在用于连接通孔的各桥处形成具有约 70 μ m 的厚度的凹口以制造电流中断装置。

[0122] 此外,在具有 12mm 的外径和 0.5mm 的厚度的聚丙烯板材料上制造了不遮挡电流中断装置的通孔的电流中断装置用衬垫,如图 11 中所示。

[0123] 将电流中断装置的外围插入到电流中断装置用衬垫中,并且通过激光焊接将安全排气口的弯曲的底部附着至电流中断装置的突出部的顶部。

[0124] 将板型绝缘构件安装在 1-3 的圆筒形壳体的开口顶部上,通过卷边来固定容器的顶部,将具有电流中断装置以及其中插入了 PTC 元件和顶盖的衬垫的安全排气口插入到卷边部分中,并且通过向内压制容器顶部而夹紧衬垫以制造电池。

[0125] [实施例 2]

[0126] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制造安全排气口的过程中第一凹口的厚度为 0.1mm,并且第二凹口的厚度为 0.06mm。

[0127] [实施例 3]

[0128] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制造安全排气口的过程中第一凹口的厚度为 7mm,并且第二凹口的厚度为 0.06mm。

[0129] [实施例 4]

[0130] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制备电解质的过程中,基于电解质的总重量,氟代碳酸亚乙酯的含量为 20 重量%。

[0131] [实施例 5]

[0132] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制备电解质的过程中,基于电解质的总重量,氟代碳酸亚乙酯的含量为 30 重量%。

[0133] [实施例 6]

[0134] 以与实施例 4 中相同的方式制造电池,不同之处在于,在电解质的制备中使用由体积比为 8 : 92 的碳酸亚丙酯 (PC) 和丙酸甲酯 (MP) 构成的非水溶剂。

[0135] [实施例 7]

[0136] 以与实施例 4 中相同的方式制造电池,不同之处在于,在制备电解质的制备中使用由体积比为 8 : 92 的碳酸亚丙酯 (PC) 和丙酸甲酯 (MP) 构成的非水溶剂。

[0137] [比较例 1]

[0138] 以与实施例 1 中相同的方式制造袋型单电池,不同之处在于,在电解质的制备中不添加添加剂。

[0139] [比较例 2]

[0140] 以与实施例 1 中相同的方式制造电池,不同之处在于,在制备电解质的过程中,基于电解质的总重量,氟代碳酸亚乙酯的含量为 5 重量%。

[0141] [比较例 3]

[0142] 以与实施例 1 中相同的方式制造电池,不同之处在于,相对于电流中断装置的总面积,电流中断装置的总面积为 18%。

[0143] [实验例 1]

[0144] 制备了在实施例 1、4 和 5 以及比较例 1 和 2 中制造的电池,并且测量了其正极界面电阻。将结果示于图 18 中。

[0145] 参考图 18,使用电化学阻抗谱(EIS)设备测量单位频率的电阻,并且将所得特性转化为半圆并示于图中。可以看出,与其中未添加添加剂的电池(比较例 1)和包含 5 重量%的添加剂的电池(比较例 2)相比,其中将氟代碳酸亚乙酯添加至电解质的实施例 1 的电池显示电阻的相对增加。特别地,可以看出,包含 20 重量%的添加剂的电池(实施例 4)和包含 30 重量%的添加剂的电池(实施例 5)显示正极界面电阻的显著增加。

[0146] [实验例 2]

[0147] 制备两个实施例 1、4 至 7 以及比较例 1 和 2 中制造的各电池,在充满电的电池中诱发外部短路并测量其电路电流。将由此获得的结果示于表 1 中。

[0148] <表 1>(单位:电流(A))

[0149]

	1	2	平均值
实施例 1	46.8	50	48.4
实施例 4	44.5	45.8	45.2
实施例 5	40.9	43.2	42.0
实施例 6	45.3	45.6	45.4
实施例 7	43.9	42.1	43.0
比较例 1	51.2	53.6	52.4
比较例 2	51	52	51.5

[0150] 从上表 1 中可以看出,与其中未添加添加剂的比较例 1 的电池和其中将 5 重量%的添加剂添加至电解质的比较例 2 的电池相比,其中将氟代碳酸亚乙酯添加至电解质的实施例 1 的电池显示在短路时的电流低。

[0151] 此外,与包含 11 重量%添加剂的实施例 1 的电池相比,包含 20 重量%添加剂的实施例 4 的电池和包含 30 重量%添加剂的实施例 5 的电池显示相当低的短路电流。在具有与实施例 4 和 5 不同的非水电解质组合物的实施例 6 和 7 中类似地显示了这种特性。

[0152] 因此,考虑到短路时产生的热与电流量成比例地增加的事实,根据本发明的电池因产生的热量少而相对稳定。

[0153] [实验例 3]

[0154] 在热稳定性方面,将实施例 1 和 4 至 7 中制造的电池与比较例 1 和 2 中的电池进行比较。将由此获得的结果示于下表 2 中。

[0155] <表 2>

[0156]

	开始温度 (°C)	热容量(J/g)			
		1峰值	2峰值	3峰值	总计
实施例1	206±1	71±4	47±4	-	118±6
实施例4	202±1	55±3	45±5	-	100±6
实施例5	204±1	50±4	40±4	-	90±4
实施例6	203±1	53±2	50±4	-	103±4
实施例7	205±1	48±5	43±5	-	91±6
比较例1	198±1	223±6			223±6
比较例2	204±1	98±11	56±5	49±6	203±6

[0157] 上表 2 中的“开始温度”表示热分解的初始温度,“热容量”表示在热分析时的峰面积。通过比较这些值来评价热稳定性。

[0158] 从实验中可以看出,在不包含氟代碳酸亚乙酯的比较例 1 的电池中,在约 210°C 下诱发一次电解质分解,而在包含氟代碳酸亚乙酯的实施例 1 的电池中,在约 225°C 和 275°C 下诱发两次电解质分解。因此可以看出,抑制了一次产生气体的浓度。

[0159] 此外,从表 2 中可以看出,实施例 1 的电池显示等于比较例 1 的电池的约 50% 的比热容,这表明由产生高热量而导致的安全问题较不严重。特别地,与实施例 1 的电池相比,分别包含 20 重量%和 30 重量%的电解质添加剂的实施例 4 和 5 的电池显示相当低的比热容,这表明实施例 4 和 5 的电池显示优异的稳定性。在使用具有 PC/MP 的组合物的非水电解质的实施例 6 和 7 中显示了这种所产生的热显著减少的现象。

[0160] 同时,与包含 11 重量%电解质添加剂的实施例 1 的电池相比,包含 5 重量%电解质添加剂的比较例 2 的电池显示具有 70% 以上的比热容增加。这意味着,当氟代碳酸亚乙酯以小于 10 重量%的量存在时,尽管添加了其,但是通过添加而获得的效果也相当低。

[0161] [实验例 4]

[0162] 对实施例 1 和 3 及比较例 3 中制造的电池进行针刺试验。将由此获得的结果示于下表 3 中。

[0163] <表 3>

[0164]

	通孔面积比	安全排气口的 切口区域	爆炸	电池容器 的变形
实施例1	30%	第一凹口	X	X
实施例2	30%	第二凹口	X	Δ
实施例3	30%	内部凹口 (增加的直径)	X	Δ
比较例3	18%	第一凹口	O	O

[0165] 从上表 3 中可以看出,当使用具有 30%的通孔面积比的电流中断装置时,电池爆炸和变形都未发生。

[0166] 另一方面,使用具有 18%的通孔面积比的电流中断装置的比较例 3 的电池在针刺试验期间发生爆炸。此外,如实施例 2 和 3 中所示,在其中将第二凹口切开并且气体未从安全排气口平稳地排出的情况中,电池未发生爆炸,但是电池容器发生变形。

[0167] 可以看到,在高容量电池中,在安全排气口的切口部分和电流中断装置的通孔的尺寸不足的情况中,气体不能充分地通过这些元件。此外,可以看到,在其中安全排气口的切口部分设置在电池内部的第二凹口的情况中,尽管使凹口的直径变宽,但是通过在切开之后安全排气口倾斜的内部凹口可能不利地产生对气体排放的阻力。

[0168] 如从上文中显而易见的,电流中断装置的通孔的面积增加,并且使用安全排气口的切口部分作为第一凹口的结构产生电池稳定性提高的效果,以便有效地从电池中排出气体。

[0169] 尽管为了进行说明性目的公开了本发明的优选实施方式,但是本领域的技术人员应理解,在不背离所附权利要求书中公开的本发明的范围和主旨的情况下,可以进行各种修改、增加和替代。

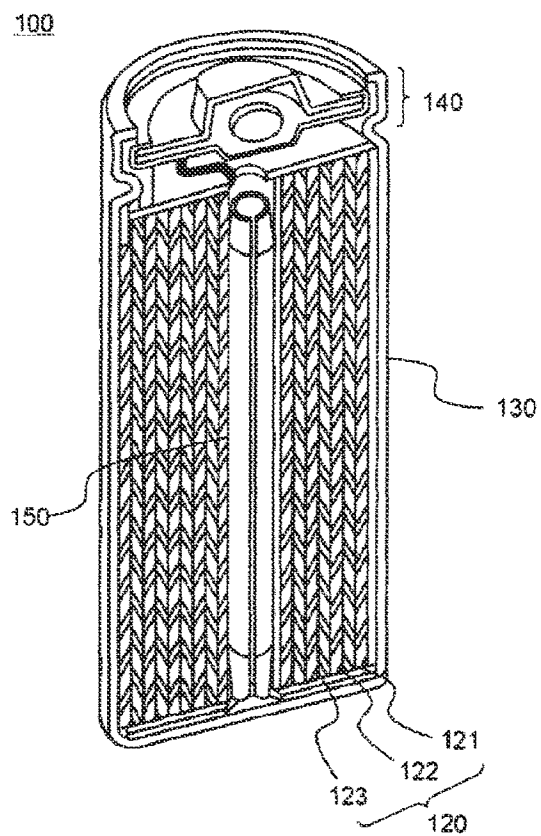


图 1

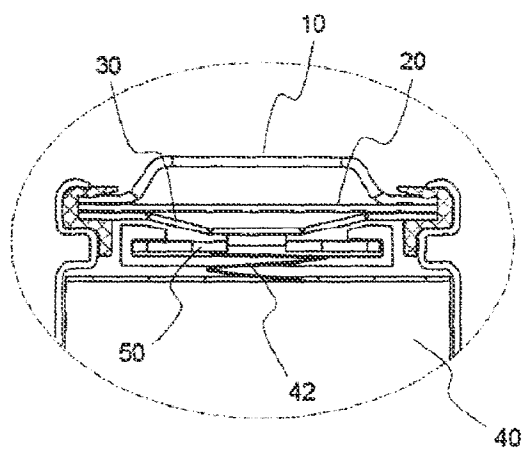


图 2

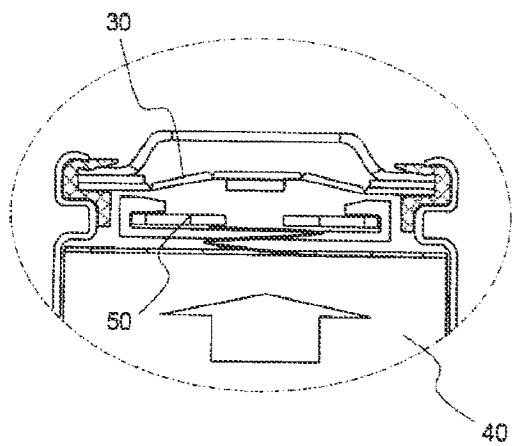


图 3

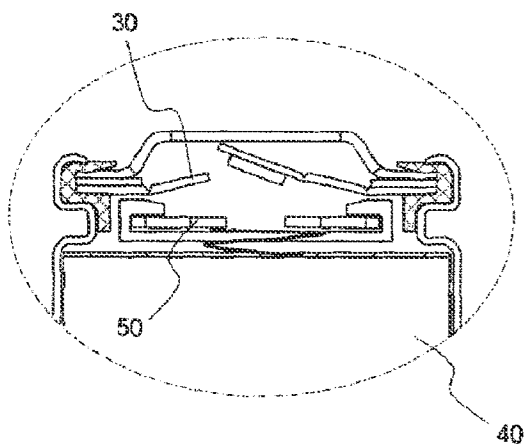


图 4

100

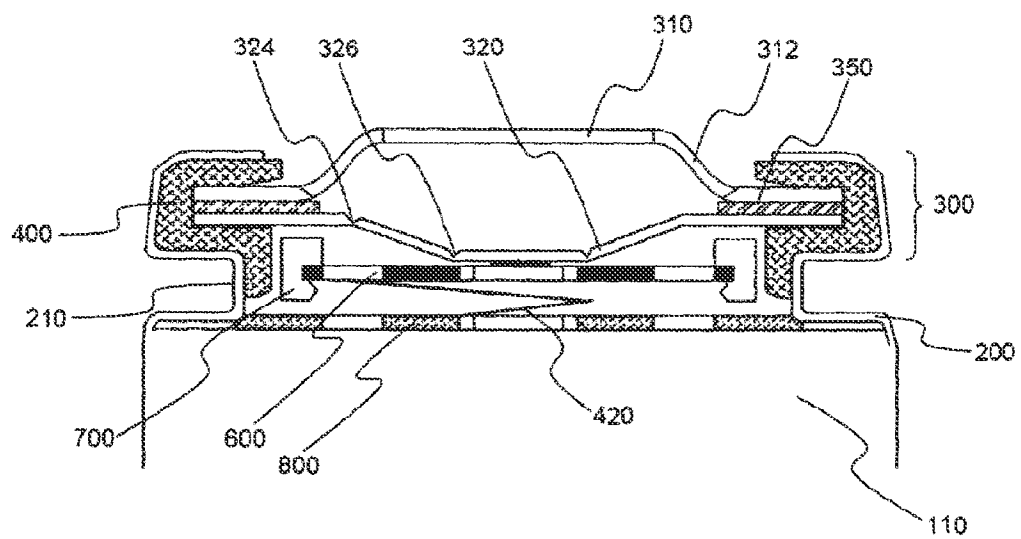


图 5

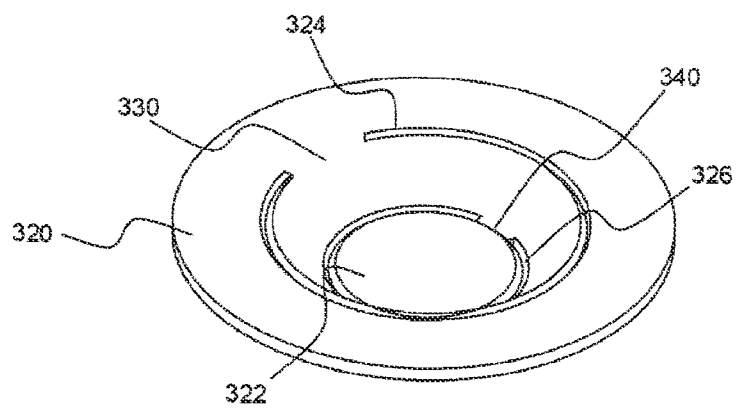


图 6

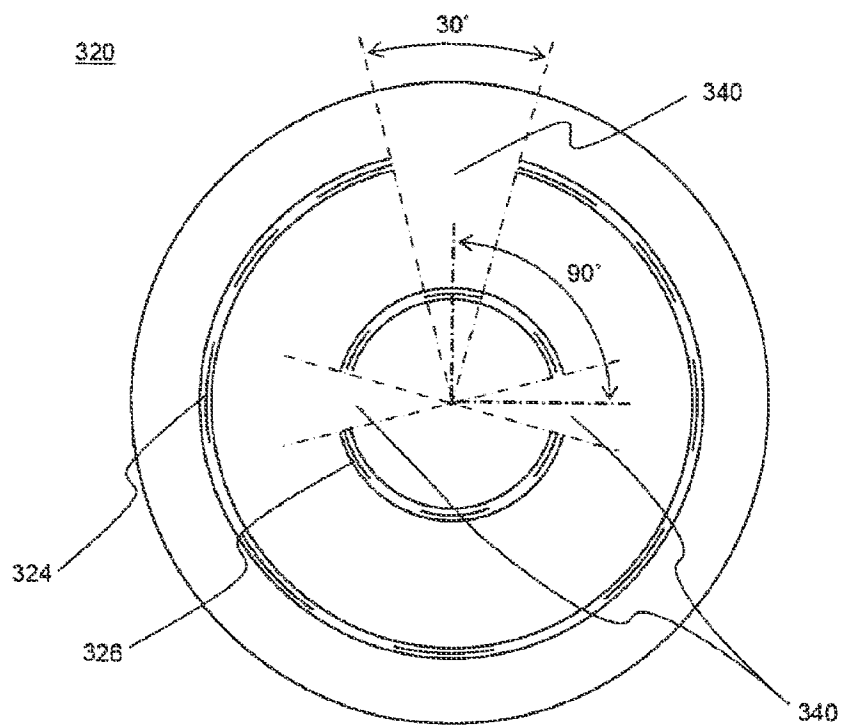


图 7

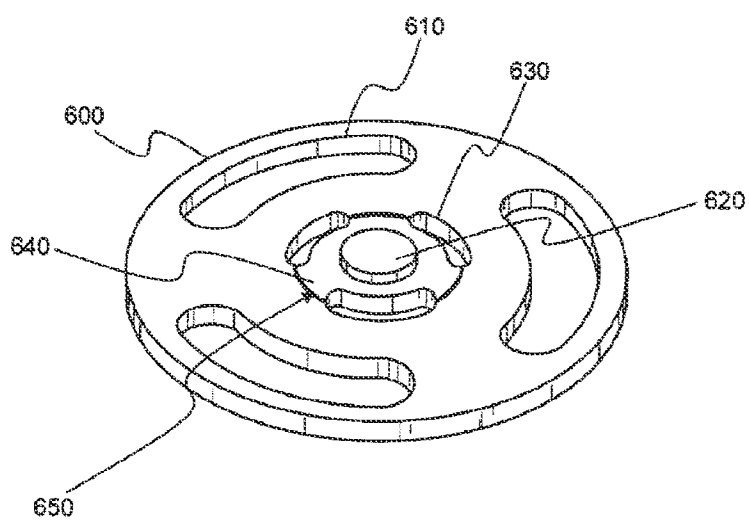


图 8

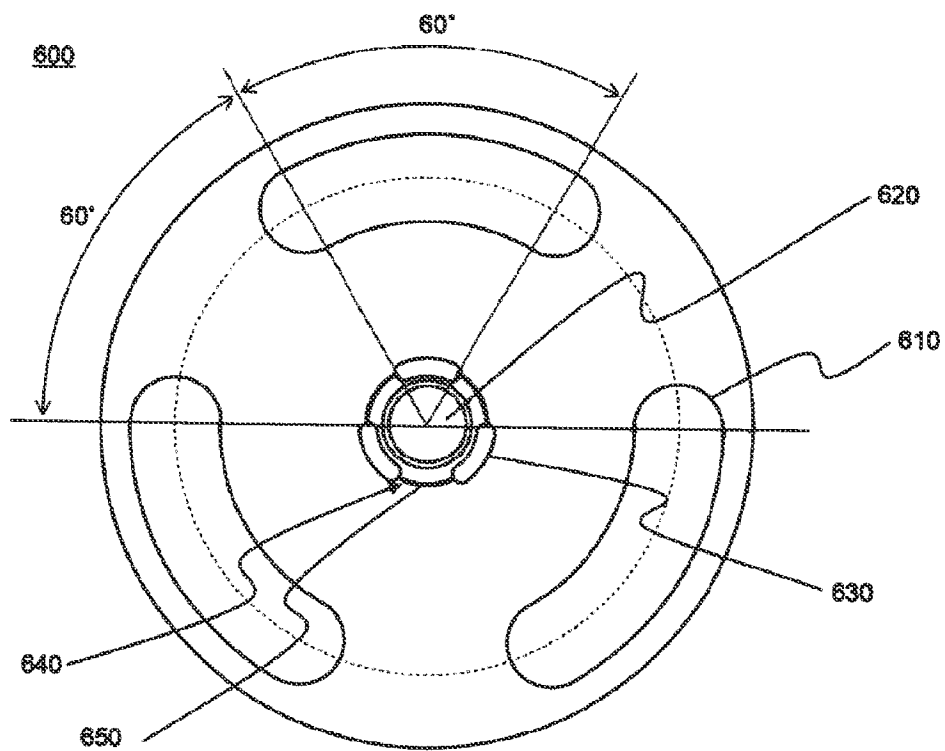


图 9

700

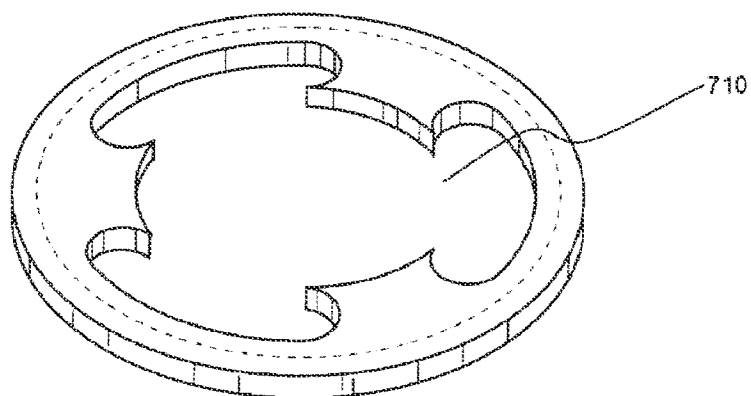


图 10

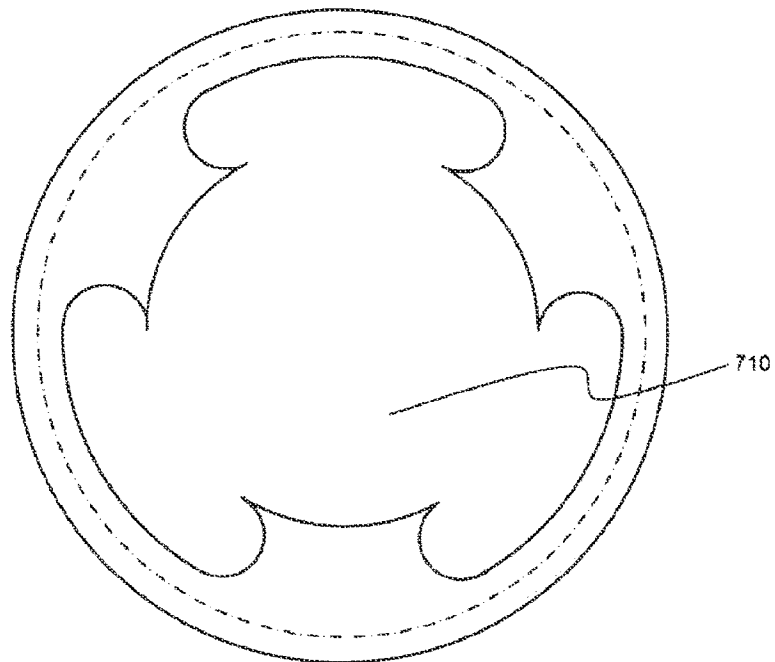
700

图 11

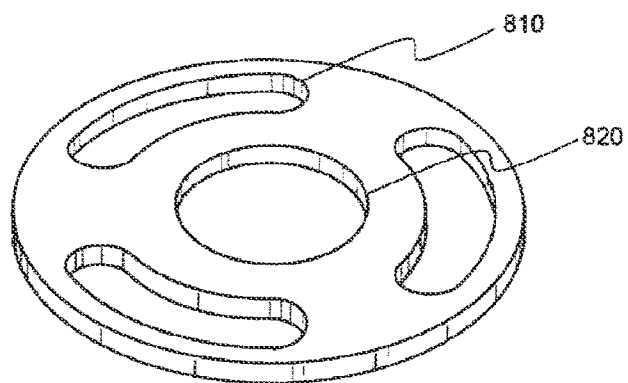
800

图 12

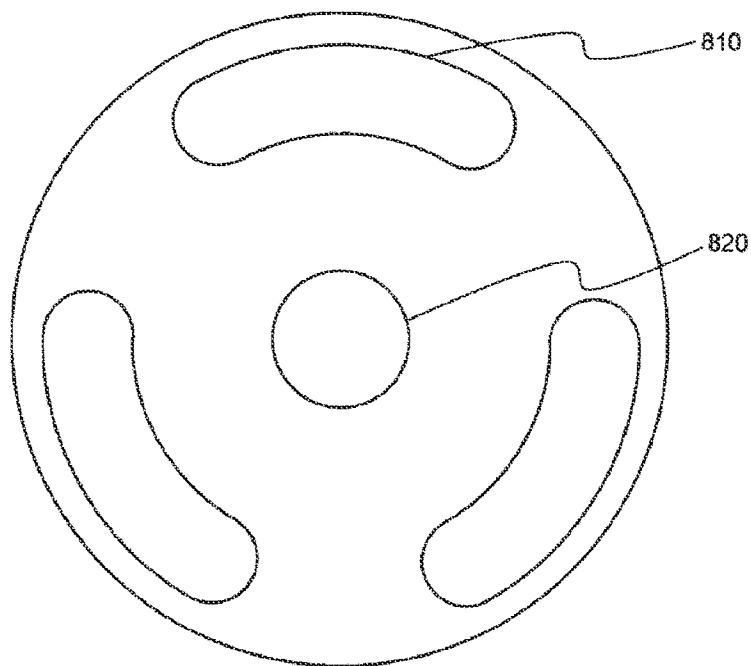
800

图 13

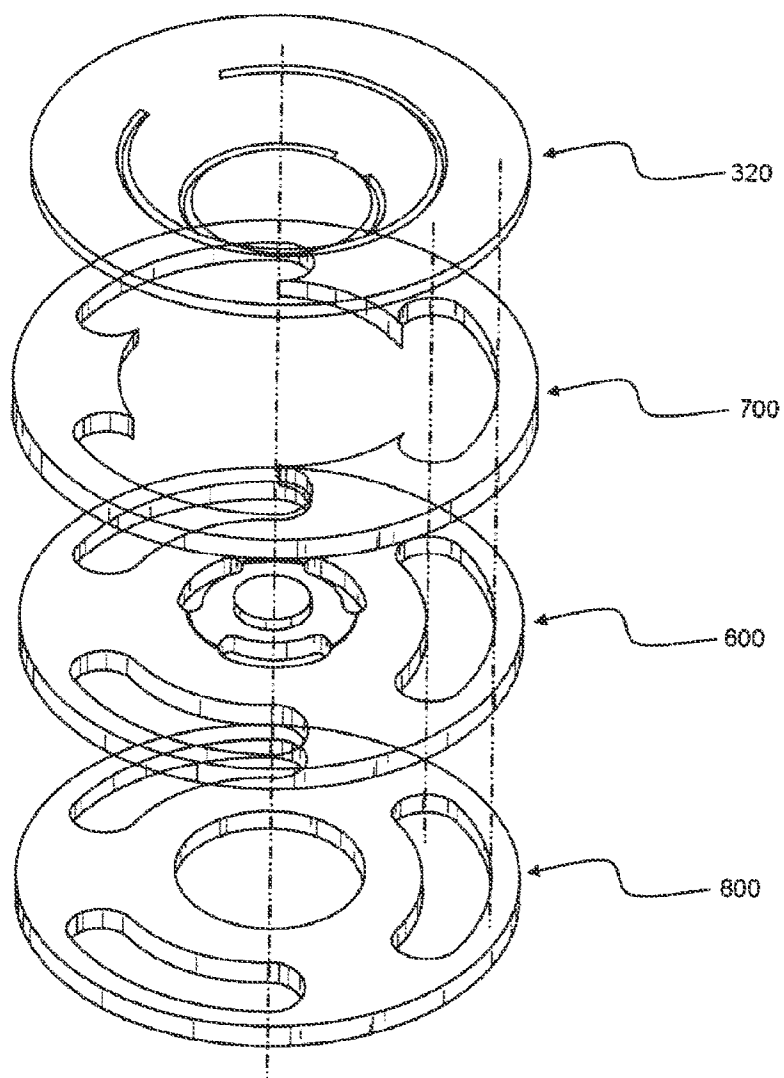


图 14

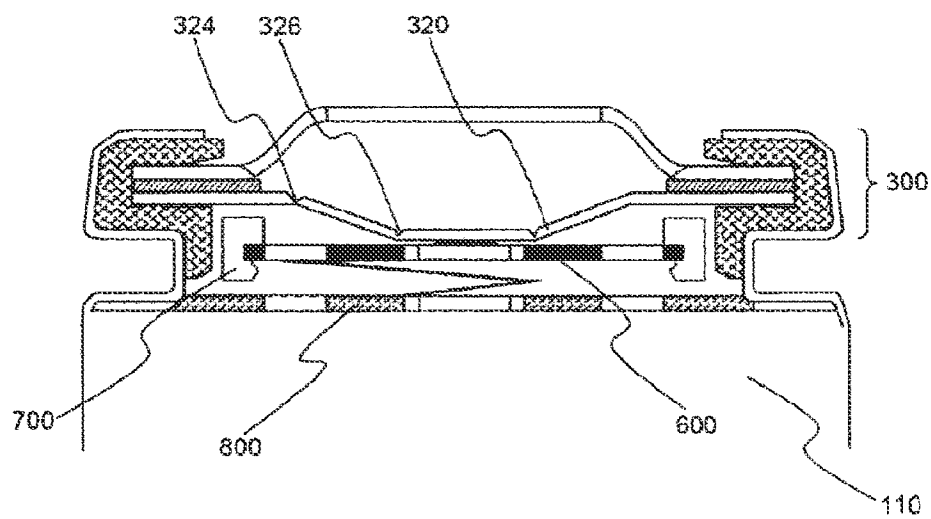


图 15

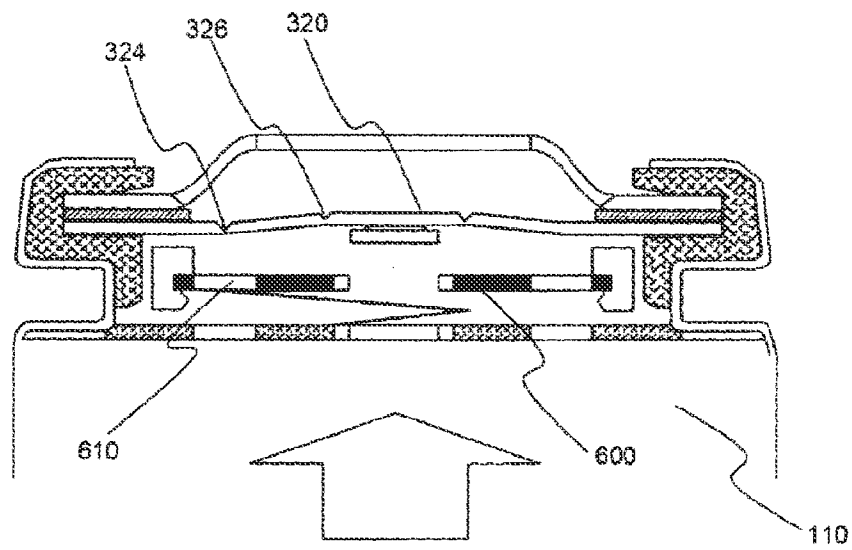


图 16

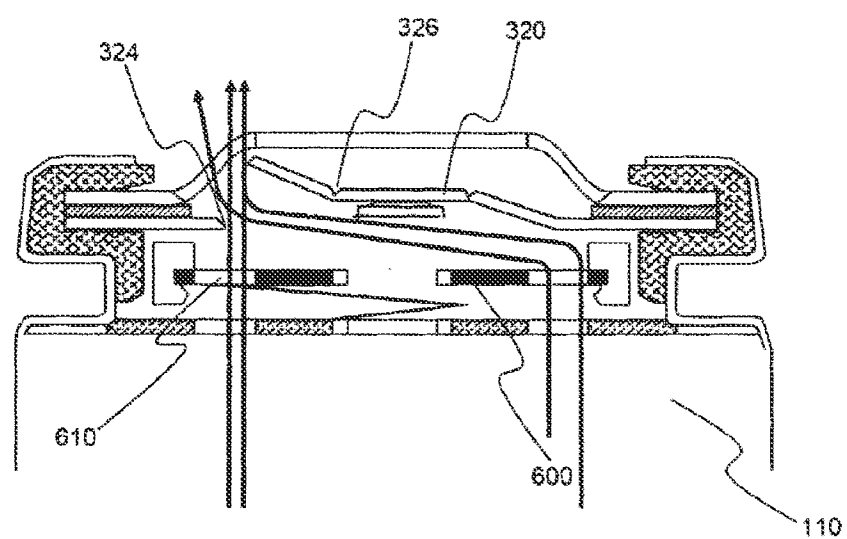


图 17

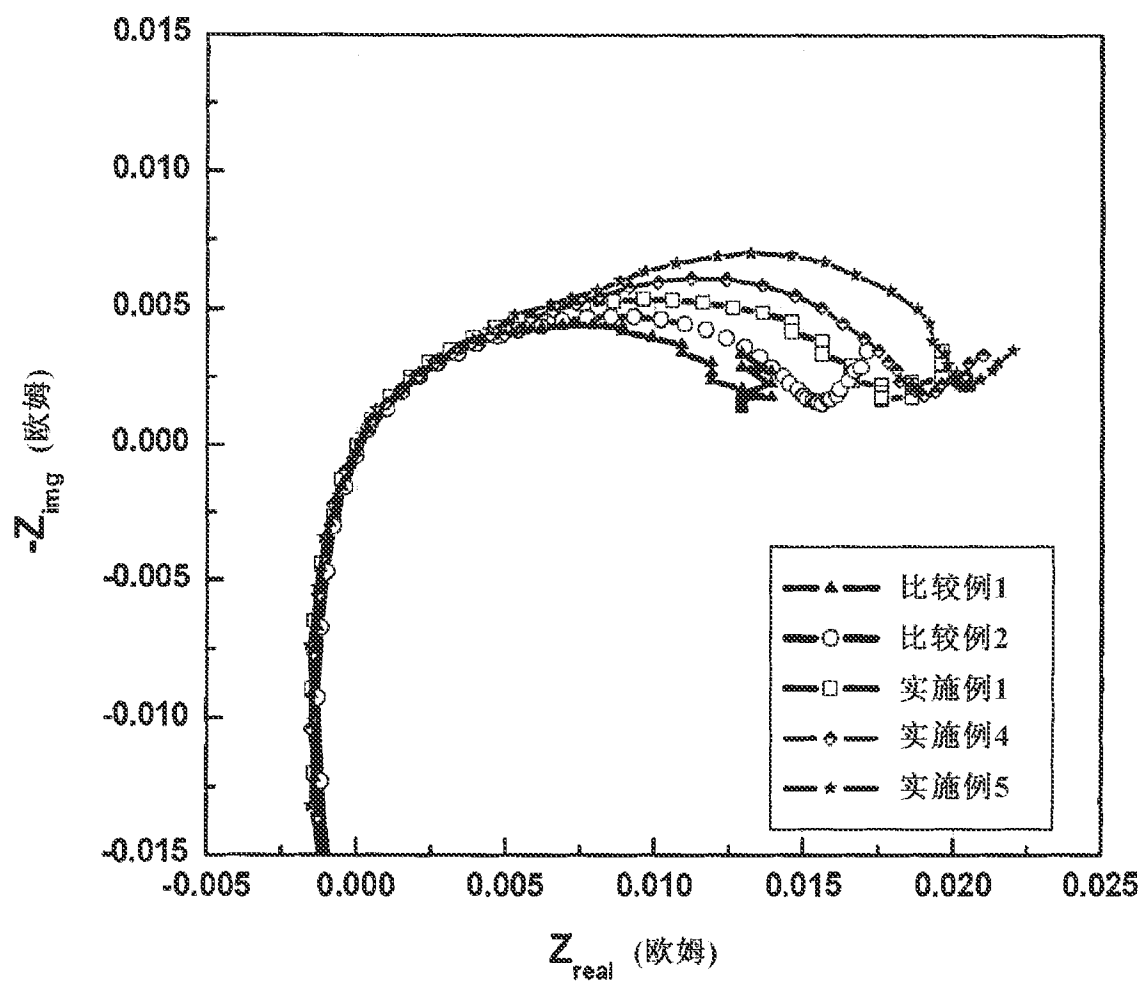


图 18



Espacenet

Bibliographic data: CN103155225 (A) — 2013-06-12

Cap assembly having novel structure and cylindrical battery containing same

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Applicant(s): LG CHEMICAL LTD ± (LG CHEM. LTD)

Classification: - **international:** H01M50/528
- **cooperative:** H01M50/166 (US); H01M50/167 (EP); H01M50/171 (EP, US); H01M50/30 (US); H01M50/3425 (EP, US); H01M50/528 (EP, US); H01M50/581 (EP, US); H01M2200/103 (EP, US); Y02E60/10 (EP); Y02P70/50 (EP)

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Abstract of CN103155225 (A)

The present invention provides a cap assembly mounted on an open upper end of a can in a jelly-roll type electrode assembly having a cathode/separator/anode structure, wherein an electrode lead of the jelly-roll is electrically connected to the lower end of a safety vent having notches which are broken to discharge high pressure gas in case high pressure is generated inside of a battery, and a safety member for intercepting current at a predetermined temperature is placed between the safety vent and the electrode lead.



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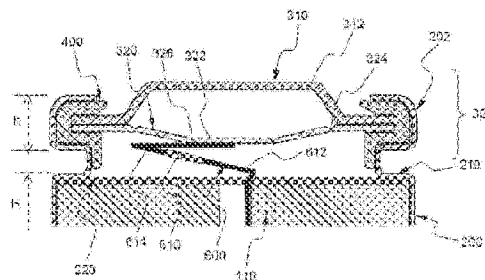
(54) 发明名称

新型结构的帽组件与利用该帽组件的圆筒形
电池

(57) 摘要

本发明提供了一种帽组件,该帽组件安装在具有阴极/隔板/阳极结构的果冻卷式电极组件中的容器的开放上端上,其中,果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端,在电池的內部产生高压的情况下,该切口破裂以排出高压气体,并且用于在预定的温度下中断电流的安全元件被设置在安全排放孔与电极引线之间。

100a



1. 一种布置在电池的圆筒形容器的开放的上端处的帽组件,在所述电池中,阴极/隔板/阳极结构的电极组件(果冻卷状物)被安装在所述圆筒形容器中,其中

所述果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端,并且用于在预定的温度下中断电流的安全元件设置在所述安全排放孔与所述电极引线之间,其中,当在所述电池中产生高压时,所述切口破裂以排出高压气体。

2. 根据权利要求1所述的帽组件,其中,用于防止所述果冻卷状物与所述安全元件之间接触的绝缘板被安装在所述果冻卷状物的顶部。

3. 根据权利要求1所述的帽组件,其中,所述安全元件包括热熔丝。

4. 根据权利要求1所述的帽组件,其中,所述预定的温度是80℃至110℃。

5. 根据权利要求1所述的帽组件,其中,所述预定的温度是90℃至100℃。

6. 根据权利要求1所述的帽组件,其中,所述电极引线被从所述电极引线的联接有所述安全元件的区域的前端向外弯曲,使得所述电极引线从所述果冻卷状物的中央被向外引导,并且然后所述电极引线被从所述电极引线的联接有所述安全元件的所述区域的后端向内弯曲。

7. 根据权利要求6所述的帽组件,其中,所述安全元件形成在所述电极引线的上端的一部分处。

8. 根据权利要求1所述的帽组件,其中,所述安全排放孔由具有0.15mm至0.4mm的厚度的铝片形成。

9. 根据权利要求1所述的帽组件,其中,在所述安全排放孔的中央处设有被构造成向下凹入的凹陷,第一切口和第二切口分别形成在限定所述凹陷的上弯曲区域和下弯曲区域处,所述第一切口形成闭合曲线,形成在所述第一切口的下方的所述第二切口被构造成一侧开放的开放曲线的形状,并且所述第二切口形成成为比所述第一切口深。

10. 根据权利要求1所述的帽组件,其中,所述帽组件被构造成具有如下结构,在该结构中,顶帽和所述安全排放孔被堆叠,至少一个气体排出口形成在所述顶帽处,并且垫圈被安装在所述帽组件的外周处。

11. 一种包括根据权利要求1至10中的任一项所述的帽组件的圆筒形电池。

12. 根据权利要求11所述的圆筒形电池,其中,所述电池包括锂二次电池。

新型结构的帽组件与利用该帽组件的圆筒形电池

技术领域

[0001] 本发明涉及一种新型结构的帽组件,并且,尤其涉及一种布置在电池的圆筒形容器的开放的上端处的帽组件,在该电池中,阴极/隔板/阳极结构的电极组件(果冻卷状物)被安装在圆筒形容器中,其中,果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端,当在电池中产生高压时,该切口破裂以排出高压气体,并且用于在预定的温度下中断电流的安全元件设置在安全排放孔与电极引线之间。

背景技术

[0002] 随着移动装置已经日益改进并且对该移动装置的需求已经增加,对作为用于移动装置的能源的二次电池的需求也已经大大增加。在该二次电池之中的是具有高能量密度和高放电电压的锂二次电池,在该锂二次电池中已经进行了许多研究并且现在该锂二次电池被商业化且广泛地使用。

[0003] 取决于电池壳体的形状,二次电池可以被划分为具有安装在圆筒形金属容器中的电极组件的圆筒形电池、具有安装在棱柱形金属容器中的电极组件的棱柱形电池或具有安装在由铝层压片形成的袋形壳体中的电极组件的袋形电池。圆筒形电池的优点在于,圆筒形电池具有相对大的容量并且在结构上是稳定的。

[0004] 另外,安装在电池壳体中的电极组件充当具有阴极/隔板/阳极堆结构的功率产生元件,该功率产生元件能够被充电和放电。电极组件可以被划分为:果冻卷式电极组件,该果冻卷式电极组件被构造成具有如下结构,被涂敷活性材料的长片式阴极和长片式阳极被卷绕,而隔板被布置在阴极与阳极之间;或堆叠式电极组件,该堆叠式电极组件被构造成具有如下结构,具有预定尺寸的多个阴极和阳极被依次堆叠,而隔板被分别布置在阴极与阳极之间。果冻卷式电极组件的优点在于,果冻卷式电极组件易于制造并且具有高的能量密度每单位质量。

[0005] 图1是典型地示出一般圆筒形电池的竖直截面透视图。

[0006] 参考图1,通过如下过程制造圆筒形电池100:将果冻卷式(卷绕式)电极组件120插入圆筒形壳体130中,将电解质注入圆筒形壳体130中,并且将具有电极端子(未示出)例如阴极端子的顶帽140联接到圆筒形壳体130的开放的上端。

[0007] 果冻卷式电极组件120被构造成具有如下结构,在该结构中,在隔板123被布置在阴极121与阳极122之间的状态下,阴极121和阳极122被卷绕成圆形。圆筒形中心销150被布置在卷的中央,即,电极组件120的中央。中心销150由金属材料制成以呈现预定的强度。中心销150被构造成具有通过卷起金属片形成的中空圆筒形结构。中心销150用于固定并且支撑电极组件120。此外,当对二次电池充电和放电时或当操作二次电池时,中心销150充当通过二次电池的内部反应产生的排出的气体的通道。

[0008] 同时,锂二次电池的缺点在于,锂二次电池具有低安全性。例如,当电池被过度充电至约4.5V或以上时,阴极活性材料被分解,在阳极发生锂金属的树枝状生长,并且电解质被分解。这时,从电池产生热,结果是,上文提及的分解和若干子分解迅速地进行,并且最

后,电池可能着火并且爆炸。

[0009] 为了解决上文提及的问题,因此,一般的圆筒形二次电池包括电流中断装置(CID),该电流中断装置被安装在限定在果冻卷式电极组件与顶帽之间的空间中以当二次电池发生故障时中断电流并且释放内部压力。

[0010] 图2至图4中示出由该CID执行的一系列操作。

[0011] 参考该附图,顶帽10突出以形成阴极端子。顶帽10具有排气口。在顶帽10的下方依次布置正温度系数(PTC)元件20,该正温度系数元件20用于当电池的内部温度升高时通过电池电阻的显著增加来中断电流;安全排放孔30,该安全排放孔30被构造成在正常状态下具有向下凹陷的形状以及用于当电池的内部压力增加时突出并且破裂以用于排出气体;以及电流中断装置50,该电流中断装置50在其上端的一侧处被联接到安全排放孔30并且在其下端的一侧处被连接到电极组件40的阴极。另外,电流中断装置50的外周被电流中断装置垫圈52包围以固定电流中断装置50。

[0012] 因此,在正常操作条件下,电极组件40的阴极经由电极引线42、电流中断装置50、安全排放孔30以及PTC元件20连接到顶帽10以实现电气传导。

[0013] 然而,当由于各种原因诸如过度充电从电极组件40产生气体时,电池的内部压力增加,并且安全排放孔30的形状被反向,如图3中所示。即,安全排放孔30向上突出。这时,安全排放孔30从电流中断装置50分离以中断电流。因此,防止过度充电进一步进行,从而实现安全性。然而,当电池的内部压力继续增加时,安全排放孔30破裂,如图4中所示,结果是,加压气体通过顶帽10的排气口经由破裂的安全排放孔30被排出,从而防止电池爆炸。

[0014] 同时,除上述的过度充电特性之外,圆筒形二次电池必须呈现高温存储特性。

[0015] 高温存储特性需要电池的如下独特特性,在该电池中,当电池被维持在预定的温度下(例如,75℃)持续长时间段时,电池的电压不会下降。具体地,甚至在测试之后高温存储特性需要电池在正常电压和电流下的操作。

[0016] 当电池被充电到比正常电压高的水平时,过度充电特性需要CID用于有效地中断电压和电流的正常运行,从而防止电池在由于电池的内部压力和温度增加使电池爆炸之前爆炸。

[0017] 在8至14kgf/cm²的短路压力范围内调节注入的电解质的量的方法主要被用于满足高温存储特性和过度充电特性。

[0018] 如果将少量电解质注入圆筒形电池容器中,则尽管高温存储特性变得有利,但过度充电特性变得不利。即,如果电解质的量是小的,则电池单元的内部空间增加。因此,如果在电池单元被过度充电的状态下在电池单元中产生气体,则在用于使CID破裂的充足的气体产生之前,电池单元的内部温度增加,结果使电池单元爆炸。

[0019] 另一方面,如果将大量的电解质注入圆筒形电池容器中,则尽管过度充电特性变得有利,但高温存储特性变得不利。即,如果电解质的量是大的,则电池单元的内部空间减小。因此,如果在电池单元在高温下的状态下在电池单元中产生气体,则CID破裂,结果电池单元不能使用。

[0020] 因此,非常需要新型结构的帽组件以及包括诸如帽组件的圆筒形电池,该帽组件能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。

发明内容

[0021] 技术问题

[0022] 因此,已经完成本发明来解决以上问题以及尚未解决的其它的技术问题。

[0023] 为了解决如上文所述的问题,作为各种广泛和深入研究的结果,本申请的发明人已经发现,当用于在预定的温度下中断电流的安全元件被联接在帽组件的安全排放孔与电极引线之间时,能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。基于该发现已经完成了本发明。

[0024] 技术方案

[0025] 根据本发明的一个方面,以上的和其它的目的可以通过提供一种布置在电池的圆筒形容器的开放的上端处的帽组件来实现,在该电池中,阴极/隔板/阳极结构的电极组件(果冻卷状物)被安装在圆筒形容器中,其中,果冻卷状物的电极引线被电连接到具有切口的安全排放孔的下端,当在电池中产生高压时,该切口破裂以排出高压气体,并且用于在预定的温度下中断电流的安全元件设置在安全排放孔与电极引线之间。

[0026] 即,在根据本发明的帽组件中,帽组件的安全排放孔和果冻卷状物的电极引线经由用于在预定的温度下中断电流的安全元件被电连接到彼此。因此,能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。

[0027] 具体地,在根据本发明的帽组件中,安全元件用于在预定的温度下中断电流,并且因此,虽然电池单元的内部压力稍微增加,但是电流不中断,而不像常规帽组件那样,在常规帽组件中,当电解质的量是大的时,电池单元的内部空间减小,结果当产生气体时使电流中断装置(CID)破裂。因此,根据本发明的帽组件有助于高温存储特性。

[0028] 另外,甚至在电解质的量较小的情况下,当电池单元的内部温度由于过度充电而连续地增加时,电流被安全元件中断,并且因此,电池单元的温度在爆炸之前达到预定的温度。因此,能够抑制温度的进一步上升,从而有效地防止电池单元的爆炸。

[0029] 此外,根据本发明的帽组件不需要电流中断装置以及用于电流中断装置的垫圈,并且因此,能够降低制造成本。另外,充分的是,注入适当的量的电解质到电池单元被正常操作的程度,并且因此,能够提高注入效率。

[0030] 此外,安全元件位于在安全排放孔与电极引线之间使得安全元件被布置成与果冻卷状物相邻。因此,安全元件能够较迅速地反应以改变果冻卷状物的温度以在预定的温度下中断电流。

[0031] 优选地,用于防止在果冻卷状物与安全元件之间的接触的绝缘板被安装在果冻卷状物的顶部以防止由于在安全元件与果冻卷状物之间的接触而发生短路。

[0032] 在优选的示例中,安全元件可以包括热熔丝。

[0033] 当电池被热冲击、过度充电或电压下降时,热熔丝中断电流,从而提高电池的安全性。另外,热熔丝容易地解决基于过程的问题,诸如短路压力的增加和减小。

[0034] 预定的温度可以是 80℃至 110℃,优选是 80℃至 100℃。如果温度太低,则在高温存储条件下可能使电流中断。另一方面,如果温度太高,则在过度充电条件下可能使电流不中断。

[0035] 电极引线可以被从电极引线的安全元件联接到的区域的前端向外弯曲,使得电极

引线被从果冻卷状物的中央向外引导,并且然后电极引线被从电极引线的安全元件联接到的区域的后端向内弯曲。因此,防止安全元件弯曲和损坏。

[0036] 在以上结构中,安全元件可以被联接到电极引线的上端的一部分。例如,安全元件可以通过焊接被一体地联接到电极引线的上端的一部分。

[0037] 安全排放孔是一种安全装置,当由于电池的异常操作或组成电池的部件的劣化而使电池的内部压力增加时,该安全装置排来自电池的气体,从而确保电池的安全。例如,当在电池中产生气体并且电池的内部压力超过临界值时,安全排放孔破裂,并且因此,气体穿过破裂的安全排放孔并然后被从电池通过形成在顶帽处的一个或多个气体排出口排出。

[0038] 安全排放孔的材料不被特别地限制。优选地,安全排放孔由具有 0.15mm 至 0.4mm 的厚度的铝片形成,使得当电池的内部压力超过临界值时安全排放孔破裂。

[0039] 在优选的示例中,安全排放孔在其中央处可以设有被构造成向下凹入的凹陷,第一切口和第二切口可以分别形成在限定凹陷的上弯曲区域和下弯曲区域处,第一切口可以形成闭合曲线,在第一切口的下方形成的第二切口可以被构造成一侧开放的开放曲线的形状,并且第二切口可以形成为比第一切口深。

[0040] 由于第二切口形成为比第一切口深,所以当其压力超过临界压力的加压气体被施加到安全排放孔时,第二切口被切断。另一方面,当其压力小于临界压力的加压气体被施加到安全排放孔时,第一切口与第二切口协作以使凹陷向上升高。

[0041] 在优选的示例中,帽组件可以被构造成具有如下结构,在该结构中,顶帽和安全排放孔被堆叠,至少一个气体排出口形成在该顶帽处,并且垫圈可以被安装在帽组件的外周。

[0042] 因此,电极引线的顶部通过焊接被联接到安全排放孔,安全元件被联接到该电极引线,并且因此,能够通过连续的过程制造帽组件。

[0043] 另外,与包括如图 2 中所示的顶帽、安全排放孔、PTC 元件、CID 和电流中断装置垫圈的常规帽组件相比,被构造成具有上文的结构的帽组件不需要正温度系数(PTC)元件、电流中断装置(CID)和电流中断装置垫圈。

[0044] 由于根据本发明的帽组件不需要如上文所述的电流中断装置和电流中断装置垫圈,所以能够降低帽组件的制造成本。另外,由于根据本发明的帽组件不需要 PTC 元件,所以能够减小与 PTC 元件的高度成正比的压褶部分(criming part)的高度,结果卷边部分(beaded part)的高度增加,从而增加果冻卷状物的电池容量。

[0045] 根据本发明的另一个方面,提供了一种包括帽组件的圆筒形电池。

[0046] 根据本发明的圆筒形二次电池通过如下过程制造:在果冻卷式电极组件被安装在圆筒形金属容器中的状态下,将帽组件联接到圆筒形金属容器的开放的上端,将电极组件的阳极焊接到容器的下端,并且在电极组件和电解质被包含在容器中的状态下,将电极组件的阴极焊接到联接到容器的上端的帽组件以密闭地密封电池。

[0047] 根据本发明的电池可以是具有高能量密度、高放电电压以及高输出稳定性的锂二次电池。以下将详细描述根据本发明的锂二次电池的其它部件。

[0048] 一般地,锂二次电池包括阴极、阳极、隔板以及含有锂盐的非水电解质。

[0049] 例如通过将阴极活性材料、导电材料以及粘合剂的混合物涂敷到阴极集电器并且对所涂敷的混合物进行干燥可以制造阴极。根据需要,可以进一步添加填料。通过将阳极材料涂敷到阳极集电器并且对所涂敷的阳极材料进行干燥可以制造阳极。根据需要,可以

进一步添加上文提及的成分。

[0050] 隔板被布置在阳极与阴极之间。隔板可以由呈现高离子渗透性和高机械强度的绝缘薄膜形成。

[0051] 含有锂盐的非水电解质由非水电解质和锂盐组成。液体非水电解质、固体非水电解质或无机固体非水电解质可以被用作非水电解质。

[0052] 在本发明所属的领域中,集电器、电极活性材料、导电材料、粘合剂、填料、隔板、电解质以及锂盐是公知的,因此,将省略其详细描述。

[0053] 可以使用本发明所属的领域中公知的一般方法来制造根据本发明的锂二次电池。即,通过将多孔隔板布置在阴极与阳极之间并且将电解质注入多孔隔板中可以制造锂二次电池。

[0054] 例如,通过将由先前描述的锂过渡金属氧化物活性材料、导电材料以及粘合剂组成的浆料涂敷到集电器并且对浆料进行干燥可以制造阴极。以相同的方式,例如,通过将由先前描述的碳活性材料、导电材料以及粘合剂组成的浆料涂敷到薄的集电器并且对浆料进行干燥可以制造阳极。

[0055] 有利的效果

[0056] 如从上文的描述明显的是,在根据本发明的帽组件中,用于在在预定的温度下中断电流的安全元件被连接在安全排放孔与电极引线之间。因此,能够同时满足高温存储特性和过度充电特性,而无需调节所注入的电解质的量。

附图说明

[0057] 根据结合附图进行的下列详细描述,将更清楚地理解本发明的上述和其它目的、特征以及其它优点,其中:

[0058] 图 1 是示出一般圆筒形电池的竖直截面透视图;

[0059] 图 2 至图 4 是示出通过电流中断装置(CID)的操作来中断电流并且对来自圆筒形电池的高压气体进行排放的一系列过程的竖直截面图;

[0060] 图 5 是典型地示出根据本发明的实施例的圆筒形电池的局部截面图;

[0061] 图 6 是示出在图 5 的圆筒形电池中使用的安全排放孔的透视图;并且

[0062] 图 7 是典型地示出图 5 的安全元件的截面图。

具体实施方式

[0063] 现在,将参考附图详细地描述本发明的优选实施例。然而,应注意,本发明的范围并不限于示出的实施例。

[0064] 图 5 是典型地示出根据本发明的实施例的圆筒形电池的局部截面图,并且图 6 是典型地示出在图 5 的圆筒形电池中使用的安全排放孔的透视图。

[0065] 参考这些附图,根据本发明的实施例的圆筒形电池 100a 通过如下过程制造:将果冻卷状物 110 插入容器 200 中,将电解质注入容器 200 中并且将帽组件 300 安装到容器 200 的开放的上端。

[0066] 帽组件 300 被构造具有如下结构,在该结构中,顶帽 310 和用于减小电池的內部压力的安全排放孔 320 在相互紧密接触的同时被布置在气密的垫圈 400 中,该气密的垫圈

400 被安装在容器的压褶部分 202 的内侧并且在容器 200 的卷边部分 210 的上部分的内侧。顶帽 310 的中央向上突出以充当用于与外部电路连接的阴极端子。多个通孔 312 围绕顶帽 310 的突起形成,使得压缩气体通过通孔 312 被从容器 200 排出。

[0067] 另外,帽组件 300 被构造成具有如下结构,在该结构中,顶帽 310 和安全排放孔 320 被堆叠,充当气体排出口的通孔 312 形成在顶帽 310 处。垫圈 400 被安装在帽组件 300 的外周处。

[0068] 因此,与图 2 的帽组件相比,图 5 的帽组件 300 不需要正温度系数(PTC)元件 200。因此,与 PTC 元件 200 的高度成正比例的压褶部分 202 的高度减小,结果卷边部分 210 的高度 H 增加,从而增加果冻卷状物 110 的电池容量。

[0069] 安全排放孔 320 是传输电流的薄膜结构。使安全排放孔 320 的中央凹入以形成凹入的中央部分 322。具有不同的深度的两个切口 324 和 326 分别形成在中央部分 322 的上弯曲区域和下弯曲区域处。

[0070] 另外,果冻卷状物 110 的电极引线 600 被连接到安全排放孔 320 的下端。用于在 80℃至 110℃的温度下中断电流的安全元件 610 被联接到电极引线 600。

[0071] 由于安全元件 610 被布置成与果冻卷状物 100 相邻,所以安全元件 610 能够较迅速地反应以改变果冻卷状物 100 的温度来中断电流。

[0072] 用于防止在果冻卷状物 110 与安全元件 610 之间接触的绝缘板 220 被安装在果冻卷状物 110 的顶部,从而防止由于在果冻卷状物 110 与安全元件 610 之间接触而发生短路。

[0073] 另外,电极引线 600 被从电极引线 600 的联接有安全元件 610 的区域的前端向外弯曲,使得电极引线 600 被从果冻卷状物的中央向外引导;并且然后电极引线 600 被从电极引线 600 的联接有安全元件 610 的区域的后端向内弯曲。

[0074] 同时,形成在安全排放孔 320 的上部分处的第一切口 324 形成闭合曲线,并且形成在安全排放孔 320 的下部分处的第二切口 326 被构造成其一侧是开放的开放曲线的形状。另外,第二切口 326 形成为比第一切口 324 深。因此,第二切口 326 的联接力小于第一切口 324 的联接力。

[0075] 因此,当容器 200 的内部压力超过临界压力时,安全排放孔 320 的第二切口 326 破裂,结果加压气体被从容器 200 通过顶帽 310 的通孔 312 排出。

[0076] 图 7 是典型地示出图 5 的安全元件的截面图。

[0077] 参考图 7,作为安全元件的热熔丝 610 被构造成使得热熔丝 610 的中间区域 A 形成薄片形状。因此,例如,当电池的内部温度是 80℃或以下时,电流不被中断,从而满足高温存储特性。另一方面,当电池单元的内部温度超过 80℃时,热熔丝 610 的相对薄的中间区域 A 破裂以使从果冻卷状物 110 流到安全排放孔 320 的电流中断,从而防止电池由于过度充电而爆炸。

[0078] 除了上文提及的结构之外,可以使用其它不同的结构来在预定的温度下中断电流。

[0079] 虽然出于说明性目的已经公开了本发明的优选实施例,但是本领域的技术人员应理解,在不脱离如在所附权利要求中公开的本发明的范围和精神的情况下,能够进行各种变形、添加和替换。

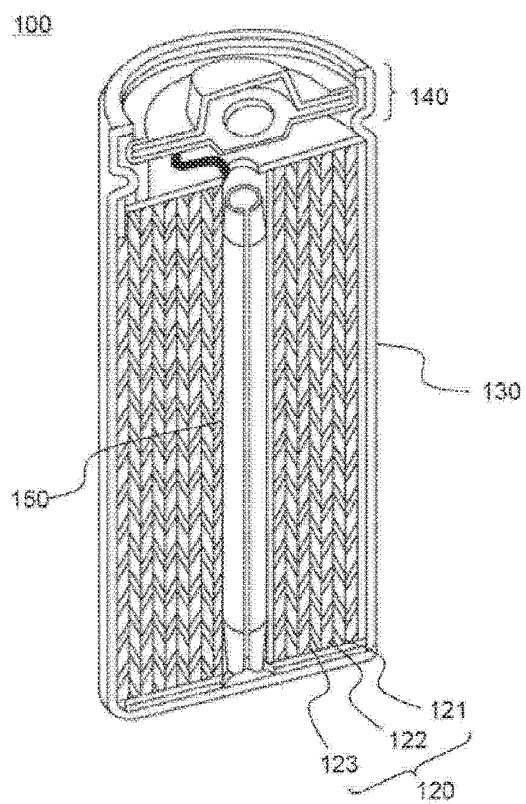


图 1

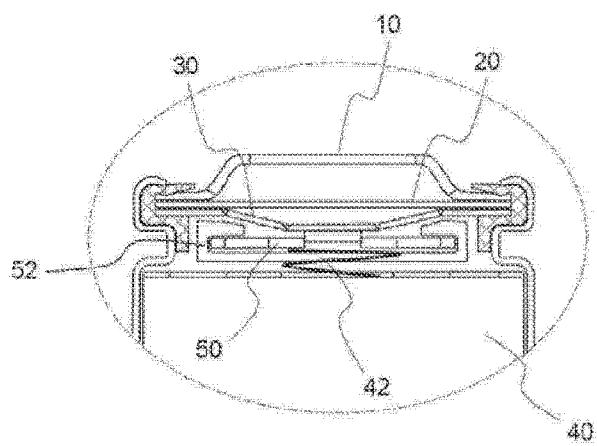


图 2

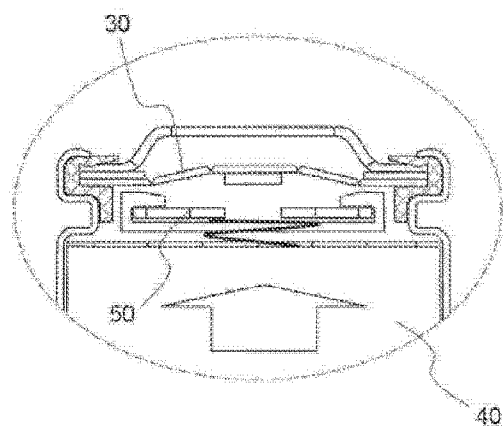


图 3

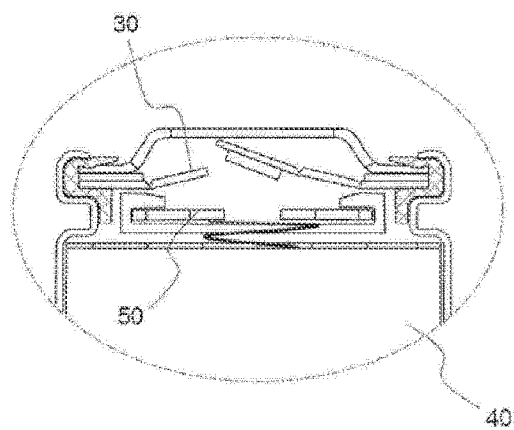


图 4

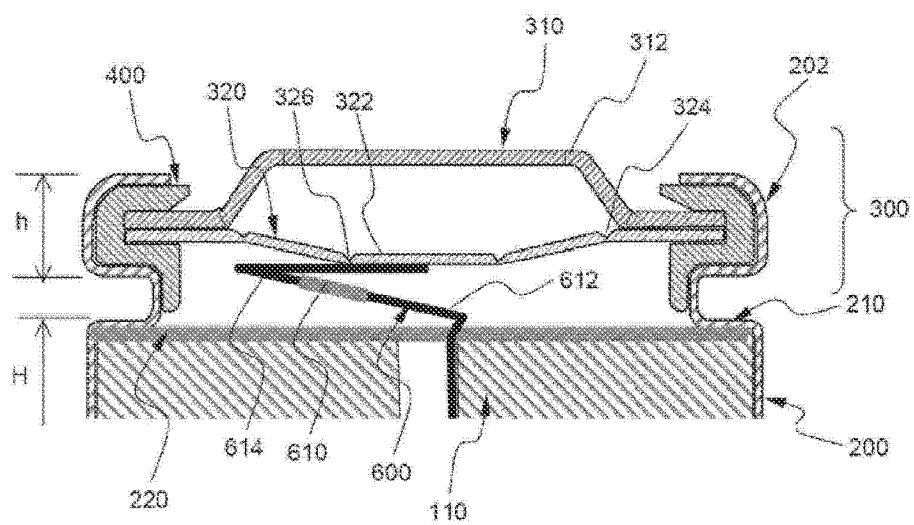


图 5

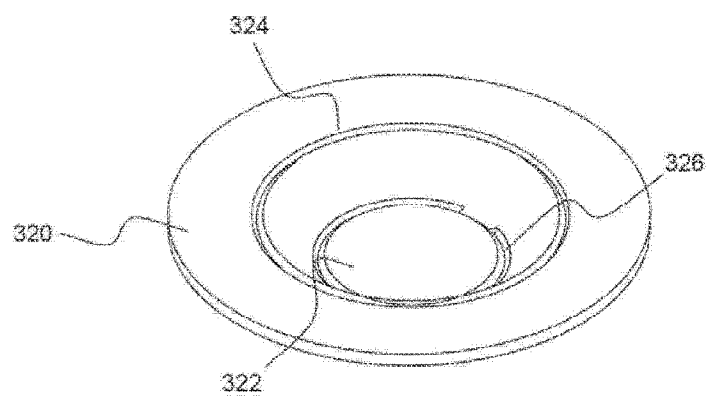


图 6

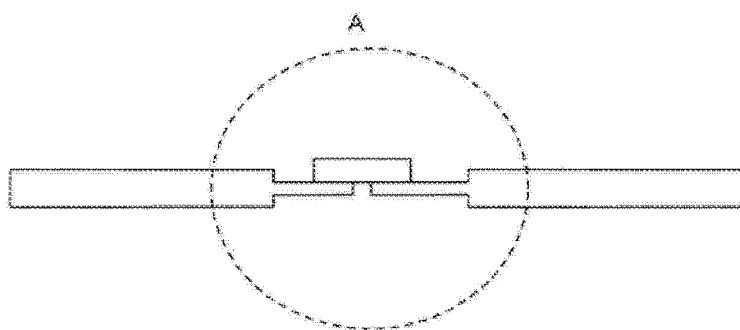


图 7



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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Priority

2. Applicant's claim for the benefit of a prior-filed application under 35 U.S.C. 119(e) or under 35 U.S.C. 120, 121, or 365(c) is acknowledged. Receipt is acknowledged of papers submitted under 35 U.S.C. 119(a)-(d) or (f), which papers have been placed of record in the file.

Information Disclosure Statement

3. Information disclosure statements (IDS), submitted November 1, 2021, March 4, 2022, and September 14, 2022, have been received and considered by the examiner.

Claim Interpretation

4. All "wherein" clauses are given patentable weight unless otherwise noted. Please see MPEP 2111.04 regarding optional claim language.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

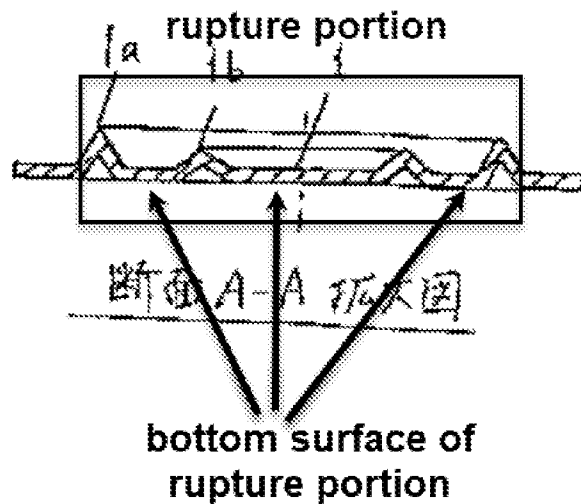
(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

5. Claims 1-3 and 6 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Mitamura JPH06215746.

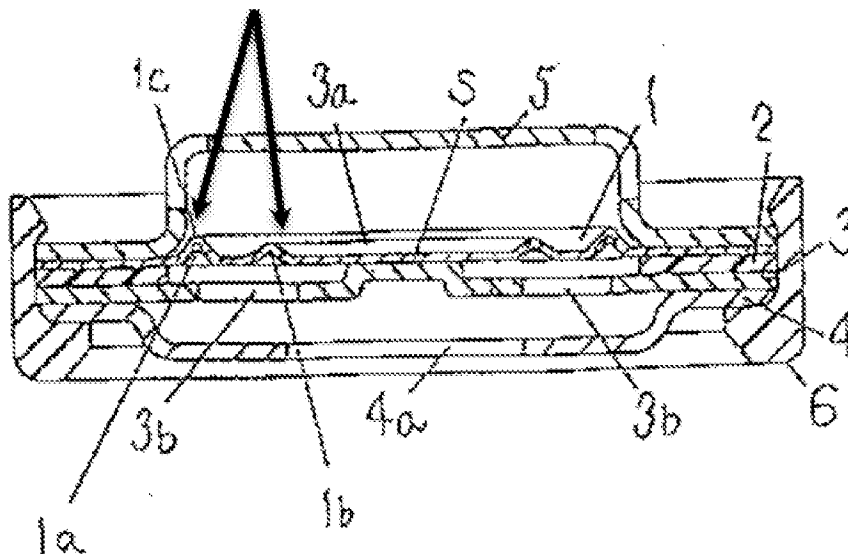
Regarding Claim 1, Mitamura discloses a secondary battery comprising an electrode assembly (electrode plate group), a can member (battery case) **8** configured to accommodate the electrode assembly therein, and a top cap assembly (sealing plate) configured to cover an opening of the can member **8**, wherein the top cap assembly comprises a top cap (external terminal plate) **5** positioned at an uppermost portion and a safety vent part (safety valve) **1** positioned below the top cap **5**, wherein the safety vent part **1** comprises a rupture portion (the circle having **1a** as its boundary) designed to be ruptured when the internal pressure within the can member **8** increases to equal or exceed a predetermined pressure, a rupture notch **1a** that defines a boundary of the rupture portion, and a bending notch portion **1b** which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing an inside of the can member **8**, wherein the bending notch portion **1b** is bent when the internal pressure within the can member **8** increases (see at least Figs 1-4b; paras 0019-0024). The Office notes that Fig. 3 shows the original (pre-venting) state where **1b** is in its original position, Fig. 4a shows a first elevated pressure situation where **1b** bends initially, and Fig. 4b shows a second (higher) elevated pressure situation where the rupture takes place at rupture notch portion **1a**.⁹⁷ Since the outer periphery of the metal safety valve **1** is crimped and fixed by the battery case **8**, (para 0021: “when the battery internal pressure reaches a predetermined value, the central portion is lifted upward and bent as shown in FIG. 4A. The portion **1b** is reversely deformed, and the welded portions are peeled off by this stress, and the connection with the perforated metal terminal plate **3** is broken, thereby interrupting the current”).

Regarding Claims 2 and 3, Mitamura discloses wherein the rupture notch **1a** is formed as a recess within a lower surface of the safety vent part and the bending notch portion **1b** is formed as a recess within the lower surface of the safety vent part and of the rupture portion (meeting **Claim 3**), wherein a

recessed depth of the bending notch part is less than a recessed depth of the rupture notch portion (see annotated figure below, as well as Fig. 2 showing A-A perspective). This is especially true when considering the thinned portion 1c, which increases the recessed depth of 1a even farther.



1a and 1b are clearly designed with different heights (1b being shorter, i.e. having a less recessed depth) as indicated by the parallel lines connecting respective 1a notches and 1b notches.



The portion below of the Fig 2 of Mitamura shows the rupture portion, and it is clear that the rupture notch **1a** is in the bottom surface of the rupture portion.



Regarding Claim 6, Fig. 2 (stack of top cap assembly) shows that the rupture notch portion **1a** and the bending notch portion **1b** have continuous circular shapes.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

6. Claims 4-5, 7-9, and 12 are rejected under 35 U.S.C. 103 as being unpatentable over Mitamura JPH06215746, as applied to Claim 1, and further in view of Kim US PG Publication 2012/0114979 (hereinafter “Kim ‘979”).

Regarding Claims 4 and 5, Mitamura discloses the claimed secondary battery as described in the rejection of Claim 1, which is incorporated herein in its entirety. Mitamura fails to specifically disclose wherein the top cap assembly further comprises a current interrupt device provided with an interrupt portion which is attached to the rupture portion which is configured to be ruptured to interrupt current when the internal pressure within the can member increases and wherein the interrupt portion of the current interrupt portion is attached to the rupture portion inside a circumference of the bending notch portion. However, in the same field of endeavor of cylindrical secondary batteries having safety vents in the top cap, Kim ‘979 discloses a top cap assembly **300** comprising a current interrupt device **600** which is attached to a rupture portion inside a circumference of a bending notch portion **324** and inside a rupture notch portion **326** (or the reverse, see Fig. 17 where **324** is ruptured and is therefore rupture notch portion) and which is configured to be ruptured to interrupt current when the internal pressure within the can member **200** increases, which improves safety, in conjunction with discharging pressurized gas as necessary via the ruptured safety vent (see at least Figs.5-6, 17; paras 0085, 0101-0104). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the battery of Mitamura such that the top cap assembly further comprises a current interrupt device provided with an interrupt portion which is attached to the rupture portion which is configured to be ruptured to interrupt current when the internal pressure within the can member increases and wherein the interrupt portion of the current interrupt portion is attached to the rupture portion inside a circumference of the bending notch portion because Kim ‘979 teaches that this feature provides additional safety via interrupting current.

Regarding Claims 7 and 9, Mitamura fails to specifically disclose wherein the rupture notch portion and the bending notch portion have a discontinuous circular or elliptical shape. However, Kim '979 teaches that a rupture notch portion (second notch) **326** and bending notch portion (first notch) **324** both have an open curve structure to allow for rupture without full separation for either notch portion (Figs 6-7; paras 0089-0090). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the rupture notch portion or the bending notch portion of Mitamura with a discontinuous circular or elliptical shape because Kim '979 teaches that this structure is known for a safety vent notch portion of a safety vent in a cylindrical secondary battery to provide for bending or rupturing without full separation of the parts, and the use of a known technique to improve similar devices (methods or products) in the same way is likely to be obvious. See *KSR International Co. v. Teleflex Inc.*, 550 U.S. ___, ___, 82 USPQ2d 1385, 1395 – 97 (2007) (see MPEP § 2143, C.). Further, the change in form or shape, without any new or unexpected results, is an obvious engineering design. See *In re Dailey*, 149 USPQ 47 (CCPA 1976) (see MPEP § 2144.04).

Regarding Claim 8, Kim '979 teaches that the discontinuous shape of the rupture portion (see e.g. Figs. 7, 17) results in more than one segment of the second notch **326**, and in the case where this notch only bends and does not rupture, then there are multiple bending notch portions (each section of the discontinuous circle is a bending notch portion). IT would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the multiple bending notch portions of Mitamura and Kim '979 having different lengths in order to more finely control the rupture or bending action since the change in form or shape, without any new or unexpected results, is an obvious engineering design. See *In re Dailey*, 149 USPQ 47 (CCPA 1976) (see MPEP § 2144.04). A change in proportion or relative dimension is obvious in the absence of unexpected results. In *Gardner v. TEC Systems, Inc.*, 725 F.2d 1338, 220 USPQ 777 (Fed. Cir. 1984), *cert. denied*, 469 U.S. 830, 225 USPQ 232 (1984), the Federal Circuit held that, where the only difference between the prior art and the claims was a recitation of relative dimensions of the claimed device and a device having the claimed relative

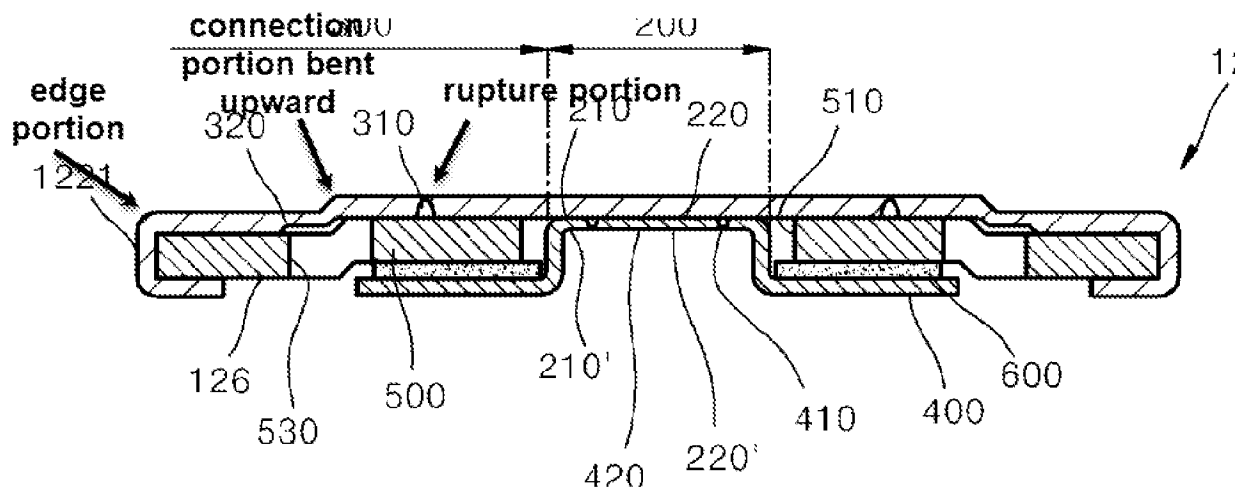
dimensions would not perform differently than the prior art device, the claimed device was not patentably distinct from the prior art device.

Regarding Claim 12, Mitamura modified by Kim ‘979 does not specifically disclose wherein a spaced distance between an edge end of the safety vent part and the rupture notch portion is equal to a spaced distance between the rupture notch portion and the vending notch portion. However, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to space these parts evenly from one another such that the spaced distance between an edge end of the safety vent part and the rupture notch portion is equal to a spaced distance between the rupture notch portion and the vending notch portion of Mitamura modified by Kim ‘979 in order to best fit the notches with the inner cell structure and to ensure even response to pressure changes since a change in proportion or relative dimension is obvious in the absence of unexpected results. In *Gardner v. TEC Systems, Inc.*, 725 F.2d 1338, 220 USPQ 777 (Fed. Cir. 1984), *cert. denied*, 469 U.S. 830, 225 USPQ 232 (1984), the Federal Circuit held that, where the only difference between the prior art and the claims was a recitation of relative dimensions of the claimed device and a device having the claimed relative dimensions would not perform differently than the prior art device, the claimed device was not patentably distinct from the prior art device.

7. Claim 10 is rejected under 35 U.S.C. 103 as being unpatentable over Mitamura JPH06215746, as applied to Claim 1, and further in view of Kim WO2018/030836 (hereinafter “Kim ‘836”).

Regarding Claim 10, Mitamura discloses the claimed secondary battery as described in the rejection of Claim 1, which is incorporated herein in its entirety. Mitamura fails to specifically disclose wherein a connection portion between an edge portion and the rupture portion of the safety vent part is bent upward. However, in the same field of endeavor of cylindrical secondary batteries having safety vents in the top cap, Kim ‘836 teaches that the safety vent includes a connection portion between an edge portion and the

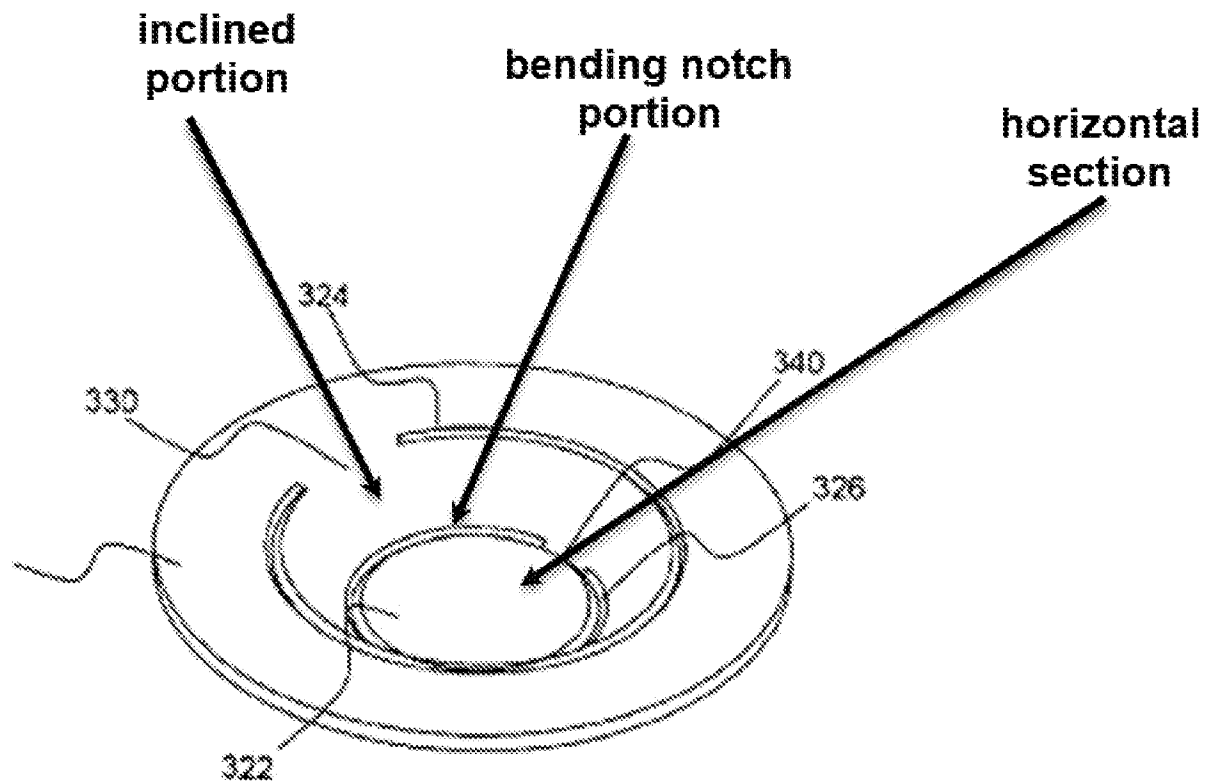
rupture portion and that said connection portion is bent upward (see annotated figure below) and it is clear that this upward bend is formed to fit with other components of the cap, such as **700** (see e.g. Figs 2-3; paras 0063, 0071). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the battery of Mitamura such that a connection portion between an edge portion and the rupture portion of the safety vent part is bent upward because Kim '836 teaches this structure and shows that it fits with other parts of the top. Applying a known technique to a known device (method or product) ready for improvement to yield predictable results is likely to be obvious. See *KSR International Co. v. Teleflex Inc.*, 550 U.S. ___, ___, 82 USPQ2d 1385, 1395 – 97 (2007) (see MPEP § 2143, D.). The change in form or shape, without any new or unexpected results, is an obvious engineering design. See *In re Dailey*, 149 USPQ 47 (CCPA 1976) (see MPEP § 2144.04).



8. Claim 11 is rejected under 35 U.S.C. 103 as being unpatentable over Mitamura JPH06215746 and Kim WO2018/030836 (hereinafter “Kim ‘836”), as applied to Claim 10, and further in view of Kim US PG Publication 2012/0114979 (hereinafter “Kim ‘979”).

Regarding Claim 11, Mitamura modified by Kim '836 discloses the claimed secondary battery as described in the rejection of Claim 10, which is incorporated herein in its entirety. Mitamura modified by

Kim '836 fails to specifically disclose wherein the rupture portion comprises an inclined section inclined so that a height thereof gradually decreases from the connection portion toward the bending notch portion and a horizontal section having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion, wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section. However, previously cited Kim '979 teaches a safety vent having a rupture portion wherein an inclined portion is inclined so that a height thereof gradually decreases from a connection portion toward a bending notch portion **326** (Fig 17 shows an embodiment wherein **326** is a bending notch portion while **324** which is a rupture notch portion) and a horizontal section **322** having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion (see Figs 5-7) wherein the bending notch portion **324** is positioned on a boundary between the inclined portion and the horizontal section.



Kim '979 teaches that this design allows the safety vent to function by providing the lower portion **322** (horizontal section) that is thrust upward when the pressure in the battery rises and allows for current interruption via CID **600** thereby improving safety (see paras 0084-0090, Figs 5-7 and 14). Therefore, it would have been obvious to a person having ordinary skill in the art before the effective filing date of the instant application to form the battery of Mitamura modified by Kim '836 such that the rupture portion comprises an inclined section inclined so that a height thereof gradually decreases from the connection portion toward the bending notch portion and a horizontal section having a horizontal shape and extending from an end of the inclined section toward the center of the rupture portion, wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section because Kim '979 teaches that this design allows the safety vent to function by providing the horizontal section that is thrust upward when the pressure in the battery rises and allows for current interruption via CID. Applying a known technique to a known device (method or product) ready for improvement to yield predictable results is likely to be obvious. See *KSR International Co. v. Teleflex Inc.*, 550 U.S. ___, ___, 82 USPQ2d 1385, 1395 – 97 (2007) (see MPEP § 2143, D.).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached on M-Th 6 or 7 to 3 or 4; Alt Fri afternoons.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/LISA S PARK/
Primary Examiner, Art Unit 1729

Notice of References Cited	Application/Control No. 17/259,395		Applicant(s)/Patent Under Reexamination Min et al.	
	Examiner LISA S PARK		Art Unit 1729	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20120114979-A1	05-2012	KIM; Ji Young	H01M50/3425	429/7
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N	JPH06215746	01-1993	JP	Matimura	
	O	WO2018/030836	02-2018	WIPO	Kim	
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
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<i>Search Notes</i> 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M50/3425 or H01M50/107 or H01M50/578 or H01M50/574 or H01M2200/20 or H01M50/56 or H01M50/152 or Y02E60/10 or H01M50/116 (text-limited)	01/31/2023	/L.S.P./

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Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search history)	01/31/2023	/L.S.P./
Analysis of Global Dossier file	02/02/2023	/L.S.P./
PE2E Applicant/Assignee search	02/02/2023	/L.S.P./
PE2E foreign database search	02/02/2023	/L.S.P./

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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Doc code: IDS

PTO/SB/08a (01-22)

Doc description: Information Disclosure Statement (IDS) Filed

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Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	OPAP
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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(Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	OPAP
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Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	OPAP
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

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Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2022-03-04
Name/Print	Jordan T. Riviello	Registration Number	72,515

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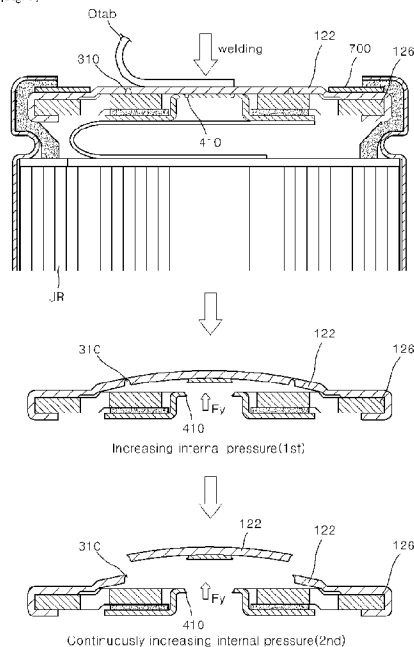
(74) Agent: **Ji, Jeong Hun**; 4F, 40-4, Teheran-ro 14-gil, Gangnam-gu, Seoul 06239 (KR).

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(54) Title: CAP ASSEMBLY FOR SECONDARY BATTERY AND SECONDARY BATTERY INCLUDING THE CAP ASSEMBLY

[Fig. 3]



(57) Abstract: Disclosed are a cap assembly for a secondary battery and a secondary battery including the cap assembly. The cap assembly includes: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion including a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and including a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween. The cap assembly can be used to increase the capacity of the power generating device in the secondary battery compared to when used in other secondary batteries of the same size, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and the safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.

Published:

- *with international search report (Art. 21(3))*
- *before the expiration of the time limit for amending the claims and to be republished in the event of receipt of amendments (Rule 48.2(h))*

Description

Title of Invention: CAP ASSEMBLY FOR SECONDARY BATTERY AND SECONDARY BATTERY INCLUDING THE CAP ASSEMBLY

Technical Field

- [1] The present invention relates to a cap assembly for a secondary battery and a secondary battery including the cap assembly. More specifically, the present invention relates to a cap assembly for a secondary battery that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery, and a secondary battery including the cap assembly.

[2]

Background Art

- [3] Secondary batteries are classified into cylindrical, prismatic, and pouch types depending on the shape of battery cases they employ. The cylindrical batteries include an electrode assembly accommodated in a cylindrical metal can. The prismatic batteries include an electrode assembly accommodated in a prismatic metal can. The pouch type batteries include an electrode assembly accommodated in a pouch type case made of an aluminum laminate sheet.
- [4] Electrode assemblies accommodated in battery cases are power generating devices capable of repeated charge/discharge cycles that have a cathode/separator/anode laminate structure. Electrode assemblies are classified into jelly-roll and stack types by their structure. The jelly-roll type structure is constructed by interposing a separator between an anode and a cathode, each of which is in the form of a long sheet coated with an active material, and winding the laminate in a roll form. The stack type structure is constructed by sequentially stacking a plurality of electrode units, each of which includes a cathode having a predetermined size, an anode having a predetermined size, and a separator interposed therebetween. The jelly-roll type electrode assemblies are most widely used in secondary batteries due to their ease of construction and high energy density per unit weight. The jelly-roll type electrode assemblies are usually employed in cylindrical batteries.
- [5] Secondary batteries employing jelly-roll type electrode assemblies are likely to

deform because the jelly-roll type electrode assemblies undergo repeated swelling and shrinkage during charge/discharge. This deformation may cause internal short circuits. Heat generated by internal short circuits decomposes organic solvents to produce gases. The gases increase the internal gas pressure of batteries to burst the batteries.

- [6] An increase in the internal gas pressure of batteries may take place also when internal short circuits are caused by an external impact.
- [7] Many attempts have been made to solve the safety problems of batteries. For example, known is a cap assembly for a cylindrical battery having a structure in which a safety vent, safety devices, and a top cap are fixed by a gasket. High-pressure gases are released through the safety vent. The safety devices include a PTC device adapted to interrupt the flow of electric current at high temperature and a current interrupt device (CID) adapted to interrupt the flow of electric current when the internal pressure of the battery increases. The top cap forms a protruding terminal adapted to protect the safety devices. Examples of such attempts are disclosed in many patent publications, for example, Japanese Patent Publication Nos. 2006-286561, 2005-100927, and 2002-373711 and Korean Patent Publication Nos. 2012-0103394 and 1999-67165. In each of the cap assemblies disclosed in these publications, the gasket serves to seal the interface between the can and the cap assembly while surrounding the outer circumferences of the safety devices, but the structure of the top cap limits the capacity of the power generating device, permits the release of a limited amount of gases when the safety vent ruptures, causes an increase in contact resistance, and increases in the construction cost of the cap assembly.
- [8] In a cylindrical secondary battery 100 illustrated in Fig. 9, a power generating jelly roll JR is accommodated in a battery can 110 such that a tab is connected to a sub-plate 128. This connection enables power supply through the battery can and a terminal of a cap-up 124. An open end (e) of the battery can is clamped to fix a cap assembly 120. A safety vent 122 of the cap assembly 120 is bent to surround the cap-up 124. Alternatively, the cap assembly may have a structure in which the safety vent is not bent and the cap-up is exposed to the top so as to be in direct contact with a gasket 130. When a module consisting of the secondary battery and one or more other secondary batteries is used in an electric vehicle or an energy storage system (ESS), energy or force generated by electrically shorting of the other secondary batteries or a power grid can melt or structurally weaken the gasket 130 interposed between the clamped battery can 110 and the cap-up 124 or the safety vent 122, and as a result, the battery can 110 comes into contact with the safety vent, causing short circuits. The short circuits increase the risk of leakage of an electrolyte from the secondary battery or safety accidents, such as explosion.
- [9]

Disclosure of Invention

Technical Problem

- [10] A first object of the present invention is to provide a cap assembly for a secondary battery that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.
- [11] A second object of the present invention is to provide a secondary battery including a cap assembly that is used to increase the capacity of a power generating device in the secondary battery compared to when used in other secondary batteries of the same size, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.
- [12]

Solution to Problem

- [13] A first aspect of the present invention provides a cap assembly for a secondary battery including: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion including a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and including a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween.
- [14] According to one embodiment of the present invention, the safety vent may be bent downwardly and extend to form a bent portion.
- [15] According to a further embodiment of the present invention, the breakable junction may be welded to a protrusion of the sub-plate entering the central hole formed in the support member.
- [16] According to another embodiment of the present invention, the protrusion may

include a portion that protrudes upwardly opposite to the central portion and is opposite and melt-bonded to the breakable junction.

- [17] According to another embodiment of the present invention, the notch may be formed in the shape of a long home on the outer circumference of the portion opposite to the breakable junction.
- [18] According to another embodiment of the present invention, the protrusion may include a portion that protrudes upwardly opposite to the central portion and is opposite to and in close contact with the current-carrying contact.
- [19] According to another embodiment of the present invention, the portion opposite to the current-carrying contact may be formed on the outer circumference of the notch.
- [20] According to another embodiment of the present invention, a breakable notch may be formed at the bottom of the peripheral portion such that the safety vent is bent by a rupture pressure.
- [21] According to another embodiment of the present invention, the cap assembly may further include a bending-relieving portion adjacent to the breakable notch to space the current-carrying contact from the cap-down and forming a space where a rupture pressure is applied to the breakable junction.
- [22] According to another embodiment of the present invention, the bottom width of the bending-relieving portion may be larger than that of the breakable notch.
- [23] According to another embodiment of the present invention, the cap-down may include a through-hole that is spaced apart from and formed around the central hole and through which internal gases and liquids pass.
- [24] According to another embodiment of the present invention, the insulating layer may extend beyond the end of the sub-plate and may reach the end of the cap-down.
- [25] According to another embodiment of the present invention, the insulating layer may have a through-hole opposite to the through-hole of the cap-down.
- [26] According to another embodiment of the present invention, spacer bridges may be formed across the opposite through-hole.
- [27] According to another embodiment of the present invention, a short-circuit preventing layer may be formed at a position opposite to the end of the battery can through a gasket of the safety vent.
- [28] According to another embodiment of the present invention, the short-circuit preventing layer may include a polymeric or organic material that is resistant to heat at a temperature of 300 °C or more.
- [29] According to another embodiment of the present invention, the short-circuit preventing layer may be an oxide or nitride film formed by surface treatment with high energy, such as a laser, after removal of a plating layer.
- [30] According to another embodiment of the present invention, the short-circuit

preventing layer may be made of an anodized ring-shaped aluminum material.

[31] According to another embodiment of the present invention, the aluminum material may be melt-bonded to an adherend by soldering.

[32] According to another embodiment of the present invention, the cap-down and the sub-plate between which the insulating layer is interposed may have rough sides opposite to each other.

[33] A second aspect of the present invention provides a secondary battery including the cap assembly.

[34]

Advantageous Effects of Invention

[35] The cap assembly of the present invention can be used to increase the capacity of a power generating device in a secondary battery compared to when used in other secondary batteries of the same size, does not undergo any deformation when a pack-tab is welded thereto, minimizes the contact resistance between terminals, is constructed at reduced cost, and can protect a gasket disposed between a battery can and a safety vent from being melted or lost by energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery.

[36]

Brief Description of Drawings

[37] Fig. 1 is a plan view illustrating a cap assembly of the present invention.

[38] Fig. 2 is a cross-sectional view of the cap assembly of Fig. 1.

[39] Fig. 3 is a cross-sectional view illustrating a state in which a pack-tab (Otab) is welded to a secondary battery fabricated by assembly of a cap assembly according to the present invention and a battery can.

[40] Fig. 4 is a cross-sectional view illustrating states in which a safety vent of a cap assembly according to the present invention is deformed at operating and rupture pressures.

[41] Fig. 5 is a cross-sectional view illustrating a state in which an electrode tab attached to a sub-plate of a cap assembly according to the present invention is deformed in a battery can and is in close contact with a cap-down through an insulating layer interposed therebetween.

[42] Fig. 6 is a detailed plan view of the insulating layer of Fig. 5.

[43] Fig. 7 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together without the need for forging.

[44] Fig. 8 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together

without the need for forging.

- [45] Fig. 9 is a cross-sectional view of a secondary battery mounted with a conventional cap assembly and illustrates a state in which the end of a battery is in contact with a safety vent as a result of melting or damage of a gasket by thermal energy generated in an external circuit or load and a state in which a cap-down is in contact with the safety vent as a result of melt-bonding of an insulating layer.

[46]

Best Mode for Carrying out the Invention

- [47] The present invention will now be described in more detail.

- [48] Technical terms used in this specification are used to merely illustrate specific embodiments, and should be understood that they are not intended to limit the present disclosure.

- [49] As far as not being defined differently, all terms used herein including technical or scientific terms may have the same meaning as those generally understood by an ordinary person skilled in the art to which the present disclosure belongs to, and should not be construed in an excessively comprehensive meaning or an excessively restricted meaning.

- [50] In addition, if a technical term used in the description of the present disclosure is an erroneous term that fails to clearly express the idea of the present disclosure, it should be replaced by a technical term that can be properly understood by the skilled person in the art. In addition, general terms used in the description of the present disclosure should be construed according to definitions in dictionaries or according to its front or rear context, and should not be construed to have an excessively restrained meaning. A singular representation may include a plural representation as far as it represents a definitely different meaning from the context. Terms "include" or "has" used herein should be understood that they are intended to indicate an existence of several components or several steps, disclosed in the specification, and it may also be understood that part of the components or steps may not be included or additional components or steps may further be included. In the description of the present invention, detailed explanations of related art are omitted when it is deemed that they may unnecessarily obscure the essence of the invention. In the accompanying drawings, those components are rendered the same reference numeral that are the same or are in correspondence, regardless of the figure number, and redundant explanations are omitted.

- [51] Fig. 1 is a plan view illustrating a cap assembly of the present invention, Fig. 2 is a cross-sectional view of the cap assembly of Fig. 1, Fig. 3 is a cross-sectional view illustrating a state in which a pack-tab (Otab) is welded to a secondary battery fabricated

by assembly of a cap assembly according to the present invention and a battery can, Fig. 4 is a cross-sectional view illustrating states in which a safety vent of a cap assembly according to the present invention is deformed at operating and rupture pressures, Fig. 5 is a cross-sectional view illustrating a state in which an electrode tab attached to a sub-plate of a cap assembly according to the present invention is deformed in a battery can and is in close contact with a cap-down through an insulating layer interposed therebetween, Fig. 6 is a detailed plan view of the insulating layer of Fig. 5, and Fig. 7 is a cross-sectional view illustrating a state in which a safety vent and a cap-down of a cap assembly according to the present invention are assembled together without the need for forging.

- [52] A cap assembly 120 for a secondary battery according to the present invention includes: a safety vent 122 made of a circular metal plate and consisting of a central portion 200 corresponding to the central area of the metal plate and a peripheral portion 300 surrounding the central portion, the central portion including a current-carrying contact 210 forming a first conductive path through which an electric current generated from a power generating device JR flows to the outside and a breakable junction 220 forming a second conductive path at the center of the central portion; a cap-down 126 opposite to the safety vent, having a central hole 510 through which the breakable junction is exposed, and including a support member 500 in close contact with the peripheral portion; and a sub-plate 400 joined to the breakable junction and having a notch 410 formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer 600 therebetween.
- [53] The current-carrying contact 210 in the central portion 200 functions to reduce the contact resistance due to the formation of the conductive path and imparts a support force against a pressure applied when a pack-tab is attached to the cap assembly. The breakable junction 220 included in the central portion 200 may be welded to a protrusion 420 of the sub-plate entering the central hole 510 formed in the central portion surrounded by the support member. The welding may be performed by various bonding processes. Preferably, laser welding or melt-bonding is used such that the welded portion is not detached even under high internal pressures (for example, 1st pressure and 2nd pressure) generated upon abnormal operation of the secondary battery.
- [54] That is to say, the protrusion 420 of the sub-plate 400 protrudes upward opposite to the central portion 200. Specifically, the protrusion 420 enters the central hole 510 and is welded to the breakable junction 220. For the prevention of short circuits, the protrusion 420 and the central hole should be spaced apart from each other such that they do not come into contact and in electrical communication with each other.

- [55] Furthermore, the insulating layer 600 has a central hole 620 opposite to the central hole 510 and through which the central portion 200 is exposed and a standing portion 625 in the form of a barrier protruding along the circumference of the central hole 620. Due to this structure, the risk of short circuits between the protrusion 420 of the sub-plate and the central hole 510 during assembly can be completely avoided.
- [56] The support member 500 of the cap-down 126 is opposite to and in close contact with the peripheral portion 300 of the safety vent 122. This arrangement reduces the contact resistance when power is applied from the power generating device JR to the outside, ensuring good current-carrying properties. In addition, the support member 500 serves to prevent the cap assembly from being deformed caused by an excessive load or pressure during welding of a pack-tab (Otab) to a package module in which the battery is connected in series or parallel with one or more other batteries.
- [57] Furthermore, the protrusion 420 is in close contact with and supports the central portion 200 over a large area. This arrangement is advantageous in preventing the cap assembly from being deformed when a pack-tab is welded to the secondary battery. Thus, the protrusion 420 is designed to have a portion 220' opposite and welded to the breakable junction 220 and a portion 210' opposite to the current-carrying contact around the circumference of the portion 220'. The portion 210' is formed in close contact with the current-carrying contact other than the notch 410 in the form of a recessed long home that is broken by high pressures upon abnormal operation of the secondary battery and permits the release of internal gases into the outside. Due to this design, a support force by the portion 210' is ensured and the conductive paths can be maintained, which is beneficial in terms of current-carrying properties.
- [58] The cap-down 126 includes a through-hole 530 spaced apart from but adjacent to the central hole 510 and through which internal gases and liquids are released into the outside. The through-hole 530 is filled with high-pressure gases, liquids, and a flowable electrolyte upon abnormal operation of the secondary battery, and as a result, operating and rupture pressures are applied to the safety vent 122 therethrough.
- [59] Here, the operating pressure refers to a 1_{st} pressure at which the contact and junction between the current-carrying contact 210 of the safety vent and the portion 210' opposite thereto and between the breakable junction 220 of the safety vent and the portion 220' opposite thereto are released to block the application of power to the outside. The rupture pressure refers to a 2_{nd} pressure at which the safety vent 122 ruptures so that the internal gases leak out before explosion of the secondary battery.
- [60] Accordingly, it is necessary for the operating pressure to be smaller than the rupture pressure. The operating pressure is designed to be much larger than the rupture pressure. For example, the operating pressure and the rupture pressure may be adjusted to about 16 kgf/cm² and about 30 kgf/cm², respectively, to clearly distinguish the

operation and rupture of the cap assembly. The structural behaviors of such operation and rupture will be discussed in detail. The internal pressure of the secondary battery increases continuously upon internal abnormal operation of the secondary battery. By the operating pressure, the current-carrying contact 210 is detached from the portion 210' opposite thereto and the breakable junction 220 is detached from the portion 220' opposite thereto. A flowable electrolyte, liquids, and gases passes through the through-hole 530 of the support member to increase the internal pressure and directly lifts the safety vent 122 because the breakable junction is welded to the protrusion 420. A vertical force (F_y) to detach the breakable junction in the upward and downward directions and a horizontal force (F_x) in the horizontal direction are effective in providing a force (F) to break the portion 220' opposite to the breakable junction. By the rupture force (F), the notch 410 is broken to interrupt the flow of electric current in the secondary battery.

[61] That is, the detachment of the current-carrying contact and the breakage of the breakable junction can be accomplished by a combination of the vertical force (F_y) and the horizontal force (F_x).

[62] Even after the notch 410 is broken by the operating pressure to interrupt the flow of electric current, the 2nd pressure of the secondary battery may increase continuously. Since the increasing 2nd pressure increases the danger of explosion, it is necessary to reduce the internal pressure beforehand by the removal of internal gases and liquids (outgasing).

[63] To this end, a breakable notch 310 in the form of a long home is formed in the peripheral portion of the safety vent. The breakable notch 310 is designed to be broken by the internal pressure of the secondary battery. The safety vent is in close contact with the cap-down, making it difficult to bend or break the breakable notch 310. Thus, a bending-relieving portion 320 in the shape of a long home or groove is formed adjacent to the breakable notch 310. Due to its shape, the bending-relieving portion 320 can vary the bending angle (θ) of the breakable notch, enabling efficient breakage of the breakable notch. The bending-relieving portion is designed to have a bottom area (G) larger than that (N) of the breakable notch. This design is advantageous in varying the bending angle (θ) of the breakable notch to freely form a space between the bending-relieving portion and the cap-down opposite thereto, thus being effective in deforming or breaking the breakable notch.

[64] The breakable notch 310 and the bending-relieving portion 320 may be arranged concentrically around the center of the central portion on the bottom of the peripheral portion of the safety vent. The internal pressure increases continuously even after interruption of the flow of electric current by the 1st pressure and reaches the 2nd pressure at which the breakable notch is broken, and as a result, the secondary battery can be

protected from explosion.

[65] The safety vent 122 may be bent downwardly and extend to form a bent portion 1221 that surrounds the cap-down 126. The formation of the bent portion is beneficial in ensuring adhesion of the safety vent 122 to the cap-down 126. The extension length of the bent portion on the bottom of the cap-down is limited to increase the power generation capacity of the secondary battery.

[66] The power generating device JR is a device that produces electrical energy through chemical reactions. For example, in the case of a secondary lithium-ion battery, the power generating device JR may be a jelly-roll type electrode assembly including an anode material, a cathode material, an electrolyte, and a center pin. One of the electrodes is in electrical communication with the cap assembly 120 through an electrode tab. Power generated in the electrode assembly is supplied to an external circuit or load through the electrode tab.

[67] The insulating layer 600 extends beyond the end of the sub-plate 400 and reaches the end of the cap-down. Thus, the insulating layer 600 precludes an extension of the electrode tab other than the portion welded to the sub-plate 400 from being bent up toward the cap assembly 120 and coming into contact with the cap-down 126 to prevent the occurrence of electrical shorting during fabrication of the secondary battery and use in an application.

[68] The insulating layer 600 is interposed between and insulates the cap-down and the sub-plate and can prevent the electrode tab from being in electrical communication with the cap-down. However, the insulating layer 600 may impede the flow of gases through the through-hole 530, which is solved by the arrangement of a through-hole 610 at a position opposite to the through-hole 530. Spacer bridges 612 extend across the opposite through-hole to space the electrode tab from the cap-down. The spacer bridges can be understood as bar-type bridges traversing the opposite through-hole.

[69] As explained previously, the central hole 620 of the insulating layer 600 is formed opposite to the central hole 510 such that the central portion 200 is exposed. The sub-plate 400 is insertable into the insulating layer 600 through the central hole 620. The standing portion 625 in the form of a barrier protrudes along the circumference of the central hole 620 to prevent short circuits.

[70] For better attachment and adhesion of the insulating layer 600 between the cap-down 126 and the sub-plate 400, the cap-down and the sub-plate have long homes 1265 and 4005 in the form of depressions or grooves at sides opposite to each other, respectively. The insulating layer is interposed between the cap-down and the sub-plate, followed by pressing or thermal pressing to form a filling portion 615 in which the insulating layer is filled in the long homes.

[71] A gasket 130 is disposed between the battery can 110 and the safety vent 122. The

gasket 130 should be protected from being melted or lost by the application of energy or force generated upon external electrical shorting to prevent electrical shorting of the secondary battery. To this end, a short-circuit preventing layer 700 can be interposed between the safety vent and a gasket 130 at a position opposite to the end of the battery can.

- [72] Any insulating material may be used without particular limitation for the short-circuit preventing layer 700. For example, the short-circuit preventing layer may be made of a metallic, nonferrous metallic or non-metallic material. The surface of the short-circuit preventing layer made of a metallic or nonferrous metallic material is preferably insulated.
- [73] The material for the insulating layer is insulated by wet or dry physical or chemical process in such a manner that the surface chemical composition is changed to increase insulating properties. For example, aluminum is a current-carrying material but its insulating properties increase when an oxide layer is formed on the surface. An oxide film is formed on the surface of a metallic or nonferrous metallic material, for example, by dipping in an acidic or basic solution, electrolysis, anodizing or laser irradiation.
- [74] Particularly, a dry process by laser irradiation may have difficulty in processing the insulating layer or film whose surface is plated. In this case, after removal of the plating layer, high-energy laser beams are irradiated to form an oxide or nitride film on the surface of the insulating layer or film, achieving good insulating properties.
- [75] Alternatively, the short-circuit preventing layer 700 may be formed using a polymeric material. Since the short-circuit preventing layer 700 is required to have a sufficient physical support force in a high-temperature environment where the gasket 130 is melted or structurally weakened by thermal energy or force generated upon external short circuits, the material for the short-circuit preventing layer 700 should be resistant to heat at a temperature of 300 °C or more. However, the material is not particularly limited so long as its physical properties are maintained at a temperature of 300 °C or more.
- [76] The short-circuit preventing layer 700 may include an inorganic material. For example, the short-circuit preventing layer may be formed using an inorganic material or may be an insulating layer whose surface is coated with an inorganic material, for example, a powder of inorganic particles, to form an inorganic layer (not shown).
- [77] The short-circuit preventing layer 700 may take the form of a ring made of an oxidized or anodized metallic or nonferrous metallic material, for example, an aluminum material. The insulating layer acts as a kind of spacer that maintains the interval between the end (e) of the battery can and the safety vent 122 to prevent short circuits. The short-circuit preventing layer 700 may be provided in plurality. In this

case, the number of the short-circuit preventing layers is not limited so long as they are arranged between the end (c) of the battery can and the safety vent 122. It is beneficial from the viewpoint of production processability that the short-circuit preventing layer 700 is in the form of a ring. The short-circuit preventing layer 700 may be fixed to the safety vent 122 by soldering.

- [78] On the other hand, the present invention provides a secondary battery including the cap assembly. The secondary battery of the present invention may be fabricated by assembling an anode material, a cathode material, an electrolyte, a center pin, a jelly roll, etc. together in a battery can and aligning the cap assembly at the open end of the battery can (clamping). The fabrication method of the secondary battery is generally known in the art and detailed description thereof is thus omitted.

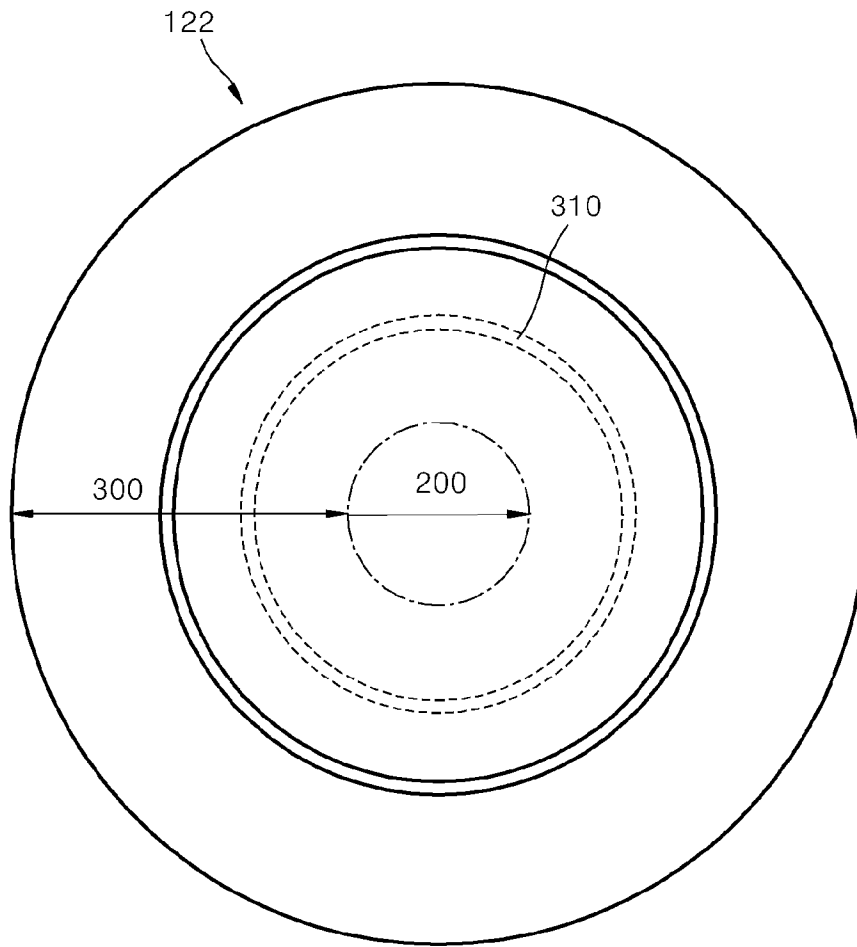
Claims

- [Claim 1] A cap assembly for a secondary battery comprising: a safety vent made of a circular metal plate and consisting of a central portion corresponding to the central area of the metal plate and a peripheral surrounding the central portion, the central portion comprising a current-carrying contact forming a first conductive path through which an electric current generated from a power generating device flows to the outside and a breakable junction forming a second conductive path at the center of the central portion; a cap-down opposite to the safety vent, having a central hole through which the breakable junction is exposed, and comprising a support member in close contact with the peripheral portion; and a sub-plate joined to the breakable junction and having a notch formed around the breakable junction, the outer circumferential surface of the sub-plate and the cap-down interposing an insulating layer therebetween.
- [Claim 2] The cap assembly according to claim 1, wherein the safety vent is bent downwardly and extends to form a bent portion.
- [Claim 3] The cap assembly according to claim 1, wherein the breakable junction is welded to a protrusion of the sub-plate entering the central hole formed in the support member.
- [Claim 4] The cap assembly according to claim 3, wherein the protrusion comprises a portion that protrudes upwardly opposite to the central portion and is opposite and melt-bonded to the breakable junction.
- [Claim 5] The cap assembly according to claim 4, wherein the notch is formed in the shape of a long home on the outer circumference of the portion opposite to the breakable junction.
- [Claim 6] The cap assembly according to claim 3, wherein the protrusion comprises a portion that protrudes upwardly opposite to the central portion and is opposite to and in close contact with the current-carrying contact.
- [Claim 7] The cap assembly according to claim 6, wherein the portion opposite to the current-carrying contact is formed on the outer circumference of the notch.
- [Claim 8] The cap assembly according to claim 1, wherein a breakable notch is formed at the bottom of the peripheral portion such that the safety vent is bent by a rupture pressure.
- [Claim 9] The cap assembly according to claim 8, further comprising a bending-

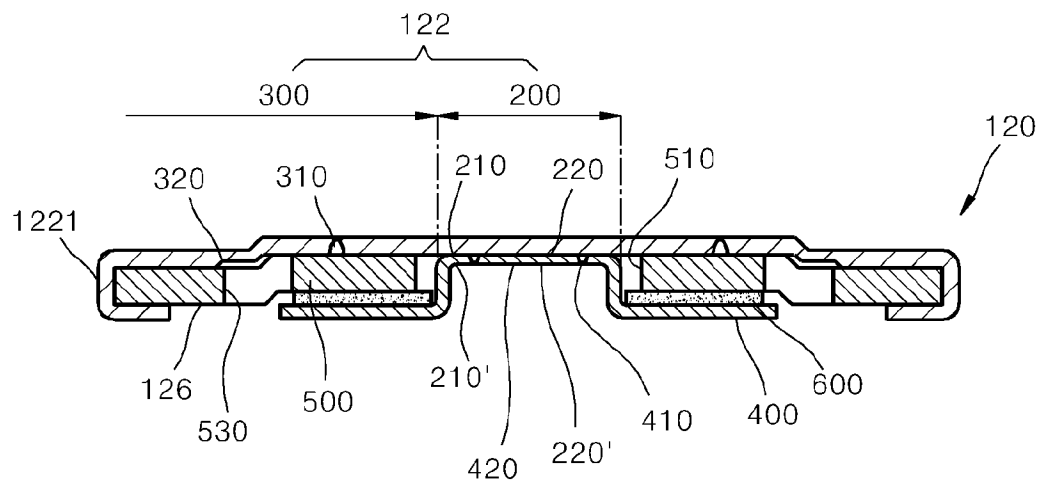
relieving portion adjacent to the breakable notch to space the current-carrying contact from the cap-down and forming a space where a rupture pressure is applied to the breakable junction.

- [Claim 10] The cap assembly according to claim 9, wherein the bottom width of the bending-relieving portion is larger than that of the breakable notch.
- [Claim 11] The cap assembly according to claim 1, wherein the cap-down comprises a through-hole that is spaced apart from and formed around the central hole and through which internal gases and liquids pass.
- [Claim 12] The cap assembly according to claim 1, wherein the insulating layer extends beyond the end of the sub-plate and reaches the end of the cap-down.
- [Claim 13] The cap assembly according to claim 12, wherein the insulating layer has a through-hole opposite to the through-hole of the cap-down.
- [Claim 14] The cap assembly according to claim 13, wherein spacer bridges are formed across the opposite through-hole.
- [Claim 15] The cap assembly according to claim 1, wherein a short-circuit preventing layer is formed at a position opposite to the end of the battery can through a gasket of the safety vent.
- [Claim 16] A secondary battery comprising the cap assembly according to any one of claims 1 to 15.

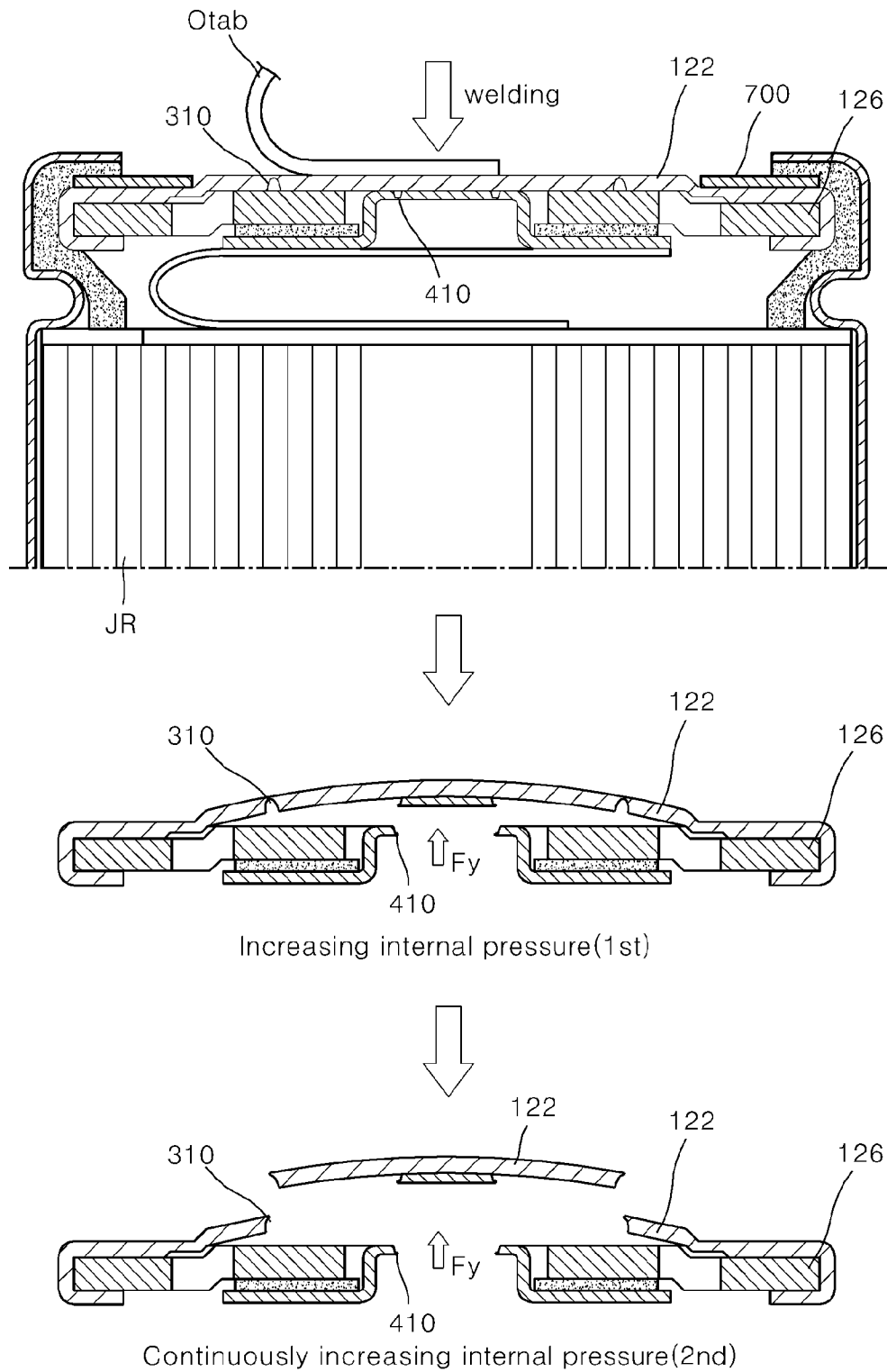
[Fig. 1]



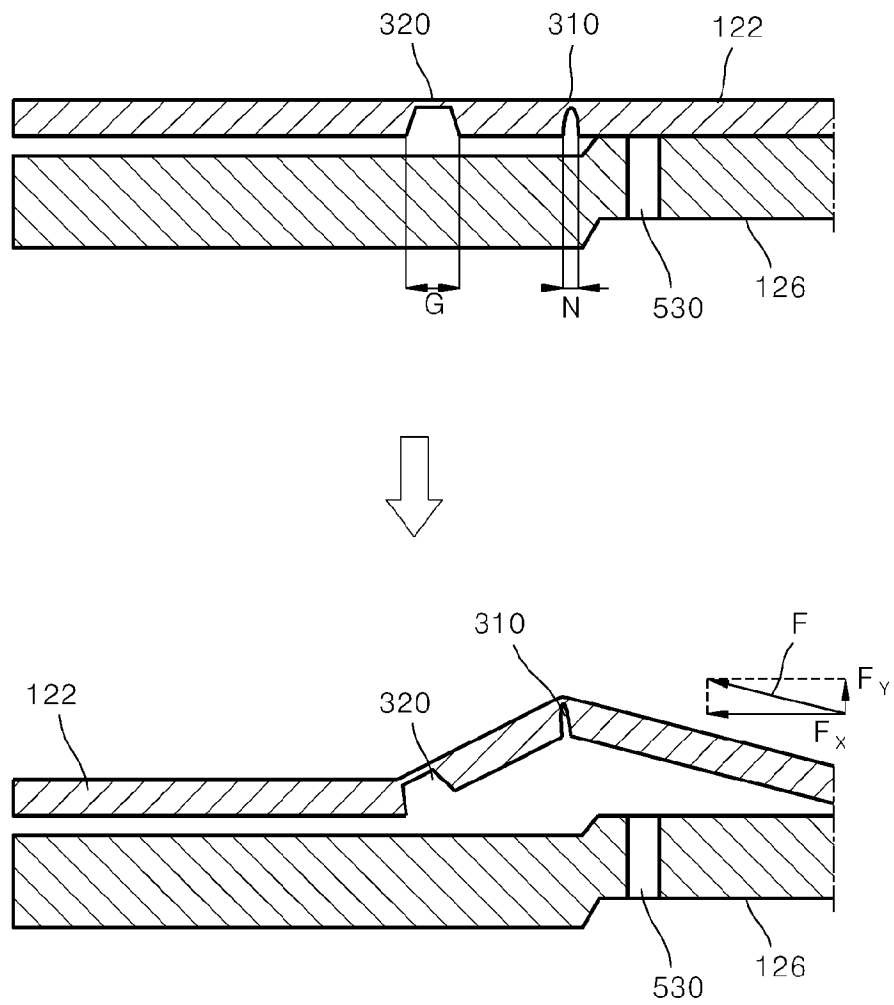
[Fig. 2]



[Fig. 3]



[Fig. 4]



600

625

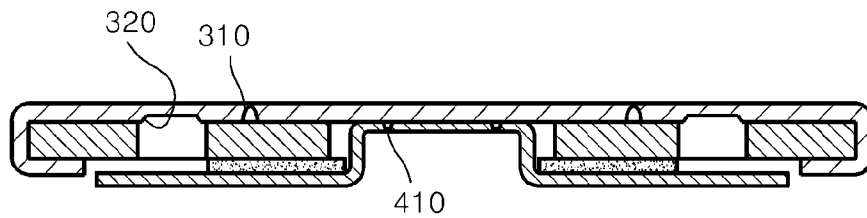
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612

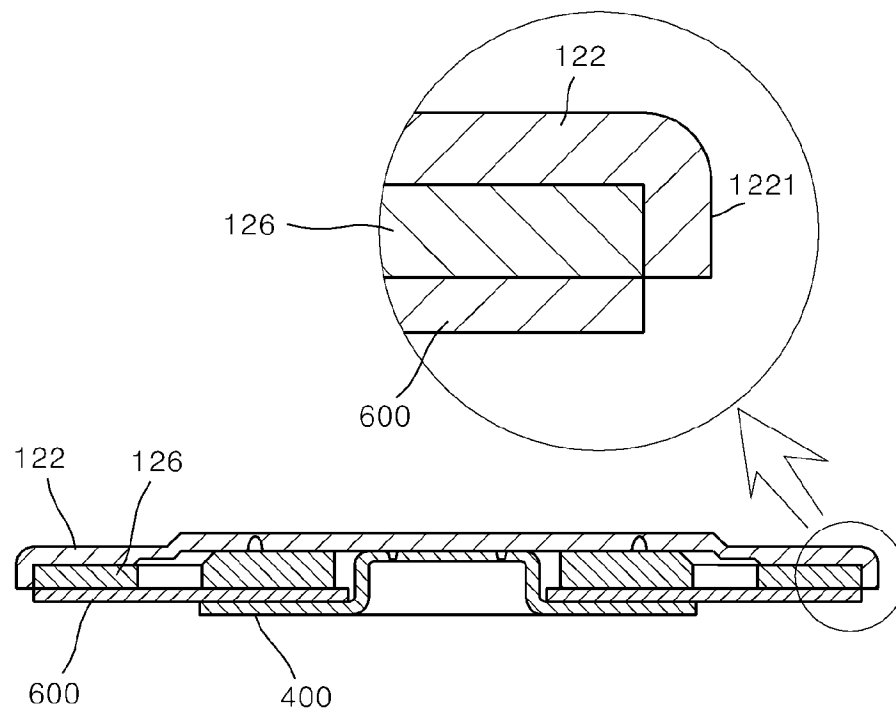
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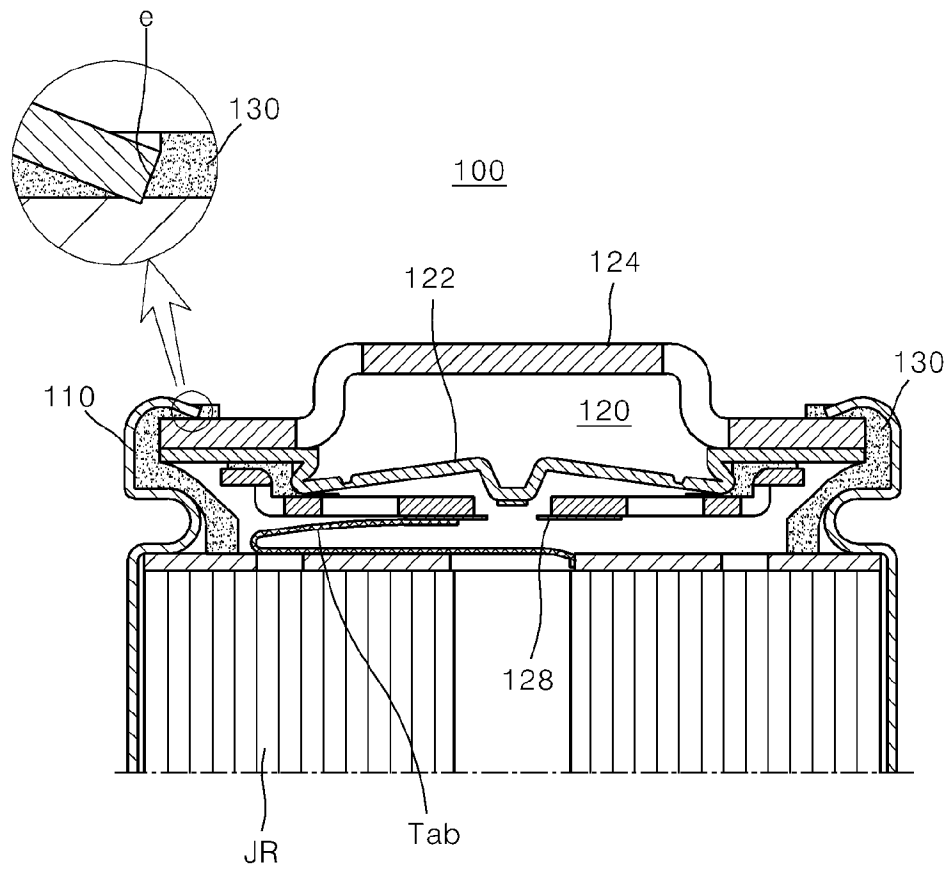
[Fig. 7]



[Fig. 8]



[Fig. 9]



INTERNATIONAL SEARCH REPORT

International application No.
PCT/KR2017/008736**A. CLASSIFICATION OF SUBJECT MATTER****H01M 2/04(2006.01)i, H01M 2/12(2006.01)i, H01M 10/04(2006.01)i**

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H01M 2/04; H01M 2/12; H01M 2/08; H01M 2/34; H01M 2/20; H01M 2/26; H01M 2/10; H01M 2/02; H01M 10/04

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Korean utility models and applications for utility models

Japanese utility models and applications for utility models

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

eKOMPASS(KIPO internal) & keywords: cap assembly, secondary battery, safety bent, shorting, notch

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KR 10-2016-0034794 A (SHIN HEUNG ENERGY & ELECTRONIC CO., LTD.) 30 March 2016 See figures 1-7; and claims 1-2.	1-16
A	US 2011 0206958 A1 (KIYAMA, A. et al.) 25 August 2011 See the entire document.	1 16
A	JP 2005 100927 A (MATSUSHITA ELECTRIC INDUSTRY CO., LTD.) 14 April 2005 See the entire document.	1 16
A	KR 10-2010-0062886 A (SAMSUNG SDI CO., LTD.) 10 June 2010 See the entire document.	1-16
A	KR 10-2006-0065378 A (LG CHEM, LTD.) 14 June 2006 See the entire document.	1-16



Further documents are listed in the continuation of Box C.



See patent family annex.

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Date of the actual completion of the international search

06 December 2017 (06.12.2017)

Date of mailing of the international search report

06 December 2017 (06.12.2017)

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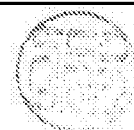
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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No.

PCT/KR2017/008736

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Doc code: IDS

PTO/SB/08a (02-18)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2020. OMB 0651-0031

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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	1	2175507	EP	A1	2010-04-14	Samsung Sdi Co., Ltd	U.S. Publication No. 20100086835 is the English language counterpart which is previously cited.	
	2	20170124343	KR	A	2017-11-10	Lg Chemical Ltd		×
	3	2018030836	WO	A1	2018-02-15	Shin Heung Energy & Electronic Co.,Ltd		□

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2021-11-01
Name/Print	Jordan T. Riviello	Registration Number	72,515

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Bibliographic data: JPH06215746 (A) — 1994-08-05

EXPLOSION-PROOF SAFETY DEVICE AND BATTERY HAVING IT

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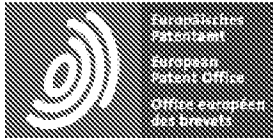
Classification: - **international:** H01M2/12; H01M2/34; (IPC1-7): H01M2/12
- **cooperative:** H01M50/3425 (EP); H01M50/574 (EP, US);
H01M50/578 (EP); Y02E60/10 (EP)

Application number: JP19930005593 19930118

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Abstract of JPH06215746 (A)

PURPOSE: To prevent the rupture of a battery, by interrupting an electric path between a terminal board and a power generating element, by a first inversion deformation when internal pressure in the battery is increased, and breaking a safety valve, concurrently with a second inversion deformation, to exhaust gas. **CONSTITUTION:** When battery internal pressure is increased by a short circuit, overcharge, and reverse charge, the battery internal pressure is transmitted to a metal-made safety valve 1 through an air vent hole provided on a metallic lead fitting plate 4, a metallic perforated terminal board 3, and an insulating ring 2. In the valve 1, since its outer peripheral part is calked by a battery case, when the battery internal pressure is reached to a given value, a center part is risen upward, a folded part 1b is reversely deformed, a fused part S is separated, and the connection with the terminal board 3 is released to interrupt current. Next, when battery internal pressure is suddenly increased, the valve 1 is ruptured concurrently with the reverse deformation of the folded part 1a of the valve 1 to exhaust gas to outside a battery through an air vent hole provided on a terminal board 5. Consequently, the rupture of the battery can be prevented.



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DESCRIPTION JPH06215746A

[0001]

¹³ [Industrial fields of application]

[0002]

¹⁷ The present invention relates to an explosion-proof safety device for use in batteries.

[0003]

²¹ [Prior art]

[0004]

²⁵ 2. Description of the Related Art In recent years, electronic devices such as AV devices and personal computers are becoming more portable and cordless.

²⁷ Lithium secondary batteries, in particular, have high expectations as batteries having a particularly high voltage and high energy density.

[0005]

³² However, sealed batteries, such as lithium secondary batteries that use lithium metal or carbon material as the negative electrode, generate gas inside the battery and increase internal pressure when misused such as short circuit, overcharge, or reverse charge. may explode and lose its function or damage peripheral devices.

[0006]

³⁹ In particular, if the overcharged state continues, the decomposition of the electrolyte and the decomposition of the active material may lead to a rapid temperature rise of the battery or an explosion.

[0007]

⁴⁵ In order to prevent the battery from exploding in such a short-circuit, overcharge, or reverse charge state, conventionally, as disclosed, for example, in Japanese Utility Model Publication No. 54-137734 and Japanese Utility Model Publication No. 59-15398, , an explosion-proof safety device has been proposed which has a flexible thin plate (diaphragm) that breaks as the internal pressure of the battery rises and discharges the internal pressure to the outside of the battery.

⁵¹ As shown in FIGS. 5(A) and 5(B), as the internal pressure rises, this safety device deforms and breaks the flexible thin plate 42 (diaphragm) outward to release gas. Although it is discharged outside the battery to prevent the battery from exploding, short-circuiting, overcharging, and reverse charging of the battery continue.

⁵⁵ Therefore, when this safety device is used in a battery with a high energy density (the amount of electrical energy that can be extracted from the battery per unit volume or unit weight), such as a lithium secondary battery, the internal pressure of the battery rises sharply especially if the overcharged state continues. However, there were cases in which gas was not exhausted due to the destruction of the flexible thin plate, leading to an explosion.

[0008]

⁶³ Japanese Patent Laid-Open Nos. 112151/1990 and 288063/1990 propose an explosion-proof safety device in which a lead and an explosion-proof valve are separated or cut off as the internal pressure of the battery rises to cut off current.

⁶⁶ As shown in FIGS. 6A and 6B, this explosion-proof safety device has an explosion-proof valve 21 that deforms outward as the internal pressure rises, and a protrusion 22a of the explosion-proof valve 21 contacts a lead blocking stripper 22. When the internal pressure reaches a predetermined value, the explosion-proof valve 21 ruptures the lead plate 23 to cut off the current and prevent the explosion.

[0009]

⁷⁴ However, in the case of the above conventional explosion-proof safety device, as can be seen in FIG. Since the lead plate 23 led out from one electrode plate is welded to the tip of the electrode, the connection strength tends to be unstable.

⁷⁷ If the connection strength varies, the battery internal pressure (operating pressure) when the current is interrupted will vary significantly.

[0010]

⁸² The connection strength can be stabilized by selecting suitable welding conditions for laser welding, ultrasonic welding, or the like.

[0011]

⁸⁷ However, lithium secondary batteries, which have a relatively high energy density among sealed batteries, use a non-aqueous electrolyte made of an organic solvent with a low flash point. The connection with the valve must be made in advance before filling the battery with the electrolyte.

⁹¹ That is, when assembling a battery, it is common to insert the electrode plate group into the battery case, connect the sealing plate to the lead plate, inject the electrolyte, and then seal.

⁹³ Therefore, even if the lead plate and the explosion-proof valve are connected with a certain connection strength, mechanical stress is applied to the connection during the assembly process before sealing, which causes variations in connection strength. Occurred.

⁹⁶ Reducing the mechanical stress applied to the connection between the lead plate and the explosion-proof valve as much as possible is likely to become an obstacle in mass production using an automatic assembly machine or the like.

⁹⁹ In addition, when the battery is subjected to mechanical vibration or external impact such as being dropped, the electrode plate group moves within the battery case, and the lead plate separates from the explosion-proof valve, resulting in loss of battery function. .

[0012]

¹⁰⁵ Therefore, the present invention facilitates mass production of batteries, and ensures the safety and reliability of batteries in short-circuit, overcharge, and reverse charge states even after shocks such as vibrations and drops are applied to the batteries. An object of the present invention is to provide an explosion-proof safety device capable of

[0013]

¹¹² The explosion-proof safety device of the present invention is electrically connected to the external terminal plate at the peripheral portion and to the internal perforated terminal plate at the central portion, respectively, so that the electricity from the power generation element to the external terminal plate is connected. a first reversal portion having a safety valve which also acts as an electrical conduction path, the safety valve being disposed centrally thereof and electrically connected to the internal perforated terminal plate in a normal state; and a second reversing portion disposed outside the portion, wherein the first reversing portion reversely deforms at a relatively low battery internal pressure to cut off the electrical connection between the internal perforated terminal plate and the safety valve. In addition, the second reversing portion is reversely deformed by a relatively high battery internal

pressure, and at least a part of the second reversing portion is broken, and the gas in the battery is discharged. It is.

[0014]

¹²⁷ It is desirable that the reversing portion is formed by at least two folds, and that a thin portion is formed in the bending portion that forms the second reversing portion.

[0015]

¹³² A battery of the present invention is characterized by comprising the above explosion-proof safety device.

[0016]

¹³⁷ [Operation] With the above configuration, when the internal pressure of the battery rises due to a short circuit, overcharge, or reverse charge, when the internal pressure of the battery reaches the lower set value, the bent portion corresponding to the lower set value causes reversal deformation, which cuts off the electrical conduction path and prevents rapid temperature rise of the battery. , the bent portion corresponding to the higher setting is reversed and destroyed at the same time, allowing the gas inside the battery to dissipate and prevent an explosion.

[0017]

¹⁴⁷ In addition, since the explosion-proof safety device itself has the function of interrupting electrical connection, it is possible to strengthen the electrical connection between the lead plate and the sealing plate, which facilitates the mass production of batteries, and further reduces the risk of vibration, dropping, etc. of batteries. Even if the electrode plate group moves within the battery case due to the impact of , the lead plate does not separate from the sealing plate, and a battery with high safety and reliability can be obtained.

[0018]

¹⁵⁶ BEST MODE FOR CARRYING OUT THE INVENTION The present invention will now be described with reference to specific embodiments together with the drawings.

[0019]

¹⁶¹ 1, 2 and 3 are diagrams showing an assembled sealing plate including an explosion-proof safety device according to an embodiment of the present invention.

¹⁶³ This assembly sealing plate consists of a flexible metal safety valve 1, an insulating ring 2,

and a metal internal perforated terminal plate 3 having a projection 3a provided in the center and a ventilation hole 3b formed around the projection 3a. , a metal lead mounting plate 4 , an external terminal plate 5 and an insulating gasket 6 .

- 167 The metal safety valve 1 is integrally formed with annular bent portions 1a and 1b, and the central portion of the metal safety valve 1 is a protrusion of a perforated metal terminal plate 3 projecting through an insulating ring 2. It is welded to the upper surface of the portion 3a, and the metal safety valve 1 and the terminal plate 3 are electrically connected by forming the welded portion s.
- 172 The metal safety valve 1 and the projecting portion 3a of the perforated metal terminal plate 3 are preferably welded together by ultrasonic welding, resistance welding, laser welding, or the like.
- 175 The bent portion 1a of the metal safety valve 1 may be provided with a thin portion 1c as shown in FIG.

[0020]

- 180 The metal lead mounting plate 4 is provided with a vent hole 4a, and the internal pressure of the battery is released through the vent hole 4a, the inner opening of the insulating ring 2, the vent hole 3b of the perforated metal terminal plate 3, and the metal terminal plate 3. It is configured to be fully added to the safety valve 1 .

[0021]

- 187 Next, the operation of the assembled sealing plate according to the present embodiment to cut off current in response to an increase in the internal pressure of the battery will be described with reference to FIGS. 4(A) and 4(B).
- 190 The assembly sealing plate is welded to the lead plate 7 attached to the electrode plate, and after the electrolyte is injected, it is attached by crimping the battery case 8, which also serves as the terminal plate, to seal the battery.
- 193 When the internal pressure of the battery rises due to a short circuit, overcharging, or reverse charging, the internal pressure of the battery passes through the metal lead mounting plate 4, the perforated metal terminal plate 3, and the metal safety valve 1 through the ventilation holes provided in the insulating ring 2. be communicated to
- 197 Since the outer periphery of the metal safety valve 1 is crimped and fixed by the battery case 8, when the battery internal pressure reaches a predetermined value, the central portion is lifted upward and bent as shown in FIG. 4A. The portion 1b is reversely deformed, and the welded portion s is peeled off by this stress, and the connection with the perforated metal terminal plate 3 is broken, thereby interrupting the current.

[0022]

- 205 It should be noted that current cut-off between metal safety valve 1 and metal perforated

terminal plate 3 due to an increase in internal pressure of the battery described above is caused by short circuit, overcharge, or breakage of metal safety valve 1 due to internal pressure in order to release the battery from a reverse charged state. For this reason, the welding strength between the metal safety valve 1 and the projecting portion 3a of the perforated metal terminal plate 3 should be made smaller than the breaking strength of the metal safety valve 1. It is configured.

212 In this embodiment, the bent portion 1b of the metal safety valve 1 is bent at an obtuse angle and the bent portion 1a is bent at an acute angle so that the bent portion 1b is reversely deformed at a lower pressure. ing.

[0023]

218 In addition, since the metal safety valve 1 and the perforated metal terminal plate 3 are sufficiently separated by the reversal deformation, the metal safety valve 1 and the perforated metal terminal plate 3 come into contact with each other due to fluctuations in the internal pressure of the battery after the current is cut off, causing a chattering phenomenon. I can't wake you up.

[0024]

226 3, that is, if the bent portion 1a of the metal safety valve 1 is provided with the thin portion 1c, as described above, in the short-circuit, overcharge, and reverse charge states, some other factor 4B, the thin portion 1c provided at the bent portion 1a of the metal safety valve 1 breaks simultaneously with the reverse deformation of the bent portion 1a. Thus, the gas can be reliably discharged to the outside of the battery through the ventilation holes 5a provided in the terminal plate 5.

[0025]

235 When the explosion-proof safety device of the present invention is used, since the explosion-proof safety device itself has a current interrupting mechanism, the welding strength to the lead plate can be made relatively high, so troubles such as the dislodgment of the assembly sealing plate during the battery assembly process can be prevented. It is possible to avoid this and facilitate the mass production of batteries.

240 Also, even if the battery is subjected to shock such as vibration or drop, it is possible to avoid troubles such as battery operation failure due to detachment of leads from the assembly sealing plate.

[0026]

246 Effect of the Invention According to the explosion-proof safety device of the present invention, it is possible to effectively prevent heat generation and explosion of the battery

even in the case of short circuit, overcharge, and reverse charge. Therefore, it is possible to obtain a battery equipped with a highly reliable explosion-proof safety device in which malfunction is unlikely to occur.

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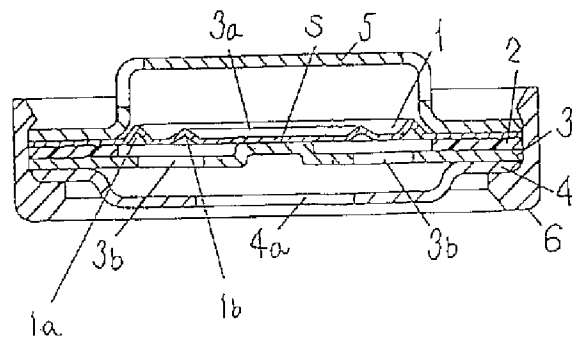
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(54)【発明の名称】 防爆安全装置および防爆安全装置を備えた電池

(57)【要約】

【目的】 電池が短絡、過充電、逆充電状態にあっても安全性と信頼性を確保できる防爆安全装置を提供することを目的とする。

【構成】 二重の反転変形構造を有し、電池内の内圧が上昇した時に、まず第一の反転変形によって端子板と発電要素間の電氣的導通経路を遮断し、次に第二の反転変形と同時に安全弁を破壊して電池内部の発生ガスを円滑に外部に排出し、電池の破裂を未然に防止する。



【特許請求の範囲】

【請求項1】 周縁部において外部端子板と、中央部において内部有孔端子板とそれぞれ電氣的に接続され、発電要素から外部端子板への電氣的導通経路としても作用する安全弁を備え、この安全弁が、その中央側に配置され、前記内部有孔端子板と通常状態においては電氣的に接続されている第1反転部と、該第1反転部の外方に配置された第2反転部とを備え、前記第1反転部が、比較的低い設定電池内部圧力で反転変形して、前記内部有孔端子板と安全弁との電氣的接続を遮断するようになっているとともに、前記第2反転部が、比較的高い設定電池内部圧力で反転変形するとともに、少なくとも一部が破断し、電池内ガスを排出するようになっている防爆安全装置。

【請求項2】 前記反転部が、少なくとも2重の折り曲がり部によって形成されている請求項1の防爆安全装置。

【請求項3】 前記第2反転部を形成する折れ曲がり部は、薄肉部が形成されている請求項2の防爆安全装置。

【請求項4】 請求項1ないし3のいずれかの防爆安全装置を備えた電池。

【発明の詳細な説明】

【0001】

【産業上の利用分野】

【0002】本発明は、電池に用いる防爆安全装置に関するものである。

【0003】

【従来の技術】

【0004】近年、AV機器あるいはパソコン等の電子機器のポータブル化、コードレス化が急速に進んでおり、これらの駆動用電源として小型、軽量で高エネルギー密度を有する密閉型電池への要望が高い。特にリチウム二次電池は、とりわけ高電圧、高エネルギー密度を有する電池として期待が大きい。

【0005】ところが、密閉型電池、例えばリチウム金属やカーボン材料を負極として使用するリチウム二次電池は、短絡、過充電、逆充電等の誤使用された時に、電池内にガスが発生し、内圧が上昇して破裂しその機能を失ったり、周辺機器に損傷を与えることがあった。

【0006】特に、過充電状態が継続すると、電解液の分解および活物質の分解により、電池が急激に温度上昇したり、爆発を招く恐れがあるという問題があった。

【0007】このような、短絡、過充電、逆充電状態における電池の爆発を未然に防止するため、従来は、例えば実開昭54-137734号公報、実公昭59-15398号公報に示されているように、電池内圧上昇に伴い破壊して内圧を電池外に排出する可撓性薄板（ダイヤフラム）を有する防爆安全装置が提案されている。この安全装置は、図5の（A）および（B）に示したように内圧の上昇に伴い、この内圧により可撓性薄板42（ダ

イアフラム）が外方に変形を生じて破壊してガスを電池外に排出し電池の爆発を未然に防止するものであるが、電池の短絡、過充電、逆充電状態は継続される。したがって、この安全装置をエネルギー密度（単位体積もしくは単位重量当りの電池から取り出せる電気エネルギー量）の高い電池、例えばリチウム二次電池に用いた場合、特に過充電状態が継続すると電池内圧が急激に上昇し、可撓性薄板の破壊によるガスの排出が追従せず、爆発にいたる場合があった。

【0008】その改良案として特開平2-112151号、特開平2-288063号には、電池内圧上昇に伴いリードと防爆弁を剥離もしくは切断して電流を遮断する防爆安全装置を提案している。この防爆安全装置は、図6の（A）および（B）に示したように内圧の上昇に伴い外方に変形を生じる防爆弁21を、その突起部22aがリード遮断用ストリッパ22に接触するようにして取り付け、所定の内圧に達したとき、防爆弁21の作用によりリード板23を破断するようにして電流を遮断し、爆発を防止するようにしたものである。

【0009】

【発明が解決しようとする課題】しかしながら、上記の従来の防爆安全装置の場合、図6で解るようにリード遮断用ストリッパ22の穴を介して突出した金属薄板製の防爆弁21の突起部22aの先端に一方の極板から導出したリード板23を溶接しているため、接続強度が不安定となりやすい。接続強度にバラツキが発生すると、電流遮断されるときに電池内圧（作動圧）の著しいバラツキとなる。

【0010】接続強度を安定化することは、レーザー溶接、超音波溶着等で適切な溶接条件を選定することで可能である。

【0011】ところが、密閉型電池のなかでも比較的エネルギー密度の高いリチウム二次電池の場合、電解液として、低引火点の有機溶媒からなる非水電解液を用いるため、火花の発生を伴うリード板と防爆弁との接続は、電池に電解液を注液する前にあらかじめ実施する必要がある。つまり、電池を組み立てる場合、極板群を電池ケースに挿入した後に、封口板をリード板に接続し、電解液を注液し、しかる後に封口することが一般的である。そこで、一定の接続強度でリード板と防爆弁を接続しても、その後封口するまでに、注液工程を経るため組み立て工程において接続部に機械的ストレスが加わり、結果的に接続強度のバラツキが発生した。リード板と防爆弁との接続部に加わる機械的ストレスを可能な限り少なくすることは自動組立機などで大量生産する際に障害となりやすい。また、電池に機械的振動もしくは、落下等外部から衝撃を加えた場合、電池ケース内において極板群が移動し、リード板が防爆弁より剥離し、電池としての機能を失うという問題があった。

【0012】そこで、本発明は、電池の大量生産が容易

で、電池に振動、落下等の衝撃が加わった後においても、短絡、過充電、逆充電状態における電池の安全性と信頼性を確保することのできる防爆安全装置を提供することを目的とするものである。

【0013】

【課題を解決するための手段】本発明の防爆安全装置は、周縁部において外部端子板と、中央部において内部有孔端子板とそれぞれ電氣的に接続され、発電要素から外部端子板への電氣的導通経路としても作用する安全弁を備え、この安全弁が、その中央側に配置され、前記内部有孔端子板と通常状態においては電氣的に接続されている第1反転部と、該第1反転部の外方に配置された第2反転部とを備え、前記第1反転部が、比較的低い電池内部圧力で反転変形して、前記内部有孔端子板と安全弁との電氣的接続を遮断するようになっているとともに、前記第2反転部が、比較的高い電池内部圧力で反転変形するとともに、少なくとも一部が破断し、電池内ガスを排出するようになっていることを特徴とするものである。

【0014】上記反転部は、少なくとも2重の折れ曲がり部によって形成され、前記第2反転部を形成する折れ曲がり部に、薄肉部が形成されていることが望ましい。

【0015】本発明の電池は、上記の防爆安全装置を備えていることを特徴とするものである。

【0016】

【作用】以上の構成により、短絡、過充電、逆充電状態により、電池内圧が上昇した場合、まず低い方の設定値に電池の内部圧力が達した時に、低い方の設定に該当する折れ曲がり部が反転変形を生じ、このことにより電氣的導通経路を遮断し電池の急激な温度上昇を防止することができ、また、急激なガス発生によって高い方の設定値に電池の内部圧力が達した時には、高い方の設定に該当する折れ曲がり部が反転変形を生じると同時に破壊され、電池内部のガスを逸散させ爆発を未然に防ぐことが可能である。

【0017】また、防爆安全装置自体が電氣的接続遮断の機能を有するため、リード板と封口板との電氣的接続を強くすることが可能となり、電池の量産が容易であり、さらに電池に振動、落下等の衝撃が加わり極板群が電池ケース内を移動しても、リード板が封口板から剥離することもなく、安全性、信頼性の高い電池を得ることができる。

【0018】

【実施例】以下、図面とともに本発明を具体的な実施例によって説明する。

【0019】図1、図2および図3は、本発明の実施例における防爆安全装置を含む組立封口板を示す図である。この組立封口板は、可撓性の金属製安全弁1、絶縁リング2、中央に突起部3aが設けられ、この突起部3aの周囲に通気孔3bが形成された金属製内部有孔端子

板3、金属製リード取りつけ板4、外部端子板5および絶縁ガasket6より構成されている。金属製安全弁1は、環状の折り曲がり部1aおよび1bが一体に形成されており、また金属製安全弁1の中央部は、絶縁リング2を介して突出する金属製有孔端子板3の突起部3aの上面に溶着しており、溶着部sの形成により、金属製安全弁1と端子板3とは電氣的に接続されている。なお、金属製安全弁1と金属製有孔端子板3の突起部3aとの溶着は、超音波溶着もしくは抵抗溶接、レーザー溶接等により行うことが適切である。金属製安全弁1の折り曲がり部1aには図3に示したように薄肉部1cが設けられていてもよい。

【0020】また、金属製リード取りつけ板4には、通気孔4aが設けられており、電池内圧は、この通気孔4a、絶縁リング2の内開口および金属製有孔端子板3の通気孔3b、金属製安全弁1に充分加えられるよう構成されている。

【0021】次に、上記本実施例による組立封口板が、電池内圧の上昇に応じて電流を遮断する動作について、図4の(A)および(B)を参照して説明する。組立封口板は、極板に付属したリード板7と溶接され、電解液注液後に、端子板を兼ねる電池ケース8をかしめることにより取り付けられ、電池を密閉している。短絡、過充電もしくは、逆充電等により電池内圧が上昇した場合、この電池内圧が金属製リード取りつけ板4、金属製有孔端子板3、絶縁リング2に設けられた通気孔を通して金属製安全弁1に伝えられる。金属製安全弁1はその外周部が電池ケース8によりかしめられ固定されているため、電池内圧が所定の値に達すると、図4の(A)に示すように中心部が上方へ持ち上げられ折り曲がり部1bが反転変形し、この応力によって前記溶着部sが剥離され金属製有孔端子板3との接続が解かれて電流が遮断される。

【0022】なお、上述の電池内圧の上昇による金属製安全弁1と金属製有孔端子板3との電流遮断は、電池の短絡、過充電もしくは、逆充電状態を解くため内圧による金属製安全弁1の破壊される以前に確実に行う必要があるが、そのために、金属製安全弁1と金属製有孔端子板3の突起部3aとの溶着はその溶着強度を金属製安全弁1の破壊強度よりも小さくするよう構成されている。本実施例においては、金属製安全弁1の折り曲がり部1bの折り曲がり角度を鈍角に折れ曲がり部1aの折り曲がり角度を鋭角に構成し、折り曲がり部1bの方が低圧で反転変形を生じるようにしている。

【0023】さらに、反転変形によって金属製安全弁1と金属製有孔端子板3が充分隔離されるため、電流遮断後に電池内圧の変動によって金属製安全弁1と金属製有孔端子板3が接触しチャタリング現象を起こすこともない。

【0024】また、図3に示すように構成すれば、すな

わち、金属製安全弁1の折れ曲がり部1aに薄肉部1cを設ければ、上述のように、短絡、過充電、逆充電状態において、他のなんらかの要因で電池の内圧が急激に上昇した場合においても、図4の(B)に示すように前記金属製安全弁1の折れ曲がり部1aに設けた薄肉部1cが折れ曲がり部1aの反転変形と同時に破壊され、端子板5に設けた通気孔5aを通してガスを電池外部に確実に排出することができる。

【0025】本発明の防爆安全装置を用いれば、防爆安全装置自体が電流遮断機構を有するため、リード板との溶接強度を比較的高くすることができるので、電池組み立て工程における組立封口板はずれのトラブル発生を回避でき電池の量産化を容易にすることができる。また、電池に振動あるいは落下等の衝撃を加えても組立封口板からのリードはずれによる電池動作不良といったトラブルも回避できる。

【0026】

【発明の効果】本発明における防爆安全装置によれば、短絡、過充電、逆充電時においても、電池の発熱、爆発を効果的に防止することができ、振動、落下等の衝撃を加えても動作不良が発生しにくい信頼性の高い防爆安全装置を備えた電池を得ることが可能である。

【図面の簡単な説明】

【図1】本発明の実施例の防爆安全装置を含む組立封口板の断面図である。

【図2】本発明の実施例の防爆安全装置を含む組立封口板の分解斜視図である。

【図3】本発明の実施例の防爆安全装置を含む組立封口板の断面図である。

【図4】上記防爆安全装置を備えた電池の作動状態を説明するための説明図である。

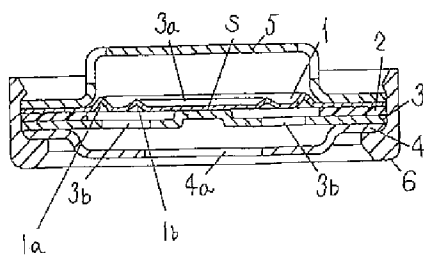
【図5】従来の密閉型電池断面図である。

【図6】従来の密閉型電池断面図である。

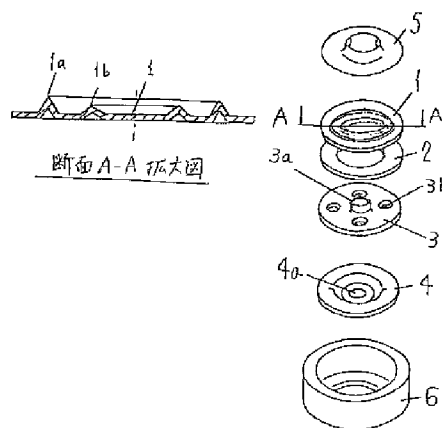
【符号の説明】

- 1 金属製安全弁
- 2 絶縁リング
- 3 金属製有孔端子板
- 4 金属製リード取り付け板
- 5 端子板
- 6 絶縁ガスカート
- 7 リード板
- 8 電池ケース
- 1a 折れ曲がり部
- 1b 折れ曲がり部
- 1c 薄肉部
- 3a 突起部
- 3b 通気孔
- 4a 通気孔
- 5a 通気孔

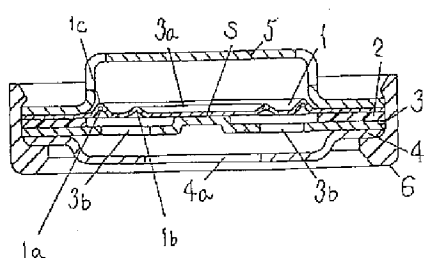
【図1】



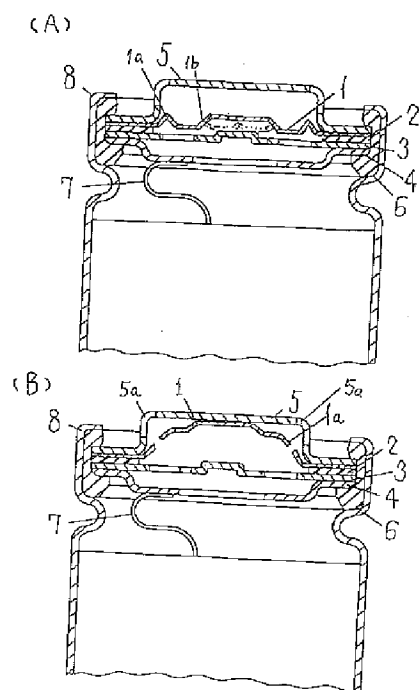
【図2】



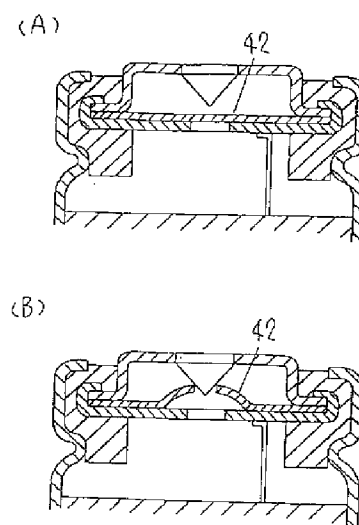
【図3】



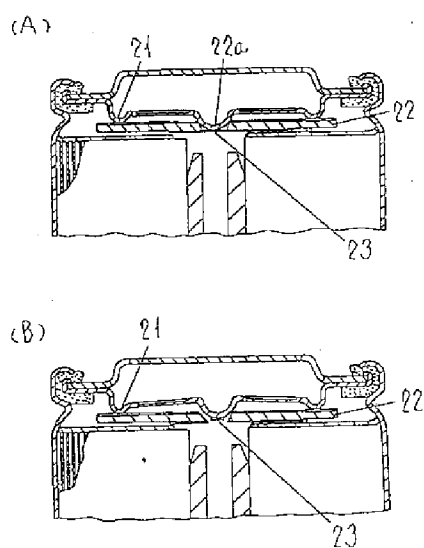
【図4】



【図5】



【図6】



フロントページの続き

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Application Number	17259395
Filing Date	2021-07-16
First Named Inventor	Geon Woo Min
Art Unit	
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

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	1	2572931	CA	A1	2008-06-21	Wu, Donald P. H		
	2	102473883	CN	A	2012-05-23	Lg Chemical Ltd	U.S. Publication No. 20120114979 A1 is an English language counterpart, which has been previously cited.	×
	3	103155225	CN	A	2013-06-12	Lg Chemical Ltd	U.S. Publication No. 20130216871 A1 is an English language counterpart.	☒

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	1	Search Report dated June 22, 2022 from the Office Action for Chinese Application No. 201980046222.8 issued June 30, 2022, 3 pages.	×

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Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	"17259395"	(US-PGPUB; USPAT)	OR	ON	ON	2023/01/31 01:55 PM
L2	1547994	(H01M50/3425 or H01M50/107 or H01M50/578 or H01M50/574 or H01M2200/20 or H01M50/56 or H01M50/152 or Y02E60/10 or H01M50/116).cpc.	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:17 PM
L3	3881	2 AND ruptur\$4 WITH (lid cap top cover)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:17 PM
L4	285	2 AND ruptur\$4 WITH (lid cap top cover) WITH notch	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:18 PM
L5	26	2 AND ruptur\$4 WITH (lid cap top cover) WITH (weak\$5 notch\$2) WITH bend\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:18 PM
L6	7	("20130216871" OR "20110008654" OR "10090497" OR "20050214641" OR "20100086835" OR "20110200854" OR "20120114979").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/01/31 02:42 PM
L7	23	2 AND ruptur\$4 WITH (lid cap top cover) WITH (weak\$5 notch\$2) WITH deform\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:46 PM
L8	155	2 AND ruptur\$4 WITH (lid cap top cover) WITH predetermin\$4 ADJ pressure	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:56 PM
L9	63	8 AND notch\$4	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/01/31 02:56 PM
L10	1	("20170294641").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/01/31 03:04 PM

L11	8	("20130273401" "6509114" "20100209765" "20140199567" "6258477" "20090148766" "20130230749" "20080314482").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 07:05 AM
L12	7596	((("LG") near3 ("CHEM") near3 ("LTD"))).AS,AANM.	(USPAT)	OR	ON	ON	2023/02/02 07:13 AM
L13	33	12 AND vent\$4 AND ruptur\$4 AND (bend\$4 bent)	(USPAT)	OR	ON	ON	2023/02/02 07:14 AM
L14	119	((("MIN") near3 ("Geon") near3 ("Woo")) OR (("LEE") near3 ("Je") near3 ("Jun")) OR (("JUNG") near3 ("Sang") near3 ("Suk"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/02/02 02:18 PM
L15	33	14 AND vent\$4 AND ruptur\$4 AND (bend\$4 bent)	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/02/02 02:18 PM
L16	26	("20090311583" "20050026033" "20110171508" "20130309529" "20020122974" "20110200854" "20170309881" "20100203367" "20180062122" "20060024576" "20170309881" "20050026033" "20150004446" "20100159310" "20150236317").PN.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2023/02/02 02:31 PM
L17	13	("20090311583" "20050026033" "20110171508" "20130309529" "20020122974" "20110200854" "20170309881" "20100203367" "20180062122" "20060024576" "20170309881" "20050026033" "20150004446" "20100159310" "20150236317").PN.	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 02:51 PM
L18	1	("20110076527").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 03:15 PM
L19	36	2 AND bending ADJ (part portion) WITH (c-shaped (open discontinuous) ADJ (shape circle))	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:21 PM
L20	36	2 AND bending ADJ (part portion) WITH (c-shaped ((open discontinuous) ADJ	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT;	OR	ON	ON	2023/02/02 03:31 PM

L21	974	(shape circle))) 2 AND bending ADJ (part portion notch) WITH (plurality first second) WITH (length width)	IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L22	991	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) WITH (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L23	9	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) WITH (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L24	3	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) WITH (longer bigger) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:40 PM
L25	0	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) near4 (longer bigger) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:41 PM
L26	4	2 AND (flex\$4 bending) ADJ (part portion notch) WITH (plurality first second) near4 (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:41 PM
L27	2	2 AND safety ADJ vent WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4) WITH (plurality first second) near4 (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:41 PM
L28	2	2 AND safety ADJ vent WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4 bent) WITH (plurality first second) near4 (differ\$4) ADJ (length width)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 03:55 PM
L29	18	2 AND safety ADJ vent WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4 bent) near4 (multiple plurality first)	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/02/02 04:16 PM
L30	3	("20120114979").pn.	(US-PGPUB; USPAT;	OR	ON	ON	2023/02/02

L31	1	("20120114979").pn.	USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	04:19 PM
L32	46	(h01m50/342.ipc. h01m50/107.ipc. h01m50/578.ipc. h01m50/574.ipc. h01m50/56.ipc. h01m50/152.ipc. h01m50/116.ipc.)	(US-PGPUB; USPAT)	OR	ON	ON	2023/02/02 04:20 PM
L33	3	32 AND vent	(JPO)	OR	ON	ON	2023/02/02 05:24 PM
							2023/02/02 05:25 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Bibliographic Data

Application No: 17/259,395

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged: /LISA S PARK/

Examiner's Signature

Initials

Title: Secondary Battery

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
07/16/2021	429	1729	LGES 3.3F-3059
RULE			

APPLICANTS

LG Chem, Ltd., Seoul, KOREA, REPUBLIC OF

INVENTORS

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Sang Suk Jung, Daejeon, KOREA, REPUBLIC OF

CONTINUING DATA

This application is a 371 of PCT/KR2019/015396 11/13/2019

FOREIGN APPLICATIONS

KOREA, REPUBLIC OF KR10-2018-0149369 11/28/2018

IF REQUIRED, FOREIGN LICENSE GRANTED**

05/08/2021

STATE OR COUNTRY

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Cranford, NJ 07016

UNITED STATES

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\$1,660



UNITED STATES PATENT AND TRADEMARK OFFICE

USPTO Automated Interview Request (AIR)

Mar 8 2023

This paper requesting to schedule and/or conduct an interview is appropriate because:

This submission is requested to be accepted as an authorization for this interview to communicate via the internet. Recognizing that Internet communications are not secure, I hereby authorize the USPTO to communicate with the undersigned concerning scheduling of the interview via video conference, instant messaging, or electronic mail, and to conduct the interview in accordance with office practice including video conferencing.

Name(s) :

Jordan T Riviello

S-signature:

/Jordan T Riviello/

Registration Number:

72515

U.S. Application Number:

17259395

Confirmation Number:

1017

E-mail Address:

jriviello@lerner david.com

Phone Number:

+1 9085186373

Proposed Time of Interview:

3-21-2023 1:00 PM ET

Alternative Proposed Time(s) of Interview:

3-22-2023 11:30 AM ET

Preferred Interview Type:

Telephonic

I am the applicant or applicant's representative for this application.

Topic for Discussion:

prior art rejections



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017
86765	7590	03/24/2023	EXAMINER	
LG			PARK, LISA S	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1729	
			NOTIFICATION DATE	DELIVERY MODE
			03/24/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOfficeAction@lernerdavid.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 17/259,395	Applicant(s) Min et al.		
	Examiner LISA S PARK	Art Unit 1729	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
LISA S PARK	Primary Examiner	Telephonic
Jordan Riviello	Attorney of Record	

Date of Interview: 21 March 2023

Issues Discussed:

Proposed Amendment(s)

Proposed amendments were discussed to overcome the rejection of record. No specific agreement was reached.

☒ Attachment

/LISA S PARK/ Primary Examiner, Art Unit 1729	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

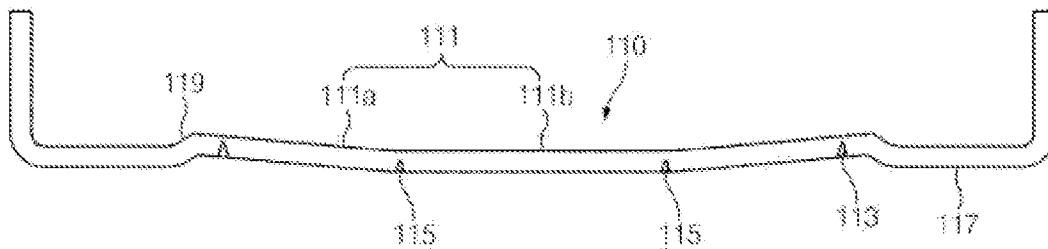
From: Riviello, Jordan T.
To: Park, Lisa
Cc: "LGES 2 SE 1736 - 4188 Correspondence"
Subject: [interview agenda] US 17/259,395 (LGES.4118)
Date: Monday, March 20, 2023 6:48:13 PM
Attachments: [image001.png](#)
[image002.png](#)

CAUTION: This email has originated from a source outside of USPTO. **PLEASE CONSIDER THE SOURCE** before responding, clicking on links, or opening attachments.

Examiner Park,

During our interview scheduled for Tuesday, March 20, 2023 at 1:00 pm EST, I would like to discuss the following information.

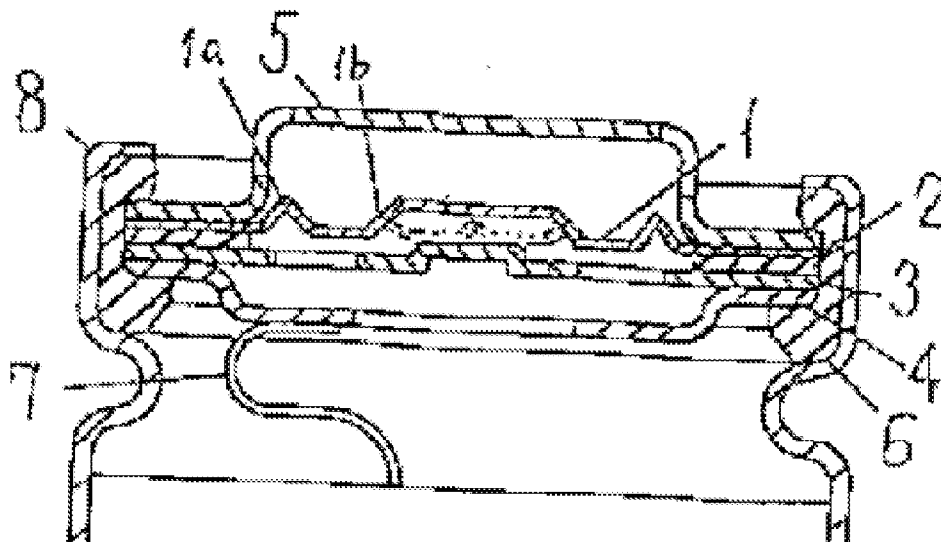
First, the rupture and bending notches are notches or grooves defined into the rupture portion. See for example FIG. 6 of the present application (reproduced below).



This mechanically changes the strength of the rupture portion and allows it to be cut at a preset venting pressure. See paragraph [0057]. In contrast, the prior art merely teaches a “rupture portion” that bends based on pressure but there is no “notch” defined into the structure. See for example FIG. 6 of the cited art where the structure is bent differently such that there is no bending notch present. With respect to the prior art, the rupturing is a result of the welding strength between the safety valve (1) and the projecting portion (3a). See page 6 of the English-language translation.

【 図 4 】

(A)



Moreover, we note that that this configuration allows the horizontal part (111b) to move from a lowermost position beneath rupture notch 113 (see fig. 2) where it can be coupled to the interrupt portion (121) and rise when the internal pressure increases to a location above the rupture notches (see FIG. 3). The prior art does not teach such movement.

I look forward to discussing the above distinctions with you in further detail during our telephone call tomorrow.

Kindly acknowledge receipt of this email.

Best regards,
Jordan

Jordan T. Riviello
jriviello@lernerjordan.com
908-518-6373 Direct


Lerner David Litfienberg Krumholz & Mentlik LLP
20 Commerce Drive, Cranford, NJ 07016
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/259,395	Filing Date 07/16/2021	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	05/09/2023		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	* 12	Minus	** 20	= 0	x \$100 = 0
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$480 = 0
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						0
AMENDMENT			CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/PAMELA V THERATT/
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".						
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronic Patent Application Fee Transmittal

Application Number:	17259395			
Filing Date:	16-Jul-2021			
Title of Invention:	Secondary Battery			
First Named Inventor/Applicant Name:	Geon Woo Min			
Filer:	James A. Vezeris/Kathleen Reynolds			
Attorney Docket Number:	LGES 3.3F-3059			
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	1251	1	220	220
Miscellaneous:				
Total in USD (\$)				220

Electronic Acknowledgement Receipt

EFS ID:	47974631
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	James A. Vezeris/Kathleen Reynolds
Filer Authorized By:	James A. Vezeris
Attorney Docket Number:	LGES 3.3F-3059
Receipt Date:	09-MAY-2023
Filing Date:	16-JUL-2021
Time Stamp:	15:13:09
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$220
RAM confirmation Number	E202359F13577437
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Extension of Time	One_Month_Request_for_Extension_of_Time_Under_37_CFR_1136a.pdf	24417	no	1
			8e106ff24a704533636322310fc0ad5b6c5012ca		
Warnings:					
Information:					
2	Miscellaneous Incoming Letter	Amendment_Transmittal.pdf	22637	no	1
			fa627fd8e10184329de5cc02540096642c9dfa42		
Warnings:					
Information:					
3	Amendment/Request for Reconsideration-After Non-Final Rejection	Amendment.pdf	45023	no	7
			ecd9b4591226bdb3f75eb9605ca49125bed9e35e		
Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	38046	no	2
			b1edb3e253596326c609f8924ee203b93317d58a		
Warnings:					
Information:					
Total Files Size (in bytes):			130123		

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PETITION FOR EXTENSION OF TIME UNDER 37 CFR 1.136(a)		Docket Number (Optional) LGES 3.3F-3059																															
Application Number 17/259,395-Conf. #1017		Filed July 16, 2021																															
For Secondary Battery																																	
Art Unit 1729		Examiner L. S. Park																															
This is a request under the provisions of 37 CFR 1.136(a) to extend the period for filing a reply in the above-identified application. The requested extension and fee are as follows (check time period desired and enter the appropriate fee below): <table style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 40%;"></th> <th style="width: 15%; text-align: center;">Fee</th> <th style="width: 15%; text-align: center;">Small Entity Fee</th> <th style="width: 15%; text-align: center;">Micro Entity Fee</th> <th style="width: 15%;"></th> </tr> </thead> <tbody> <tr> <td>☒ One month (37 CFR 1.17(a)(1))</td> <td style="text-align: center;">\$220</td> <td style="text-align: center;">\$88</td> <td style="text-align: center;">\$44</td> <td style="text-align: right;">\$ 220.00</td> </tr> <tr> <td>☐ Two months (37 CFR 1.17(a)(2))</td> <td style="text-align: center;">\$640</td> <td style="text-align: center;">\$256</td> <td style="text-align: center;">\$128</td> <td style="text-align: right;">\$ </td> </tr> <tr> <td>☐ Three months (37 CFR 1.17(a)(3))</td> <td style="text-align: center;">\$1,480</td> <td style="text-align: center;">\$592</td> <td style="text-align: center;">\$296</td> <td style="text-align: right;">\$ </td> </tr> <tr> <td>☐ Four months (37 CFR 1.17(a)(4))</td> <td style="text-align: center;">\$2,320</td> <td style="text-align: center;">\$928</td> <td style="text-align: center;">\$464</td> <td style="text-align: right;">\$ </td> </tr> <tr> <td>☐ Five months (37 CFR 1.17(a)(5))</td> <td style="text-align: center;">\$3,160</td> <td style="text-align: center;">\$1,264</td> <td style="text-align: center;">\$632</td> <td style="text-align: right;">\$ </td> </tr> </tbody> </table> ☐ Applicant asserts small entity status. See 37 CFR 1.27. ☐ Applicant certifies micro entity status. See 37 CFR 1.29. Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously. ☐ A check in the amount of the fee is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The Director has already been authorized to charge fees in this application to a Deposit Account. ☒ The Director is hereby authorized to charge any fees which may be required, or credit any overpayment, to Deposit Account Number 12-1095 . ☒ Payment made via USPTO's patent electronic filing system (Patent Center or EFS-Web). WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. I am the ☐ applicant. ☒ attorney or agent of record. Registration number 72,515 . ☐ attorney or agent acting under 37 CFR 1.34. Registration number . _____/Jordan T. Riviello/_____ Signature _____ Jordan T. Riviello Typed or printed name _____ May 9, 2023 Date _____ (908) 654-5000 Telephone Number NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms if more than one signature is required, see below*.					Fee	Small Entity Fee	Micro Entity Fee		☒ One month (37 CFR 1.17(a)(1))	\$220	\$88	\$44	\$ 220.00	☐ Two months (37 CFR 1.17(a)(2))	\$640	\$256	\$128	\$	☐ Three months (37 CFR 1.17(a)(3))	\$1,480	\$592	\$296	\$	☐ Four months (37 CFR 1.17(a)(4))	\$2,320	\$928	\$464	\$	☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,264	\$632	\$
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☐ Five months (37 CFR 1.17(a)(5))	\$3,160	\$1,264	\$632	\$																													

☐ * Total of 1 forms are submitted.

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 9, 2023

Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

AMENDMENT TRANSMITTAL LETTER			Docket No. LGES 3.3F-3059
Application No. 17/259,395-Conf. #1017	Filing Date July 16, 2021	Examiner L. S. Park	Art Unit 1729

Applicant(s): LG Chem, Ltd.

Invention: Secondary Battery

TO THE COMMISSIONER FOR PATENTS

Transmitted herewith is an amendment in the above-identified application.
The fee has been calculated and is transmitted as shown below.

CLAIMS AS AMENDED					
	Claims Remaining After Amendment	Highest Number Previously Paid	Number Extra Claims Present	Rate	
Total Claims	12	- 20 =	0	x	
Independent Claims	1	- 3 =	0	x	
Multiple Dependent Claims (check if applicable) ☐					
Other fee (please specify):					
TOTAL ADDITIONAL FEE FOR THIS AMENDMENT:					0.00

☒ Large Entity
 ☐ Small Entity

☒ No additional fee is required for this amendment.

☐ Please charge Deposit Account No. _____ in the amount of \$ _____.

☐ A check in the amount of \$ _____ to cover the filing fee is enclosed.

☐ Payment by credit card. Form PTO-2038 is attached.

☒ The Director is hereby authorized to charge and credit Deposit Account No. 12-1095 as described below. A duplicate copy of this sheet is enclosed.

☒ Credit any overpayment.

☒ Charge any additional filing or application processing fees required under 37 C.F.R. § 1.16 and 1.17.

/Jordan T. Riviello/
Dated: May 9, 2023
 Jordan T. Riviello (Atty/Agent Reg. No.: 72,515)
 LERNER DAVID LLP
 20 Commerce Dr.
 Cranford, New Jersey 07016
 (908) 654-5000

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 9, 2023

Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

I hereby certify that this paper (along with any paper referred to as being attached or enclosed) is being transmitted via the Office electronic filing system in accordance with 37 C.F.R. § 1.6(a)(4).

Dated: May 9, 2023
Electronic Signature for Jordan T. Riviello: /Jordan T. Riviello/

Docket No. LGCHEM 3.3F-3059
(PATENT)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

In re Patent Application of:
Min et al.

Application No.: 17/259,395

Confirmation No.: 1017

Filed: July 16, 2021

Art Unit: 1729

For: Secondary Battery

Examiner: L. S. Park

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

Dear Commissioner:

In response to the official action mailed February 8, 2023, Applicant submits the following amendments and remarks, along with a one-month extension of time and the requisite fee.

IN THE CLAIMS

1. (currently amended) A secondary battery comprising:
an electrode assembly;
a can member configured to accommodate the electrode assembly therein; and
a top cap assembly configured to cover an opening of the can member,
wherein the top cap assembly comprises a top cap positioned at an uppermost portion and a safety vent part positioned below the top cap,
wherein the safety vent part comprises:
a rupture portion configured to be ruptured when an internal pressure within the can member ~~increases to equal or exceeds~~ a predetermined pressure, the rupture portion including a first section and a second section;
a rupture notch portion that defines a boundary of the rupture portion; and
a bending notch portion which is disposed at a position spaced apart from the rupture notch portion toward a center of the rupture portion on a bottom surface of the rupture portion facing an inside of the can member, the bending notch portion separating the first section and the second section,
wherein the bending notch portion is bent when the internal pressure within the can member increases such that the first section is moved at least partially above the bending notch portion when the internal pressure is equal to the predetermined pressure.
2. (previously presented) The secondary battery of claim 1, wherein the rupture notch portion is formed as a recess within a lower surface of the safety vent part, and
the bending notch portion is formed as a recess within the lower surface of the safety vent part,
wherein a recessed depth of the bending notch portion is less than a recessed depth of the rupture notch portion.
3. (previously presented) The secondary battery of claim 1, wherein the rupture notch portion is provided in the bottom surface of the rupture portion.

4. (previously presented) The secondary battery of claim 1, wherein the top cap assembly further comprises a current interrupt device provided with an interrupt portion which is attached to the rupture portion and which is configured to be ruptured to interrupt current when the internal pressure within the can member increases.

5. (previously presented) The secondary battery of claim 4, wherein the interrupt portion of the current interrupt device is attached to the rupture portion inside a circumference of the bending notch portion.

6. (previously presented) The secondary battery of claim 1, wherein the bending notch portion has a continuous circular or elliptical shape.

7. (previously presented) The secondary battery of claim 1, wherein the bending notch portion has a discontinuous circular or elliptical shape.

8. (previously presented) The secondary battery of claim 1, wherein the bending notch portion comprises a plurality of bending notches, and the plurality of bending notches have different lengths.

9. (previously presented) The secondary battery of claim 1, wherein the rupture notch portion has a discontinuous circular or elliptical shape.

10. (currently amended) The secondary battery of claim 1, wherein a connection portion between an edge portion (417) and the rupture portion of the safety vent part is bent upward.

11. (currently amended) The secondary battery of claim 10, wherein ~~the rupture portion comprises:~~

the second section of the rupture portion is declined ~~an inclined section inclined~~ so that a height thereof gradually decreases from the connection portion toward the bending notch portion; and

~~the first section a horizontal section having~~has a horizontal shape and extending from an end of the ~~inclined declined~~ section toward the center of the rupture portion;

~~wherein the bending notch portion is positioned on a boundary between the inclined section and the horizontal section.~~

12. (currently amended) The secondary battery of claim 1, wherein a spaced distance between an edge end of the safety vent part and the rupture notch portion is equal to a spaced distance between the rupture notch portion (113) and the bending notch portion.

REMARKS

This Amendment is responsive to the February 8, 2023 non-final Office Action (“the Action”). No new claims have been added by way of this Amendment and no claims have been cancelled herein. Consequently, claims 1-12 remain pending in this application.

Initially, Applicant thanks the Examiner of the courtesies extended to Applicant’s representative during the telephonic interview of March 21, 2023 (“the Interview”) during which the prior art rejections and potential claim amendments for overcoming same were discussed. The Interview is evidenced by the Applicant-Initiated Interview Summary mailed on March 24, 2023 (“Interview Summary”). While no specific claim language was agreed upon during Interview, it is believed that a general agreement for advancing prosecution was reached. Independent claim 1 has been amended in view of this agreement. Support for the amendments to independent claim 1 may be found at least in FIGS. 2 and 3 of the original application. Claims 10-12 have been amended for clarity.

Claim Rejections under 35 U.S.C. §§ 102 and 103

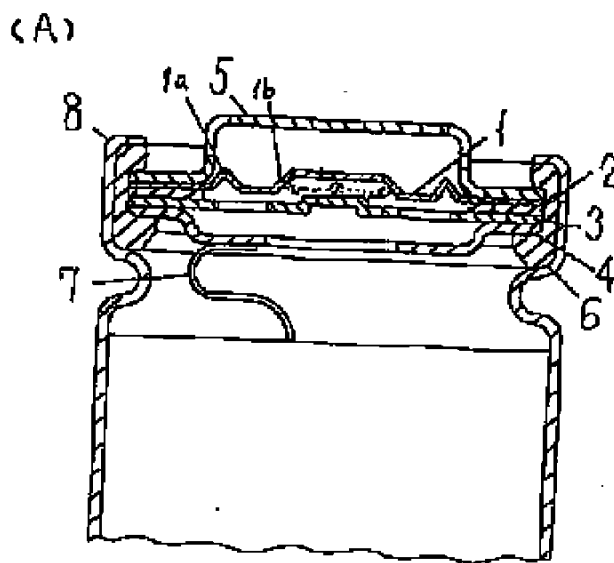
Claims 1-3 and 6 are rejected under 35 USC §102 as being anticipated by JPH 06215746 (“Mitamura”). Claims 4, 5, 7-9 and 12 are rejected under 35 USC §103 as obvious over Mitamura in view of US 2012/0114979 (“Kim ‘979”). Claim 10 is rejected under 35 USC §103 as obvious over Mitamura in view of WO 2018/030836 (“Kim ‘836”). Claim 11 is rejected under 35 USC §103 as obvious over Mitamura in view of Kim ‘979 and further in view of Kim ‘836. In view of the appended claim amendments and the following remarks, reconsideration and withdrawal of the rejections is requested.

In the Action, the bent portions (1a, 1b) of Mitamura are interpreted as the claimed “rupture notch portion” and the claimed “bending notch portion,” respectively. Although Applicant maintains that the bends are not “notches,” independent claim 1 has nevertheless been amended to expedite prosecution.

Amended independent claim 1 recites “the bending notch portion is bent when the internal pressure within the can member increases such that the first section is moved at least partially above the bending notch portion when the internal pressure is equal to the predetermined pressure.” In contrast thereto, as internal pressure within the device of Mitamura increases to the claimed predetermined pressure, the safety valve deforms. More specifically, the bent portion (1b) (equated

to the bending notch) deforms such that it no longer forms a “notch.” Consequently, in this state, the first section of Mitamura cannot be located at least partially above the bending notch portion as required by amended independent claim 1. *See* FIG. 4 of Mitamura reproduced below.

【図4】



Kim '979 and Kim '836 are cited in the Action merely in support of the assertion that the references teach one or more features recited in dependent claims 4, 5, 7-9, 11 and 12. Nevertheless, neither Kim '979 nor Kim '836 cure the deficiencies of Mitamura with respect to amended claim 1.

For these reasons, independent claim 1 is not anticipated or rendered obvious by the cited prior art and is in condition for allowance. Claims 2-12, depending directly or indirectly therefrom, are in condition for allowance for at least the same reasons.

Reconsideration of the rejections and allowance of the claims are earnestly solicited. If, however, for any reason the Examiner does not believe that such action can be taken at this time, the Examiner is encouraged to telephone Applicant's undersigned representative at (908) 654-5000 to discuss any additional objections which the Examiner might have.

All required fees, if any, are being paid via credit card. However, please charge any additional fees not already paid via credit card, and credit any overpayments, to our Deposit Account No. 12-1095.

Dated: May 9, 2023

Respectfully submitted,

Electronic signature:

/Jordan T. Riviello/

Jordan T. Riviello

Registration No.: 72,515

LERNER DAVID LLP

20 Commerce Dr.

Cranford, New Jersey 07016

(908) 654-5000

Attorney for Applicant

Issue Classification 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

CPC						
Symbol					Type	Version
H01M	/	50	/	3425	F	2021-01-01
H01M	/	50	/	107	I	2021-01-01
H01M	/	50	/	578	I	2021-01-01
H01M	/	2200	/	20	A	2013-01-01
H01M	/	50	/	574	A	2021-01-01

CPC Combination Sets							
Symbol				Type	Set	Ranking	Version
	/		/				

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	12	
/LISA S PARK/ Primary Examiner, Art Unit 1729	20 May 2023	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	4

<i>Issue Classification</i> 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

INTERNATIONAL CLASSIFICATION					
CLAIMED					
H01M	/	50	/	342	
H01M	/	50	/	107	
H01M	/	50	/	578	
H01M	/	50	/	574	
H01M	/	50	/	56	
H01M	/	50	/	152	
H01M	/	50	/	116	
NON-CLAIMED					
	/		/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	12	
/LISA S PARK/ Primary Examiner, Art Unit 1729	20 May 2023	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	4

Issue Classification 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	10	10												
2	2	11	11												
3	3	12	12												
4	4														
5	5														
6	6														
7	7														
8	8														
9	9														

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 12	
/LISA S PARK/ Primary Examiner, Art Unit 1729 (Primary Examiner) _____ (Date) 20 May 2023		O.G. Print Claim(s) 1	O.G. Print Figure 4

<i>Search Notes</i> 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

CPC - Searched*		
Symbol	Date	Examiner
H01M50/3425 or H01M50/107 or H01M50/578 or H01M50/574 or H01M2200/20 or H01M50/56 or H01M50/152 or Y02E60/10 or H01M50/116 (text-limited)	01/31/2023, 5/20/2023	/L.S.P./

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
PE2E keyword search (see attached search history)	02/02/2023, 5/20/2023	/L.S.P./
Analysis of Global Dossier file	02/02/2023	/L.S.P./
PE2E Applicant/Assignee search	02/02/2023	/L.S.P./
PE2E foreign database search	02/02/2023	/L.S.P./

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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<i>Search Notes</i> 	Application/Control No. 17/259,395	Applicant(s)/Patent Under Reexamination Min et al.
	Examiner LISA S PARK	Art Unit 1729

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
H01M50	3425, 107, 578, 574, 56, 152, 116	05/20/2023	/L.S.P./
H01M2200	20	05/20/2023	/L.S.P./
Y02E60	10	05/20/2023	/L.S.P./

/LISA S PARK/ Primary Examiner, Art Unit 1729	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L34	50	2 AND vent\$4 WITH (flex\$4 bending) ADJ (part portion notch) AND (flex\$4 bend\$4 bent) near4 (multiple plurality first) AND pressure	(US-PGPUB; USPAT; USOCR; FIT (CN, FR, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/05/20 03:17 PM

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	83	(H01M50/3425 OR H01M50/107 OR H01M50/578 OR H01M50/574 OR H01M2200/20 OR H01M50/56 OR H01M50/152 OR Y02E60/10 OR H01M50/116).cpc. AND ((break\$4 ruptur\$4) AND (bend\$4 bent) AND (section portion part) near3 (first second)).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2023/05/20 03:19 PM
N2	5	(H01M50/3425 OR H01M50/107 OR H01M50/578 OR H01M50/574 OR H01M2200/20 OR H01M50/56 OR H01M50/152 OR Y02E60/10 OR H01M50/116).cpc. AND ((break\$4 ruptur\$4) AND (bend\$4 bent) AND (section portion part) near3 (first second) AND predetermined ADJ pressure).clm.	(US-PGPUB; USPAT)	OR	ON	ON	2023/05/20 03:19 PM

Doc code: IDS

PTO/SB/08a (02-18)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2020. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Geon Woo Min	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

U.S.PATENTS

Remove

Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	10090497	B2	2018-10-02	Kim et al.	

If you wish to add additional U.S. Patent citation information please click the Add button.

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U.S.PATENT APPLICATION PUBLICATIONS

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1	20050214641	A1	2005-09-29	Kim	
	2	20100086835	A1	2010-04-08	KIM	
	3	20110200854	A1	2011-08-18	Bak et al.	
	4	20120114979	A1	2012-05-10	KIM et al.	

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Geon Woo Min
Art Unit	N/A
Examiner Name	Not Yet Assigned
Attorney Docket Number	LGCHEM 3.3F-3059

1	2001102038	JP	A	2001-04-13	Toyo Kohan Co Ltd	Machine generated English translation attached.	×
2	H06215746	JP	A	1994-08-05	Matsushita Electric Ind Co Ltd	Machine generated English translation attached.	☒
3	H08339793	JP	A	1996-12-24	Mitsubishi Cable Ind Ltd	Machine generated English translation attached.	☒
4	100578805	KR	B1	2006-05-11	Samsung Sdi Co., Ltd	U.S. Publication No. 20050214641 A1 is an English language counterpart.	☒
5	101058386	KR	B1	2011-08-22	Lg Chem. Ltd	U.S. Publication No. 20120114979 A1 is an English language counterpart.	☒
6	20100038054	KR	A	2010-04-12	Samsung Sdi Co., Ltd	U.S. Publication No. 20100086835 A1 is an English language counterpart.	☒
7	20110095118	KR	A	2011-08-24	Samsung Sdi Co., Ltd	U.S. Publication No. 20110200854 A1 is an English language counterpart.	☒
8	20150035267	KR	A	2015-04-06	Lg Chem. Ltd	Machine generated English translation attached.	☒

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NON-PATENT LITERATURE DOCUMENTS

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report for Application No. PCT/KR2019/015396, mailed February 20, 2020, pages 1-2.	×

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Geon Woo Min	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

EXAMINER SIGNATURE

Examiner Signature	/LISA S PARK/	Date Considered	05/20/2023
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Geon Woo Min	
Art Unit	N/A	
Examiner Name	Not Yet Assigned	
Attorney Docket Number	LGCHEM 3.3F-3059	

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Jordan T. Riviello/	Date (YYYY-MM-DD)	2021-01-11
Name/Print	Jordan T. Riviello	Registration Number	72,515

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

86765 7590 06/01/2023
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EXAMINER

PARK, LISA S

ART UNIT

PAPER NUMBER

1729

DATE MAILED: 06/01/2023

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017

TITLE OF INVENTION: Secondary Battery

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	09/01/2023

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

86765 7590 06/01/2023
LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017

TITLE OF INVENTION: Secondary Battery

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	09/01/2023

EXAMINER	ART UNIT	CLASS-SUBCLASS
PARK, LISA S	1729	429-056000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via Patent Center or EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017
86765	7590	06/01/2023	EXAMINER	
LG			PARK, LISA S	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1729	
DATE MAILED: 06/01/2023				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 17/259,395	Applicant(s) Min et al.	
	Examiner LISA S PARK	Art Unit 1729	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed 5/8/2023.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-12. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☒ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>20210111</u> .	6. ☒ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____.	7. ☐ Other _____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	

/LISA S PARK/ Primary Examiner, Art Unit 1729	
--	--

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

2. In response to the amendment received on 5/9/2023:
 - a. Claims 1-12 are pending in the current application. Claims 1 and 10-12 are amended.
 - b. The previous prior art rejections are overcome in light of the amendment.

Allowable Subject Matter

3. **Claims 1-12 are allowed.**

The following is an examiner's statement of reasons for allowance: none of the prior art of record, alone or in combination, appears to teach, suggest, or render obvious the invention of Claims 1-12.

Independent Claim 1 recites a very specific structure for a safety vent of a battery top cap assembly, said vent comprising a rupture portion and a bending notch portion that have structural relationships relative to one another (e.g. the first section being moved partially above the bending notch portion and the bending notch portion separating first and second sections) that are not taught by the prior art, and there is no prior art available to teach or make obvious such specific structure. Therefore, the reference fails to teach or suggest the particulars of the independent Claim 1, and it's not obvious to modify these teachings to give the instant claimed

invention. Thus none of the prior art of the record, alone or in combination, appears to teach, suggest, or render obvious the invention of independent Claim 1. Since Claims 2-12 depend on Claim 1, they are allowable for the same reason.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LISA S PARK whose telephone number is (571)270-3597. The examiner can normally be reached on M-Th 6 or 7 to 3 or 4; Alt Fri afternoons.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Ula Ruddock can be reached on 5712721481. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LISA S PARK/
Primary Examiner, Art Unit 1729

PART B—FEE(S) TRANSMITTAL

Complete and send this form, together with the applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

By fax, send to: (571) 273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

LERNER DAVID LLP
20 Commerce Dr.
Cranford, New Jersey 07016

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

Jordan T. Riviello	(Typed or printed name)
/Jordan T. Riviello/	(Signature)
July 25, 2023	(Date)

APPLICATION NO	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO	CONFIRMATION NO
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017

TITLE OF INVENTION: Secondary Battery

APPLN: TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1,200.00			\$1,200.00	09/01/2023

EXAMINER	ART UNIT	CLASS-SUBCLASS
L. S. Park	1729	429-056000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363)

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) The names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1. Lerner David LLP

2.

3.

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE or COUNTRY)

LG Energy Solution, Ltd.

Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent):

☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees Submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment (Please first reapply any previously paid fee shown above):

☒ Electronic Payment via Patent Center or EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 12-1095

5. Change of Entity Status (from status indicated above)

- ☐ Applicant certifying micro entity status. See 37 CFR 1.29.
- ☐ Applicant asserting small entity status. See 37 CFR 1.27.
- ☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid Certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken as a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken as a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature

/Jordan T. Riviello/

Date July 25, 2023

Typed or printed name

Jordan T. Riviello

Registration No.

72,515

Electronic Patent Application Fee Transmittal

Application Number:	17259395			
Filing Date:	16-Jul-2021			
Title of Invention:	Secondary Battery			
First Named Inventor/Applicant Name:	Geon Woo Min			
Filer:	Keith McWha/Melinda Cormier			
Attorney Docket Number:	LGES 3.3F-3059			
Filed as Large Entity				
Filing Fees for U.S. National Stage under 35 USC 371				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	48346471
Application Number:	17259395
International Application Number:	
Confirmation Number:	1017
Title of Invention:	Secondary Battery
First Named Inventor/Applicant Name:	Geon Woo Min
Customer Number:	86765
Filer:	Keith McWha/Melinda Cormier
Filer Authorized By:	Keith McWha
Attorney Docket Number:	LGES 3.3F-3059
Receipt Date:	25-JUL-2023
Filing Date:	16-JUL-2021
Time Stamp:	15:26:51
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E20237OF27428204
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	Issue_Fee_Transmittal.pdf	37048	no	1
			c55f4cee834c11874aad868670ad994e5cc60b6d		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	37667	no	2
			c74f59021a213d817a28c1d45c9eb42ce4fc36ab		

Warnings:

Information:

Total Files Size (in bytes):			74715
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	09/05/2023	11749866	LGES 3.3F-3059	1017

86765 7590 08/16/2023

LG

Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 194 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

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APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/259,395	07/16/2021	Geon Woo Min	LGES 3.3F-3059	1017
86765	7590	09/05/2023	EXAMINER	
LG			PARK, LISA S	
Lerner David LLP			ART UNIT	
20 Commerce Drive			PAPER NUMBER	
Cranford, NJ 07016			1729	
			NOTIFICATION DATE	
			DELIVERY MODE	
			09/05/2023	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eOfficeAction@lernerdavid.com



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APPLICATION NO.	ISSUE DATE	PATENT NO.
17/259,395	05-Sep-2023	11749866

LG
Lerner David LLP
20 Commerce Drive
Cranford, NJ 07016

EGRANT NOTIFICATION

Your electronic patent grant (eGrant) is now available, which can be accessed via Patent Center at <https://patentcenter.uspto.gov>

The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>